Linear Gaussian state-space model

Table 1: Linear Gaussian SSM: state RMSE (lower is better). $x_t = 0.8x_{t-1} + 0.5\varepsilon_t$, $y_t = x_t + 0.8\eta_t$. 50 steps, resampling when ESS $< 0.5N$; mean $\pm$ 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.52 ± 0.045	0.506 ± 0.047	0.505 ± 0.0508	0.504 ± 0.0504
DSR (τ=0.1, no clamp)	0.52 ± 0.045	0.506 ± 0.047	0.505 ± 0.0508	0.504 ± 0.0504
DSR (τ=0.1, unsorted)	0.524 ± 0.0468	0.51 ± 0.0505	0.509 ± 0.0547	0.507 ± 0.0551
DSR (τ=0.5)	0.523 ± 0.0467	0.509 ± 0.0506	0.509 ± 0.0545	0.507 ± 0.0549
DSR (τ=2.5/N)[†]	0.521 ± 0.0446	0.507 ± 0.0457	0.504 ± 0.0483	0.503 ± 0.0478
Systematic	0.52 ± 0.0464	$\mathbf{0.506 \pm 0.0464}$	0.505 ± 0.0477	0.503 ± 0.048
Multinomial	0.521 ± 0.0435	0.507 ± 0.0451	0.504 ± 0.048	0.503 ± 0.0479
Systematic, stop-gradient (PyDPF)	0.521 ± 0.0439	0.506 ± 0.0469	0.505 ± 0.048	0.503 ± 0.0477
Soft (α=0.5)	0.523 ± 0.0472	0.508 ± 0.046	0.505 ± 0.0478	$\mathbf{0.503 \pm 0.0476}$
OT (PyDPF)	0.53 ± 0.0456	0.515 ± 0.0434	–	–
OT (PFJAX)	0.529 ± 0.0456	0.515 ± 0.0434	–	–
OT (FilterFlow)	0.53 ± 0.0456	0.515 ± 0.0439	–	–
Eigen moment matching	0.52 ± 0.045	0.507 ± 0.0455	0.505 ± 0.0481	0.503 ± 0.0478
Gaussian moment matching	0.521 ± 0.0445	0.507 ± 0.0456	$\mathbf{0.504 \pm 0.0478}$	0.503 ± 0.0476
Gumbel-softmax	0.526 ± 0.0464	0.514 ± 0.048	–	–
Variational (KL)	$\mathbf{0.519 \pm 0.0446}$	0.507 ± 0.0468	0.504 ± 0.0482	0.503 ± 0.0482

[†]Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

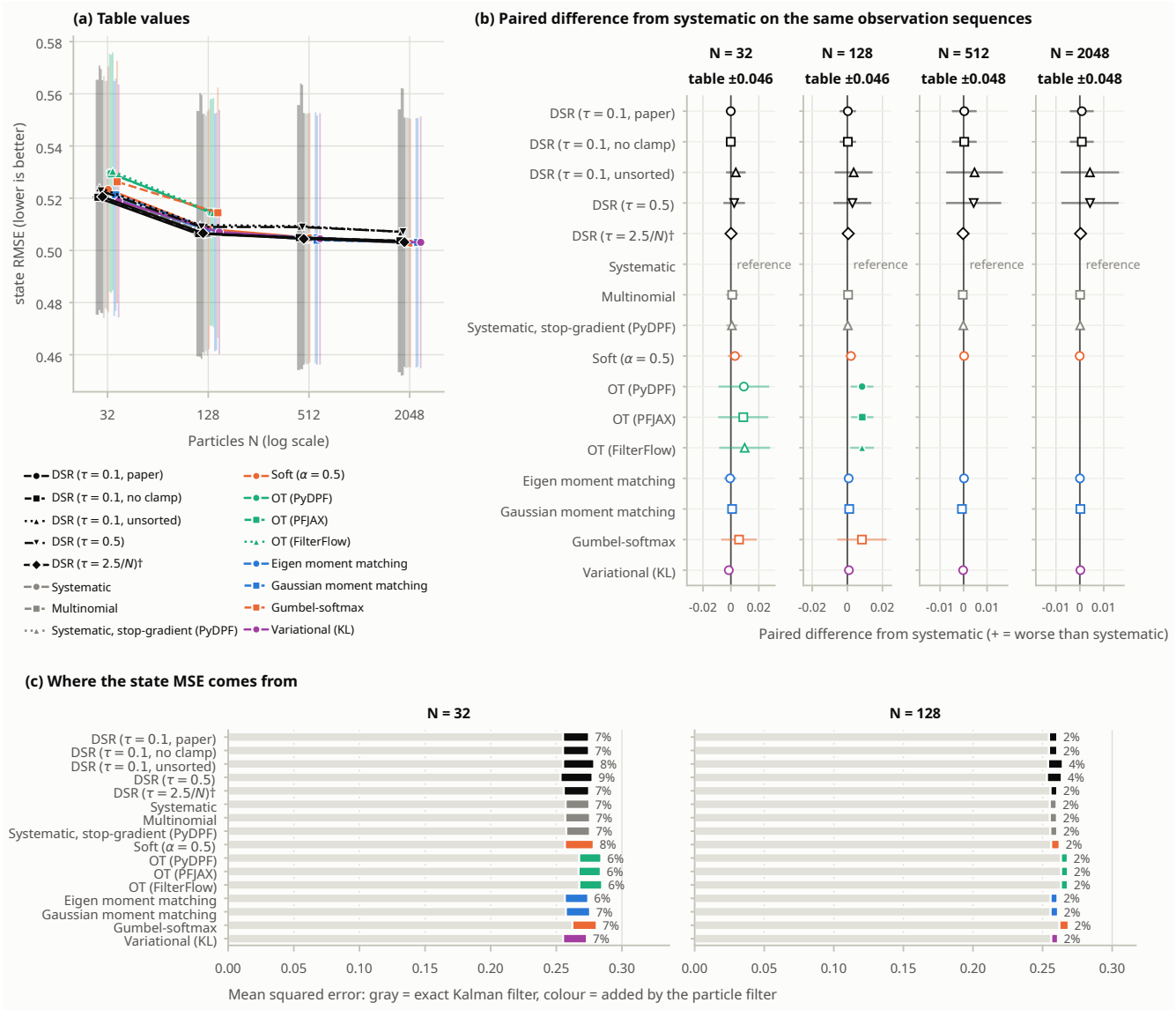


Figure 1: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here. **(c)** Mean squared state error, split into the exact Kalman filter's error (gray; it depends only on the data) and the extra error from the particle filter (colour; the squared posterior-mean RMSE of Table **??**). The particle filter contributes only 2–9% of the MSE, and RMSE adds the two in quadrature, so the state RMSE cannot separate resamplers; compare methods with (b) or with Table **??**.

Table 2: Linear Gaussian SSM: posterior-mean RMSE vs. exact Kalman filter (lower is better). $x_t = 0.8x_{t-1} + 0.5\varepsilon_t$, $y_t = x_t + 0.8\eta_t$. 50 steps, resampling when ESS $< 0.5N$; mean $\pm$ 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.138 ± 0.0123	0.0743 ± 0.00949	0.0442 ± 0.00453	0.0339 ± 0.00505
DSR (τ=0.1, no clamp)	0.138 ± 0.0123	0.0743 ± 0.00949	0.0442 ± 0.00453	0.0339 ± 0.00505
DSR (τ=0.1, unsorted)	0.151 ± 0.0129	0.1 ± 0.011	0.0803 ± 0.0124	0.0745 ± 0.0134
DSR (τ=0.5)	0.153 ± 0.0126	0.101 ± 0.0119	0.0797 ± 0.0125	0.0732 ± 0.0132
DSR (τ=2.5/N)[†]	0.136 ± 0.0115	$\mathbf{0.0668 \pm 0.00895}$	0.0326 ± 0.00252	0.0171 ± 0.00195
Systematic	0.132 ± 0.0106	0.0675 ± 0.00818	0.0324 ± 0.00279	0.017 ± 0.00158
Multinomial	0.134 ± 0.0104	0.0692 ± 0.00877	0.0336 ± 0.00292	0.0177 ± 0.00208
Systematic, stop-gradient (PyDPF)	0.131 ± 0.00987	0.0678 ± 0.00889	0.0329 ± 0.00271	0.0169 ± 0.00167
Soft (α=0.5)	0.145 ± 0.0109	0.0754 ± 0.00866	0.0357 ± 0.00267	0.0182 ± 0.00149
OT (PyDPF)	0.127 ± 0.0212	0.067 ± 0.0136	–	–
OT (PFJAX)	$\mathbf{0.127 \pm 0.0211}$	0.0672 ± 0.0136	–	–
OT (FilterFlow)	0.128 ± 0.0224	0.0679 ± 0.0134	–	–
Eigen moment matching	0.131 ± 0.00892	0.0678 ± 0.00958	$\mathbf{0.0321 \pm 0.00303}$	$\mathbf{0.0166 \pm 0.00188}$
Gaussian moment matching	0.133 ± 0.00959	0.0707 ± 0.00978	0.0335 ± 0.00296	0.0172 ± 0.00208
Gumbel-softmax	0.133 ± 0.0208	0.0793 ± 0.0123	–	–
Variational (KL)	0.132 ± 0.00927	0.067 ± 0.00873	0.0334 ± 0.00282	0.0181 ± 0.00202

[†]Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

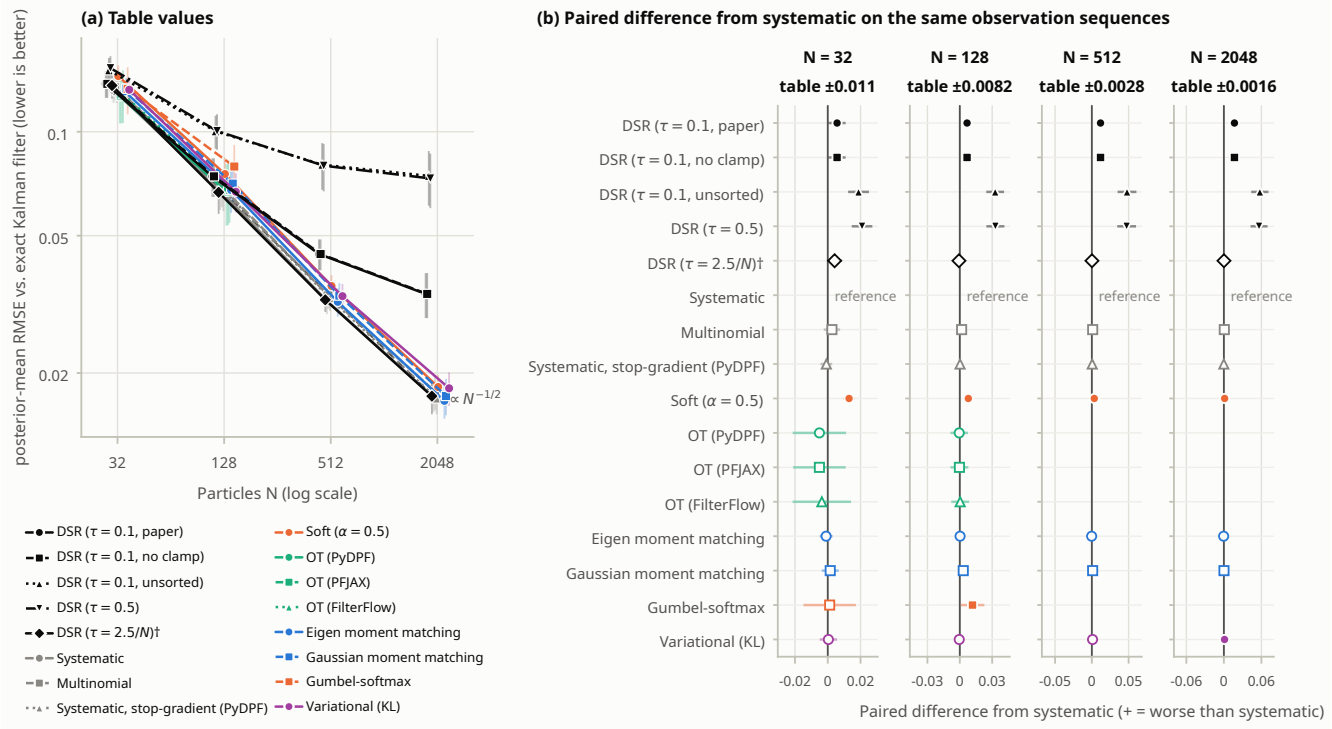


Figure 2: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 3: Linear Gaussian SSM: posterior-variance error vs. Kalman filter (lower is better). $x_t = 0.8x_{t-1} + 0.5\varepsilon_t$, $y_t = x_t + 0.8\eta_t$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.0583 ± 0.00281	0.0318 ± 0.00182	0.0201 ± 0.00134	0.0149 ± 0.00123
DSR (τ=0.1, no clamp)	0.0583 ± 0.00281	0.0318 ± 0.00182	0.0201 ± 0.00134	0.0149 ± 0.00123
DSR (τ=0.1, unsorted)	0.0648 ± 0.00327	0.0432 ± 0.00268	0.0343 ± 0.00271	0.0302 ± 0.00275
DSR (τ=0.5)	0.0663 ± 0.00316	0.0434 ± 0.0028	0.034 ± 0.00269	0.0297 ± 0.00271
DSR (τ=2.5/N)†	0.0579 ± 0.00281	$\mathbf{0.0287 \pm 0.00173}$	$\mathbf{0.0145 \pm 8.22 \times 10^{-4}}$	$0.00744 \pm 4.86 \times 10^{-4}$
Systematic	0.0574 ± 0.0026	0.0289 ± 0.00189	$0.015 \pm 8.69 \times 10^{-4}$	$0.00762 \pm 4.79 \times 10^{-4}$
Multinomial	0.058 ± 0.00265	0.0297 ± 0.00192	$0.0149 \pm 9.3 \times 10^{-4}$	$0.00762 \pm 4.52 \times 10^{-4}$
Systematic, stop-gradient (PyDPF)	0.0572 ± 0.00217	0.0298 ± 0.00168	0.0146 ± 0.00112	$0.00745 \pm 4.35 \times 10^{-4}$
Soft (α=0.5)	0.0615 ± 0.00251	0.0311 ± 0.00118	0.016 ± 0.00104	$0.00784 \pm 4.88 \times 10^{-4}$
OT (PyDPF)	0.0572 ± 0.00713	0.0287 ± 0.00338	–	–
OT (PFJAX)	0.0572 ± 0.00718	0.0287 ± 0.00337	–	–
OT (FilterFlow)	0.0575 ± 0.00675	0.029 ± 0.00346	–	–
Eigen moment matching	0.0569 ± 0.00272	0.0288 ± 0.00175	$0.0147 \pm 9.2 \times 10^{-4}$	$\mathbf{0.0073 \pm 3.98 \times 10^{-4}}$
Gaussian moment matching	0.057 ± 0.00245	0.0292 ± 0.00186	0.0147 ± 0.00101	$0.00747 \pm 4.1 \times 10^{-4}$
Gumbel-softmax	$\mathbf{0.0558 \pm 0.00504}$	0.0332 ± 0.00349	–	–
Variational (KL)	0.0581 ± 0.00263	0.0293 ± 0.00192	0.015 ± 0.00107	$0.00821 \pm 5.91 \times 10^{-4}$

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

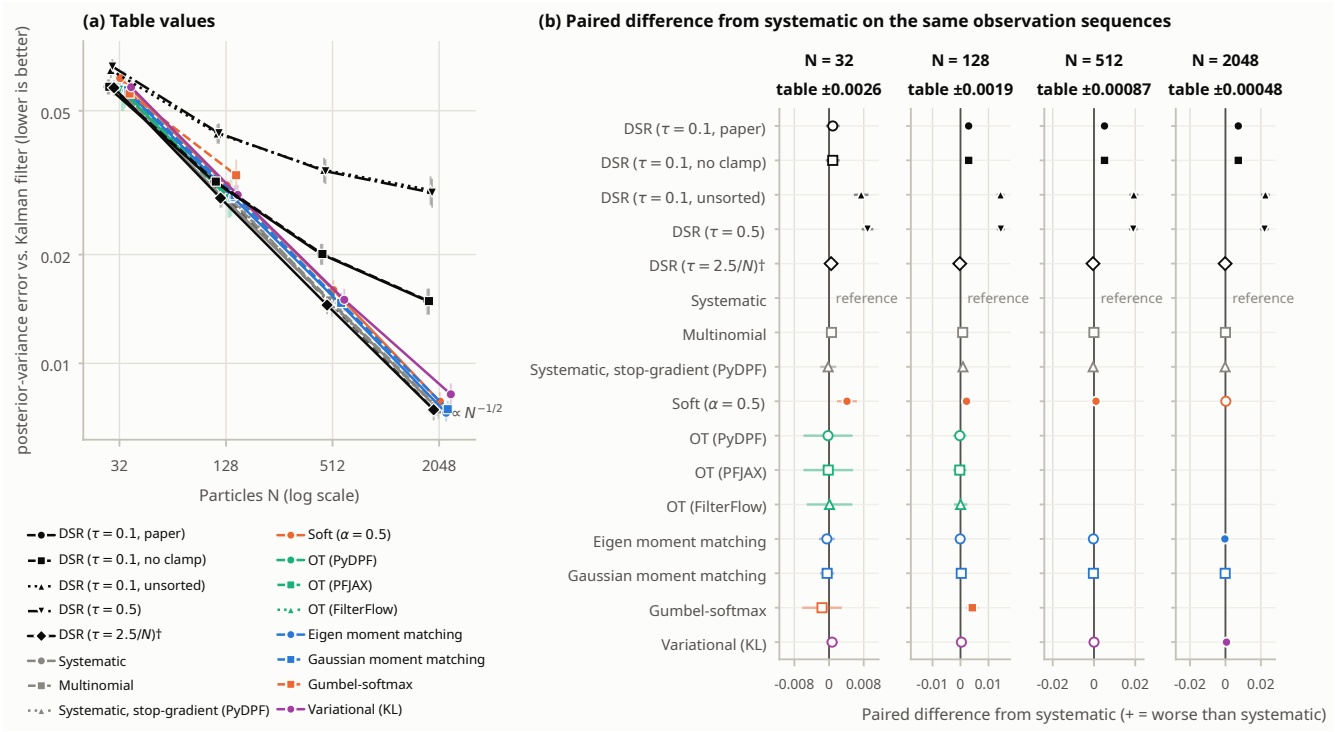


Figure 3: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 4: Linear Gaussian SSM: log-likelihood error vs. Kalman filter (closer to 0 is better). $x_t = 0.8x_{t-1} + 0.5\varepsilon_t$, $y_t = x_t + 0.8\eta_t$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	-0.847 ± 0.399	-0.19 ± 0.135	-0.0486 ± 0.155	0.0111 ± 0.167
DSR (τ=0.1, no clamp)	-0.847 ± 0.399	-0.19 ± 0.135	-0.0486 ± 0.155	0.0111 ± 0.167
DSR (τ=0.1, unsorted)	-0.854 ± 0.465	-0.265 ± 0.262	-0.195 ± 0.481	-0.117 ± 0.504
DSR (τ=0.5)	-0.887 ± 0.514	-0.235 ± 0.279	-0.195 ± 0.483	-0.114 ± 0.492
DSR (τ=2.5/N)†	-0.839 ± 0.38	-0.205 ± 0.111	-0.0415 ± 0.0859	0.00311 ± 0.0339
Systematic	-0.829 ± 0.357	-0.186 ± 0.171	-0.0508 ± 0.0724	0.0258 ± 0.0321
Multinomial	-0.903 ± 0.214	-0.207 ± 0.142	-0.0361 ± 0.0613	0.0103 ± 0.0458
Systematic, stop-gradient (PyDPF)	$\mathbf{-0.756 \pm 0.383}$	-0.171 ± 0.161	$\mathbf{-0.0196 \pm 0.059}$	-0.00821 ± 0.0429
Soft (α=0.5)	-0.93 ± 0.466	-0.333 ± 0.185	-0.0328 ± 0.0544	-0.0124 ± 0.0338
OT (PyDPF)	-1.32 ± 1.29	-0.277 ± 0.434	–	–
OT (PFJAX)	-1.28 ± 1.28	-0.276 ± 0.432	–	–
OT (FilterFlow)	-1.33 ± 1.34	-0.269 ± 0.433	–	–
Eigen moment matching	-0.852 ± 0.32	-0.23 ± 0.112	-0.0483 ± 0.0696	$\mathbf{6.98 \times 10^{-4} \pm 0.0369}$
Gaussian moment matching	-0.825 ± 0.341	$\mathbf{-0.143 \pm 0.178}$	-0.064 ± 0.0545	0.00867 ± 0.0444
Gumbel-softmax	-1.33 ± 1.5	-0.236 ± 0.451	–	–
Variational (KL)	-0.827 ± 0.316	-0.145 ± 0.125	-0.041 ± 0.0725	0.00757 ± 0.0489

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

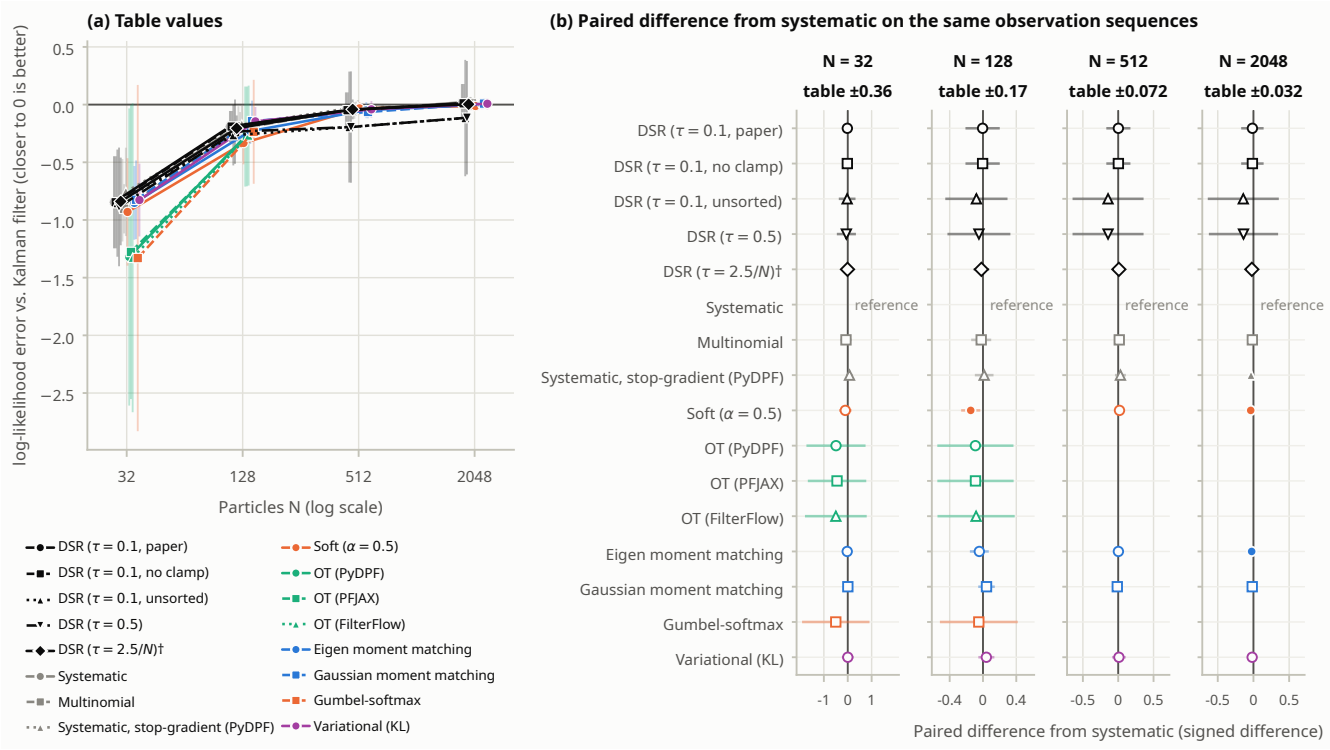


Figure 4: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 5: Linear Gaussian SSM: 95% coverage (closer to 0.95 is better). $x_t = 0.8x_{t-1} + 0.5\varepsilon_t$, $y_t = x_t + 0.8\eta_t$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.903 ± 0.028	0.923 ± 0.0227	0.926 ± 0.0242	0.928 ± 0.0233
DSR (τ=0.1, no clamp)	0.903 ± 0.028	0.923 ± 0.0227	0.926 ± 0.0242	0.928 ± 0.0233
DSR (τ=0.1, unsorted)	0.895 ± 0.0315	0.913 ± 0.0254	0.916 ± 0.0284	0.917 ± 0.0275
DSR (τ=0.5)	0.894 ± 0.0297	0.912 ± 0.025	0.916 ± 0.0281	0.917 ± 0.0279
DSR (τ=2.5/N)†	0.904 ± 0.0275	0.929 ± 0.0224	0.931 ± 0.0243	0.933 ± 0.0237
Systematic	0.912 ± 0.024	0.929 ± 0.0218	0.931 ± 0.0244	0.933 ± 0.0236
Multinomial	0.911 ± 0.0246	0.928 ± 0.0223	$\mathbf{0.931 \pm 0.0245}$	0.933 ± 0.0238
Systematic, stop-gradient (PyDPF)	0.909 ± 0.0259	$\mathbf{0.931 \pm 0.0212}$	0.931 ± 0.0247	0.932 ± 0.0242
Soft (α=0.5)	0.908 ± 0.0282	0.927 ± 0.0228	0.931 ± 0.0237	0.932 ± 0.0232
OT (PyDPF)	0.916 ± 0.0374	0.928 ± 0.0215	–	–
OT (PFJAX)	$\mathbf{0.918 \pm 0.034}$	0.93 ± 0.0205	–	–
OT (FilterFlow)	0.916 ± 0.0374	0.924 ± 0.02	–	–
Eigen moment matching	0.914 ± 0.0248	0.929 ± 0.0222	0.931 ± 0.0246	0.932 ± 0.0235
Gaussian moment matching	0.911 ± 0.023	0.929 ± 0.0222	$\mathbf{0.931 \pm 0.0236}$	$\mathbf{0.934 \pm 0.0233}$
Gumbel-softmax	0.902 ± 0.034	0.928 ± 0.0215	–	–
Variational (KL)	0.909 ± 0.0262	0.929 ± 0.0199	0.93 ± 0.0249	0.932 ± 0.0238

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

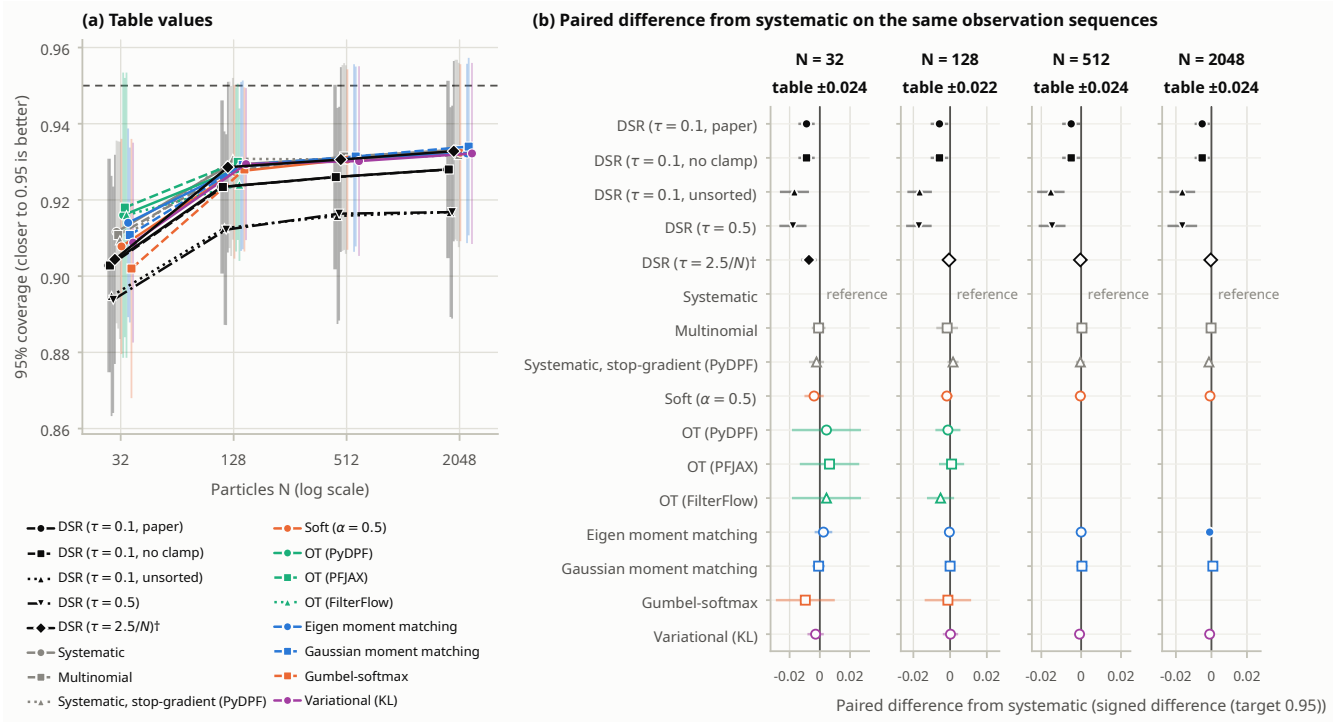


Figure 5: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Dashed line: nominal 0.95. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 6: Linear Gaussian SSM: log-likelihood variance given the data (lower is better). $x_t = 0.8x_{t-1} + 0.5\varepsilon_t$, $y_t = x_t + 0.8\eta_t$. 50 steps, resampling when ESS $< 0.5N$; mean $\pm$ 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	2.07	0.374	0.125	0.0313
DSR (τ=0.1, no clamp)	2.07	0.374	0.125	0.0313
DSR (τ=0.1, unsorted)	2.44	0.509	0.159	0.0534
DSR (τ=0.5)	2.3	0.507	0.16	0.0503
DSR (τ=2.5/N)†	2.01	0.381	0.11	0.0327
Systematic	2.22	0.353	0.098	**0.025**
Multinomial	2.29	0.406	0.116	0.0299
Systematic, stop-gradient (PyDPF)	2.18	0.327	0.0911	0.0308
Soft (α=0.5)	**1.92**	0.418	0.0898	0.0344
Eigen moment matching	2	**0.32**	**0.0861**	0.028
Gaussian moment matching	2.25	0.417	0.103	0.0274
Variational (KL)	2.29	0.358	0.105	0.0298

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

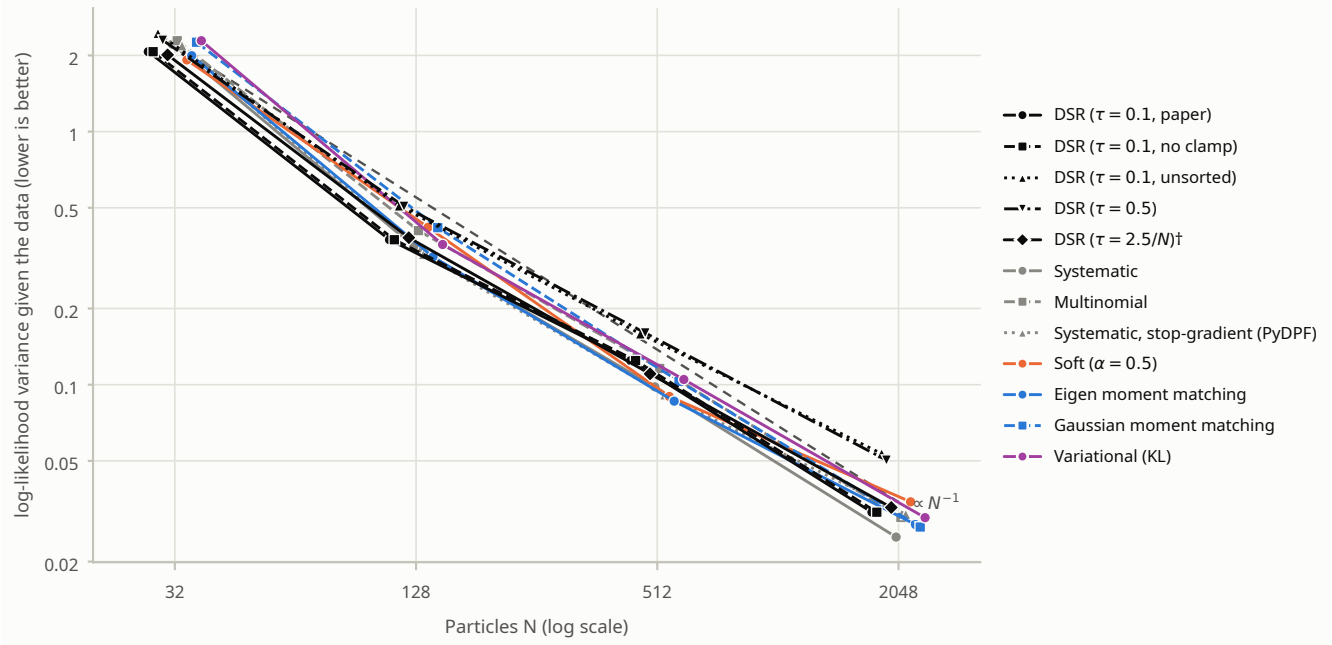


Figure 6: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 7: Linear Gaussian random walk (differentiable PF): likelihood-gradient bias $\|\mathbb{E}\hat{g} - \nabla\ell\|$ (lower is better). $x_t = x_{t-1} + 0.4\varepsilon_t$, $y_t = x_t + 0.8\eta_t$, 50 steps; gradient of the PF log-likelihood w.r.t. $\theta = \log(q, r)$ against the exact Kalman gradient, 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	19.3	8.77	7.13	7.12
DSR (τ=0.1, no clamp)	9.76	7.3	7.1	7.12
DSR (τ=0.1, unsorted)	30.9	21.5	20.5	20.9
DSR (τ=0.5)	32.3	21.5	20	20
DSR (τ=2.5/N)†	17.1	2.39	0.265	0.0636
Systematic (no gradient)	12.4	11.3	12.4	12.3
Systematic, stop-gradient (PyDPF)	**3.59**	1.75	0.836	0.225
Soft (α=0.5)	12.3	6.68	5.48	5.96
OT (PyDPF)	19.5	14.2	–	–
Eigen moment matching	3.68	**0.527**	**0.0676**	0.0808
Gaussian moment matching	3.81	0.823	0.148	**0.0309**
Gumbel-softmax	15.2	8.15	–	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

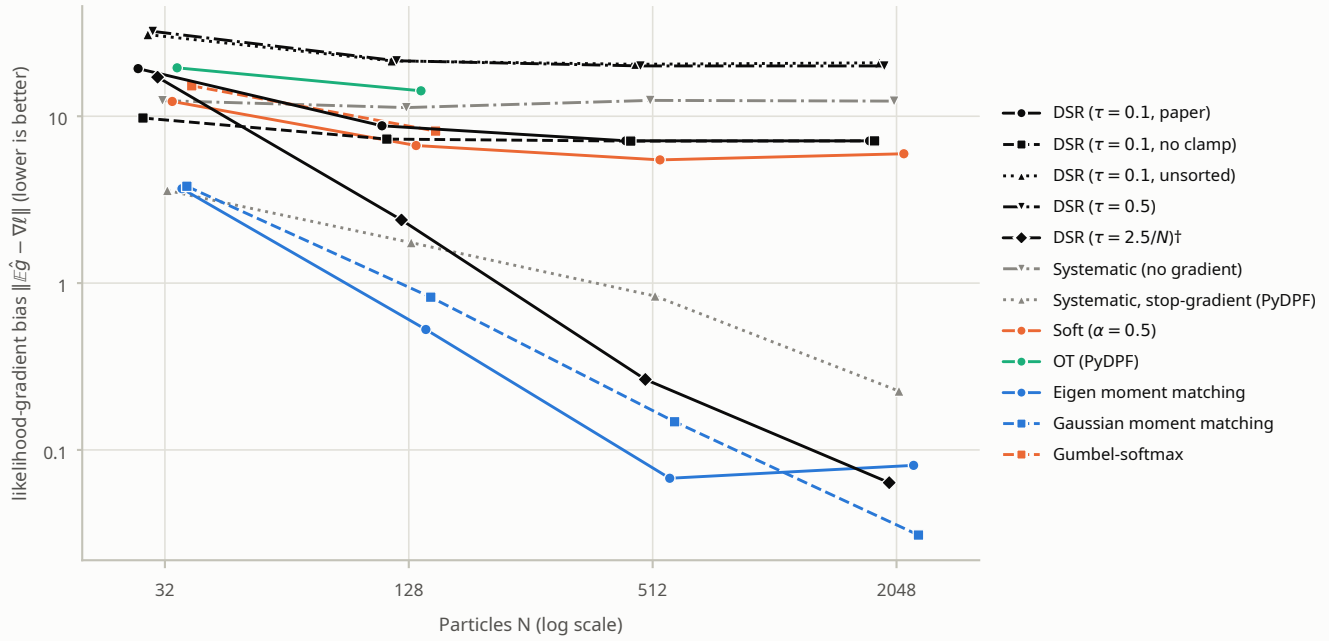


Figure 7: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis.

Table 8: Linear Gaussian random walk (differentiable PF): likelihood-gradient variance (lower is better). $x_t = x_{t-1} + 0.4\varepsilon_t$, $y_t = x_t + 0.8\eta_t$, 50 steps; gradient of the PF log-likelihood w.r.t. $\theta = \log(q, r)$ against the exact Kalman gradient, 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	61.4	11.3	1.69	0.483
DSR (τ=0.1, no clamp)	28.6	7.66	1.67	0.483
DSR (τ=0.1, unsorted)	92.7	23.7	8.69	6.11
DSR (τ=0.5)	110	24.2	8.31	5.73
DSR (τ=2.5/N)†	57.2	7.38	1.33	0.311
Systematic (no gradient)	156	41.8	10.4	2.41
Systematic, stop-gradient (PyDPF)	553	151	26.6	8.94
Soft (α=0.5)	203	51.5	13.8	2.38
OT (PyDPF)	$1.2e+03$	715	–	–
Eigen moment matching	30.9	**5.21**	**1.07**	0.268
Gaussian moment matching	**26.8**	5.6	1.2	**0.213**
Gumbel-softmax	195	20.3	–	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

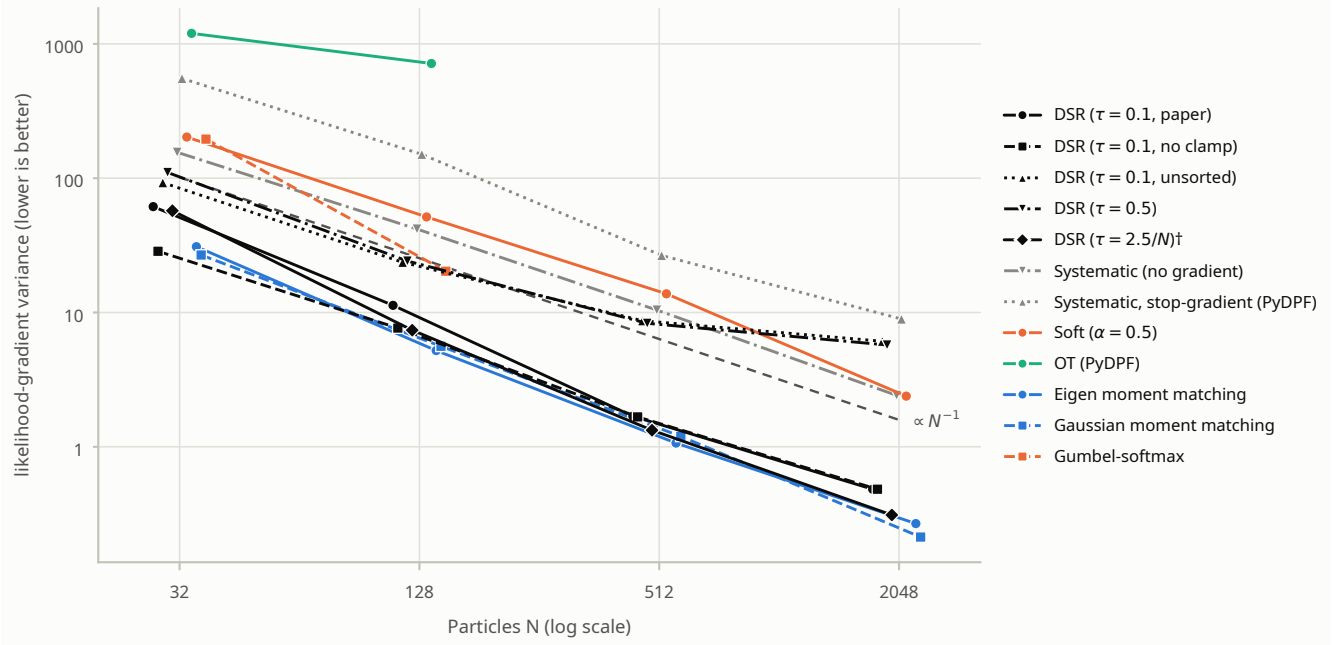


Figure 8: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 9: Linear Gaussian random walk (differentiable PF): likelihood-gradient MSE (lower is better). $x_t = x_{t-1} + 0.4\varepsilon_t$, $y_t = x_t + 0.8\eta_t$, 50 steps; gradient of the PF log-likelihood w.r.t. $\theta = \log(q, r)$ against the exact Kalman gradient, 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	433	88	52.5	51.2
DSR (τ=0.1, no clamp)	124	60.8	52	51.2
DSR (τ=0.1, unsorted)	$1.04e+03$	484	428	443
DSR (τ=0.5)	$1.15e+03$	487	409	407
DSR (τ=2.5/N)†	351	13	1.39	0.312
Systematic (no gradient)	308	168	165	154
Systematic, stop-gradient (PyDPF)	561	152	27	8.9
Soft (α=0.5)	351	95.5	43.6	37.9
OT (PyDPF)	$1.46e+03$	845	–	–
Eigen moment matching	44.2	**5.43**	**1.06**	0.272
Gaussian moment matching	**41**	6.22	1.21	**0.212**
Gumbel-softmax	408	84.6	–	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

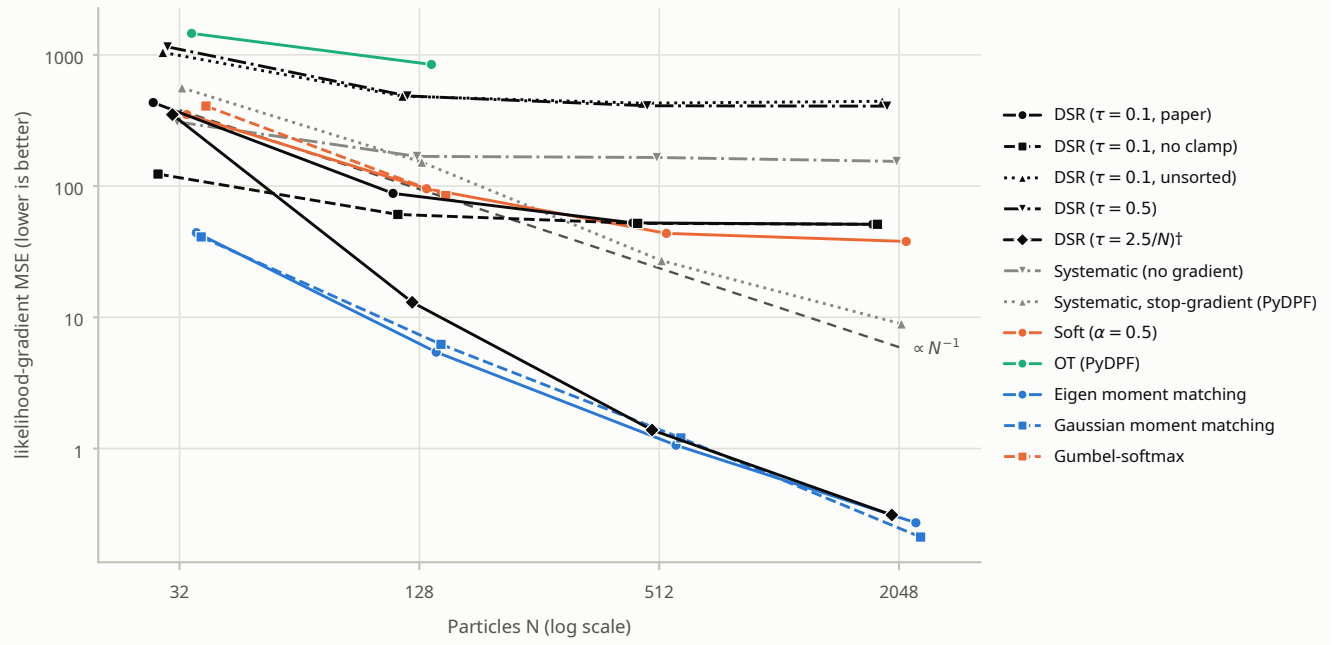


Figure 9: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 10: Linear Gaussian random walk (differentiable PF): log-likelihood bias (closer to 0 is better). $x_t = x_{t-1} + 0.4\varepsilon_t$, $y_t = x_t + 0.8\eta_t$, 50 steps; gradient of the PF log-likelihood w.r.t. $\theta = \log(q, r)$ against the exact Kalman gradient, 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	−2.99	−2.16	−1.91	−1.9
DSR (τ=0.1, no clamp)	−2.99	−2.16	−1.91	−1.9
DSR (τ=0.1, unsorted)	−6.44	−6.34	−6.57	−6.9
DSR (τ=0.5)	−7.23	−6.51	−6.34	−6.51
DSR (τ=2.5/N)†	−2.45	−0.435	−0.0687	−0.0238
Systematic (no gradient)	**−1.16**	−0.295	−0.0458	0.0151
Systematic, stop-gradient (PyDPF)	**−1.16**	−0.295	−0.0458	0.0151
Soft (α=0.5)	−1.82	−0.386	−0.0553	−0.0323
OT (PyDPF)	−1.71	−0.447	–	–
Eigen moment matching	−1.26	**−0.154**	**−0.00765**	−0.0378
Gaussian moment matching	−1.45	−0.278	−0.0375	**0.00729**
Gumbel-softmax	−3.33	−2.66	–	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

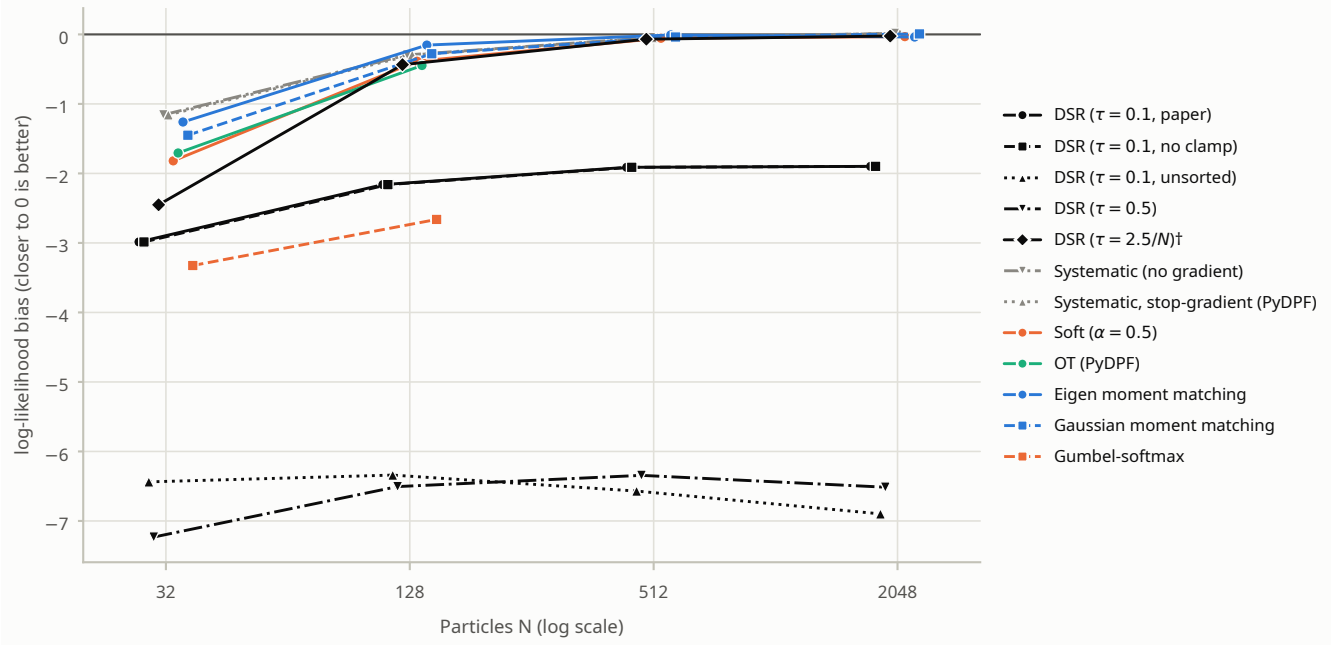


Figure 10: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family.

Table 11: Linear Gaussian random walk (differentiable PF): log-likelihood variance (lower is better). $x_t = x_{t-1} + 0.4\varepsilon_t$, $y_t = x_t + 0.8\eta_t$, 50 steps; gradient of the PF log-likelihood w.r.t. $\theta = \log(q, r)$ against the exact Kalman gradient, 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	3.63	1.14	0.239	0.0637
DSR (τ=0.1, no clamp)	3.63	1.14	0.239	0.0637
DSR (τ=0.1, unsorted)	7.59	2.49	1.63	0.754
DSR (τ=0.5)	7.78	2.99	1.54	0.943
DSR (τ=2.5/N)†	3.81	0.694	0.128	0.0318
Systematic (no gradient)	2.59	0.656	0.122	0.0314
Systematic, stop-gradient (PyDPF)	2.59	0.656	0.122	0.0314
Soft (α=0.5)	4.92	1.08	0.147	0.0407
OT (PyDPF)	4.59	**0.31**	–	–
Eigen moment matching	2.63	0.564	**0.106**	**0.0283**
Gaussian moment matching	3.22	0.86	0.185	0.0341
Gumbel-softmax	**1.69**	0.73	–	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

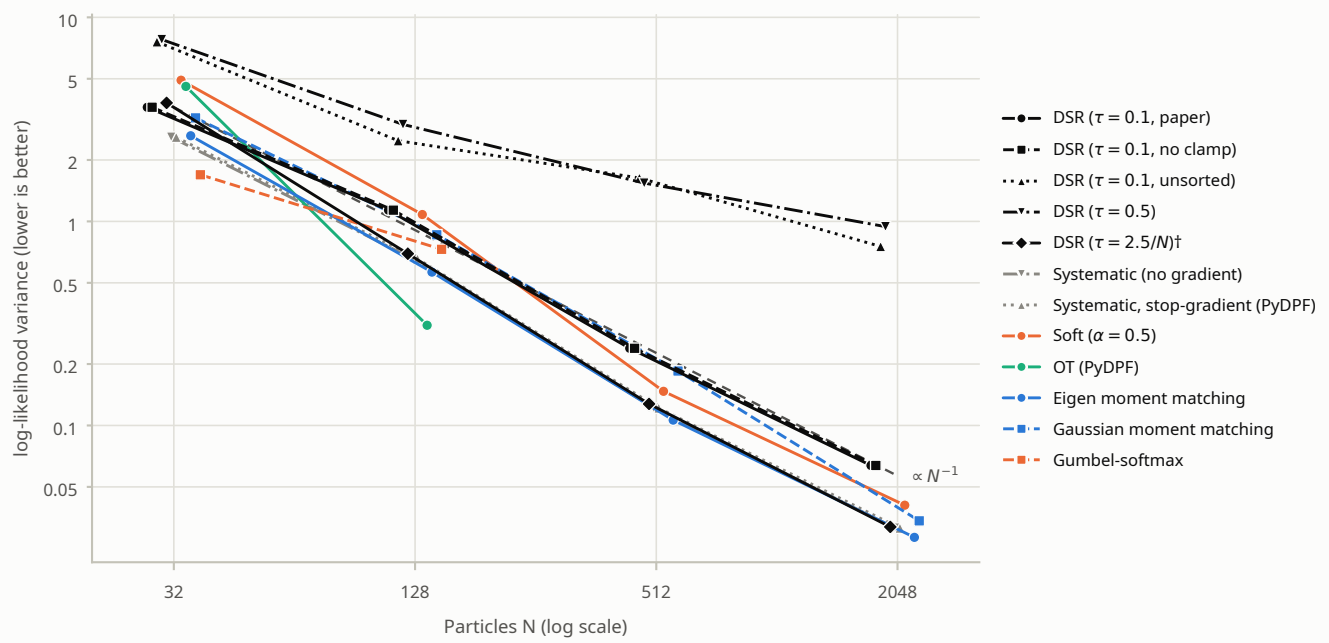


Figure 11: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Nonlinear state-space models

Table 12: UNGM: state RMSE (lower is better). $x_t = 0.5x_{t-1} + 25x_{t-1}/(1 + x_{t-1}^2) + 8\cos(1.2t) + \sqrt{10}\varepsilon_t$, $y_t = x_t^2/20 + \eta_t$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	6.22 ± 0.768	5.15 ± 0.995	4.79 ± 0.754	4.78 ± 0.653
DSR (τ=0.1, no clamp)	6.22 ± 0.768	5.15 ± 0.995	4.79 ± 0.754	4.78 ± 0.653
DSR (τ=0.1, unsorted)	7.52 ± 0.792	7.63 ± 1.2	8.64 ± 1.52	8.68 ± 1.71
DSR (τ=0.5)	8.64 ± 0.701	8.52 ± 1.41	8.81 ± 1.77	8.88 ± 1.66
DSR (τ=2.5/N)†	6.02 ± 0.987	4.98 ± 1.03	4.56 ± 0.668	4.52 ± 0.577
Systematic	6.01 ± 0.871	4.91 ± 0.929	$\mathbf{4.49 \pm 0.55}$	4.48 ± 0.603
Multinomial	5.97 ± 0.918	4.92 ± 0.88	4.57 ± 0.636	4.52 ± 0.58
Systematic, stop-gradient (PyDPF)	5.83 ± 0.996	4.88 ± 0.889	4.51 ± 0.584	4.54 ± 0.568
Soft (α=0.5)	$\mathbf{5.42 \pm 0.629}$	$\mathbf{4.83 \pm 0.749}$	4.54 ± 0.588	$\mathbf{4.44 \pm 0.581}$
OT (PyDPF)	5.6 ± 0.978	5.42 ± 1.46	–	–
OT (PFJAX)	6.1 ± 1.07	5.33 ± 1.42	–	–
OT (FilterFlow)	6.01 ± 0.972	5.48 ± 1.49	–	–
Eigen moment matching	5.99 ± 0.707	5.17 ± 1.04	4.98 ± 0.994	4.83 ± 0.828
Gaussian moment matching	6.02 ± 0.851	4.94 ± 0.855	4.8 ± 0.696	4.66 ± 0.537
Gumbel-softmax	6.51 ± 0.994	5.27 ± 1.41	–	–
Variational (KL)	5.79 ± 0.947	4.83 ± 0.858	4.56 ± 0.633	4.48 ± 0.613

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

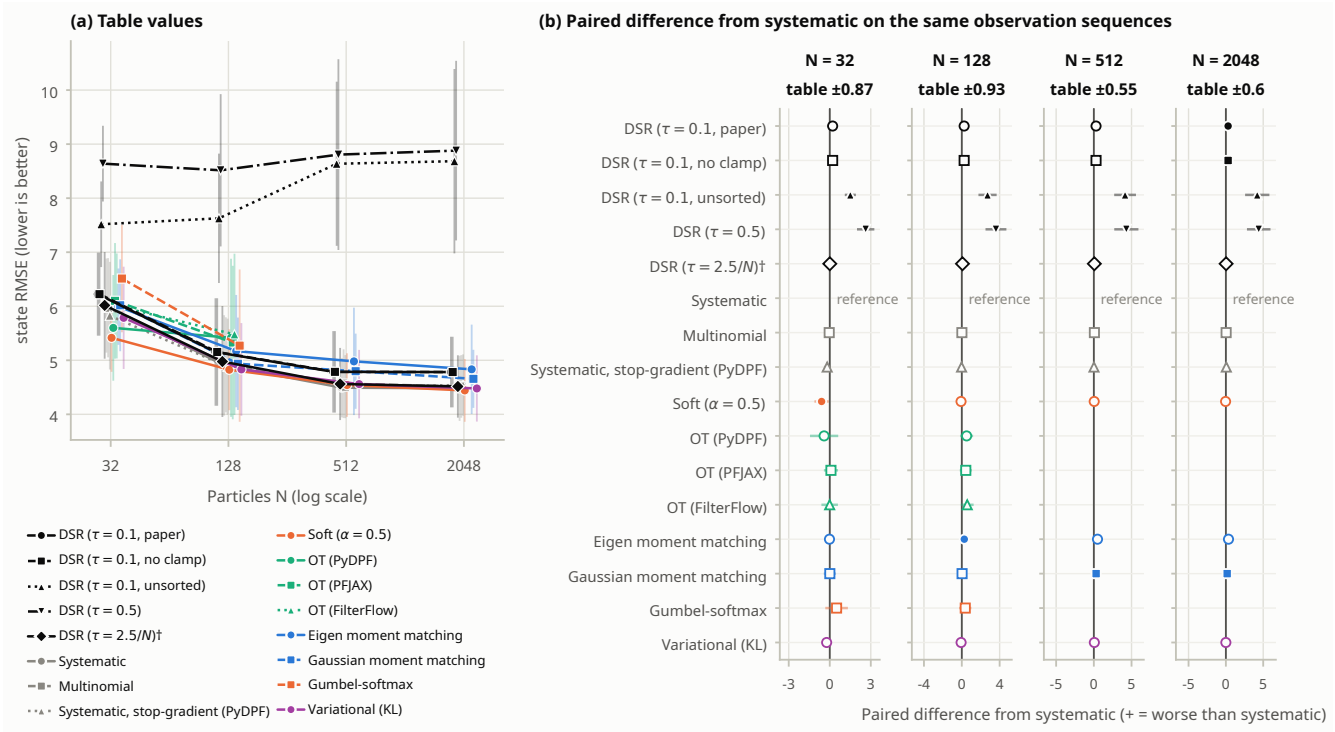


Figure 12: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 13: UNGM: 95% coverage (closer to 0.95 is better). $x_t = 0.5x_{t-1} + 25x_{t-1}/(1 + x_{t-1}^2) + 8\cos(1.2t) + \sqrt{10}\varepsilon_t$, $y_t = x_t^2/20 + \eta_t$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.813 ± 0.0413	0.926 ± 0.0319	0.955 ± 0.0208	$\mathbf{0.961 \pm 0.0167}$
DSR (τ=0.1, no clamp)	0.813 ± 0.0413	0.926 ± 0.0319	0.955 ± 0.0208	$\mathbf{0.961 \pm 0.0167}$
DSR (τ=0.1, unsorted)	0.737 ± 0.0408	0.784 ± 0.0543	0.763 ± 0.0582	0.766 ± 0.0616
DSR (τ=0.5)	0.672 ± 0.0264	0.738 ± 0.0619	0.752 ± 0.0656	0.768 ± 0.056
DSR (τ=2.5/N)†	0.822 ± 0.039	0.929 ± 0.0252	0.963 ± 0.0157	0.971 ± 0.0159
Systematic	0.821 ± 0.0366	0.93 ± 0.0275	0.966 ± 0.0174	0.971 ± 0.0149
Multinomial	0.818 ± 0.0384	0.928 ± 0.0248	0.964 ± 0.0158	0.972 ± 0.0162
Systematic, stop-gradient (PyDPF)	$\mathbf{0.828 \pm 0.0364}$	0.932 ± 0.0261	0.966 ± 0.0157	0.972 ± 0.0155
Soft (α=0.5)	0.797 ± 0.0332	0.928 ± 0.0214	0.962 ± 0.0131	0.97 ± 0.0163
OT (PyDPF)	0.814 ± 0.0448	0.914 ± 0.0458	–	–
OT (PFJAX)	0.8 ± 0.0421	0.918 ± 0.0418	–	–
OT (FilterFlow)	0.806 ± 0.0518	0.908 ± 0.0448	–	–
Eigen moment matching	0.828 ± 0.0369	0.93 ± 0.0261	$\mathbf{0.953 \pm 0.0225}$	0.963 ± 0.0185
Gaussian moment matching	0.824 ± 0.0428	$\mathbf{0.933 \pm 0.0272}$	0.962 ± 0.0158	0.97 ± 0.0162
Gumbel-softmax	0.82 ± 0.0548	0.918 ± 0.0445	–	–
Variational (KL)	0.825 ± 0.038	0.93 ± 0.0246	0.963 ± 0.0152	0.971 ± 0.0159

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

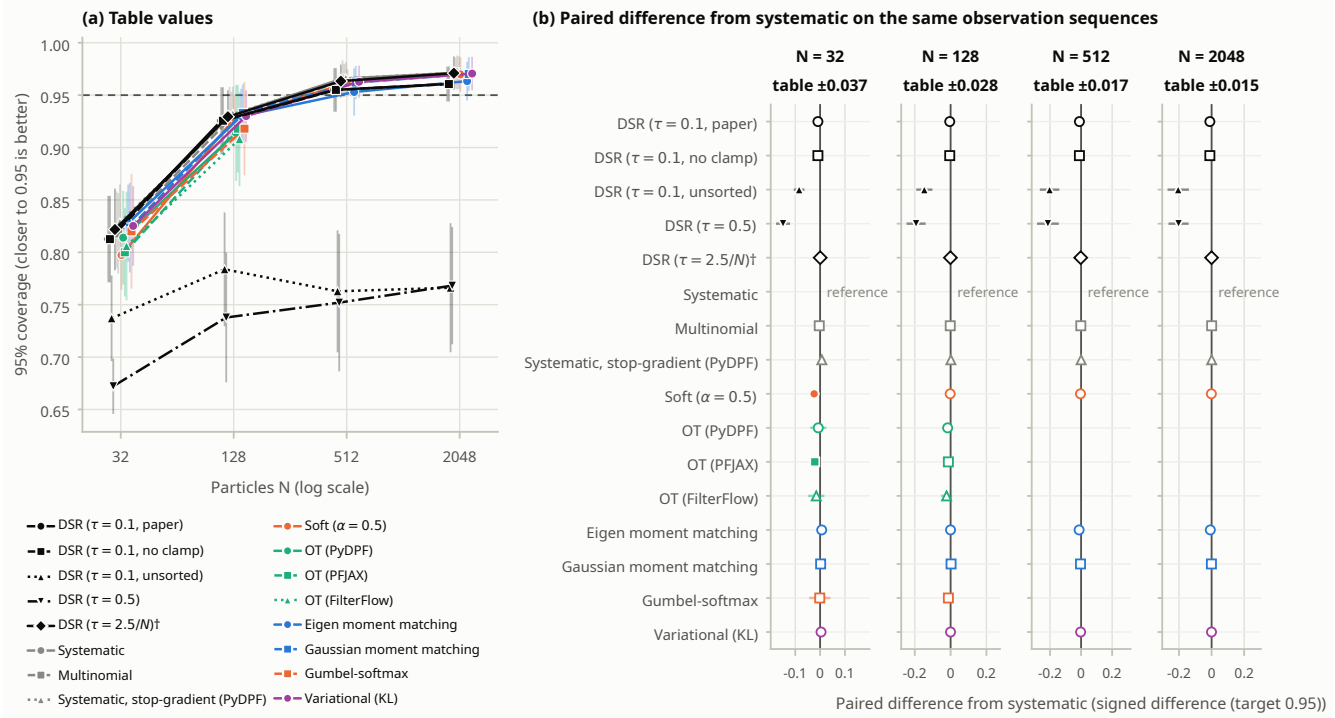


Figure 13: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Dashed line: nominal 0.95. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 14: UNGM: predictive NLL (lower is better). $x_t = 0.5x_{t-1} + 25x_{t-1}/(1 + x_{t-1}^2) + 8\cos(1.2t) + \sqrt{10}\varepsilon_t$, $y_t = x_t^2/20 + \eta_t$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	5.53 ± 1.21	3.23 ± 0.455	2.86 ± 0.169	2.77 ± 0.129
DSR (τ=0.1, no clamp)	5.53 ± 1.21	3.23 ± 0.455	2.86 ± 0.169	2.77 ± 0.129
DSR (τ=0.1, unsorted)	7.21 ± 1.49	6.09 ± 1.71	6.2 ± 1.47	5.57 ± 1.27
DSR (τ=0.5)	9.33 ± 1.72	7.06 ± 1.92	6.21 ± 1.73	5.46 ± 1.35
DSR (τ=2.5/N)†	5.33 ± 1.19	3.19 ± 0.462	2.8 ± 0.152	$\mathbf{2.7 \pm 0.098}$
Systematic	5.36 ± 1.23	3.18 ± 0.479	$\mathbf{2.78 \pm 0.175}$	2.7 ± 0.101
Multinomial	5.26 ± 1.16	3.3 ± 0.533	2.82 ± 0.161	2.7 ± 0.0971
Systematic, stop-gradient (PyDPF)	4.99 ± 1.02	3.14 ± 0.398	2.79 ± 0.176	2.7 ± 0.0988
Soft (α=0.5)	4.32 ± 0.731	3.13 ± 0.382	2.81 ± 0.185	2.71 ± 0.108
OT (PyDPF)	$\mathbf{4.18 \pm 1.27}$	3.18 ± 0.495	–	–
OT (PFJAX)	4.6 ± 1.36	3.19 ± 0.491	–	–
OT (FilterFlow)	4.53 ± 1.28	3.2 ± 0.506	–	–
Eigen moment matching	4.98 ± 1.08	3.17 ± 0.394	2.88 ± 0.2	2.77 ± 0.114
Gaussian moment matching	5.06 ± 1.13	$\mathbf{3.13 \pm 0.378}$	2.82 ± 0.164	2.71 ± 0.106
Gumbel-softmax	4.82 ± 1.44	3.16 ± 0.502	–	–
Variational (KL)	4.98 ± 1.16	3.18 ± 0.424	2.83 ± 0.171	2.71 ± 0.105

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

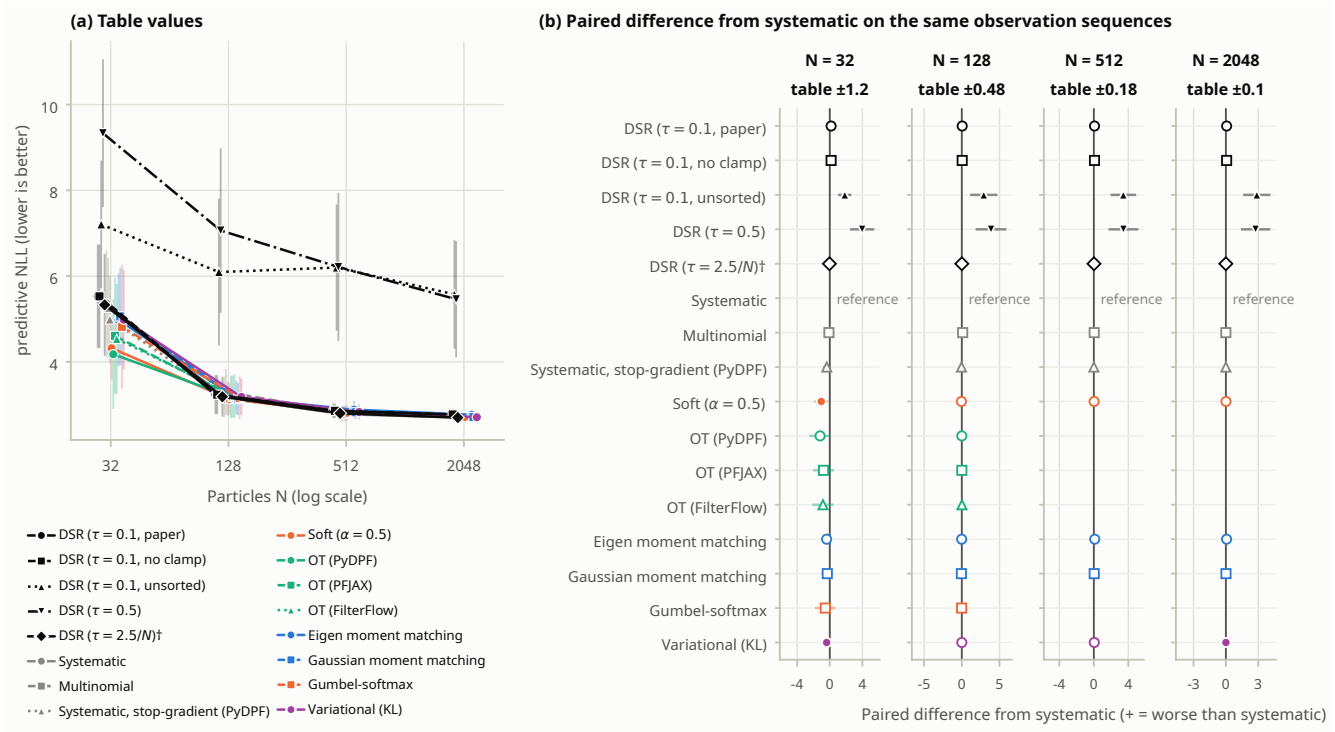


Figure 14: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 15: UNGM: log-likelihood variance given the data (lower is better). $x_t = 0.5x_{t-1} + 25x_{t-1}/(1 + x_{t-1}^2) + 8\cos(1.2t) + \sqrt{10}\varepsilon_t$, $y_t = x_t^2/20 + \eta_t$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	2.23×10^4	$1.18e+03$	346	365
DSR (τ=0.1, no clamp)	2.23×10^4	$1.18e+03$	346	365
DSR (τ=0.1, unsorted)	2.15×10^4	$9.22e+03$	$6.86e+03$	$1.49e+03$
DSR (τ=0.5)	1.9×10^4	$5.42e+03$	$6.27e+03$	883
DSR (τ=2.5/N)†	1.81×10^4	$1.03e+03$	269	**6.95**
Systematic	1.53×10^4	653	73.7	14.1
Multinomial	1.5×10^4	$4.17e+03$	339	9.62
Systematic, stop-gradient (PyDPF)	1.21×10^4	670	**64.8**	9.05
Soft (α=0.5)	**3.08e+03**	**438**	156	26.5
Eigen moment matching	1.51×10^4	810	457	146
Gaussian moment matching	1.48×10^4	845	194	15.6
Variational (KL)	1.25×10^4	756	240	11.5

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

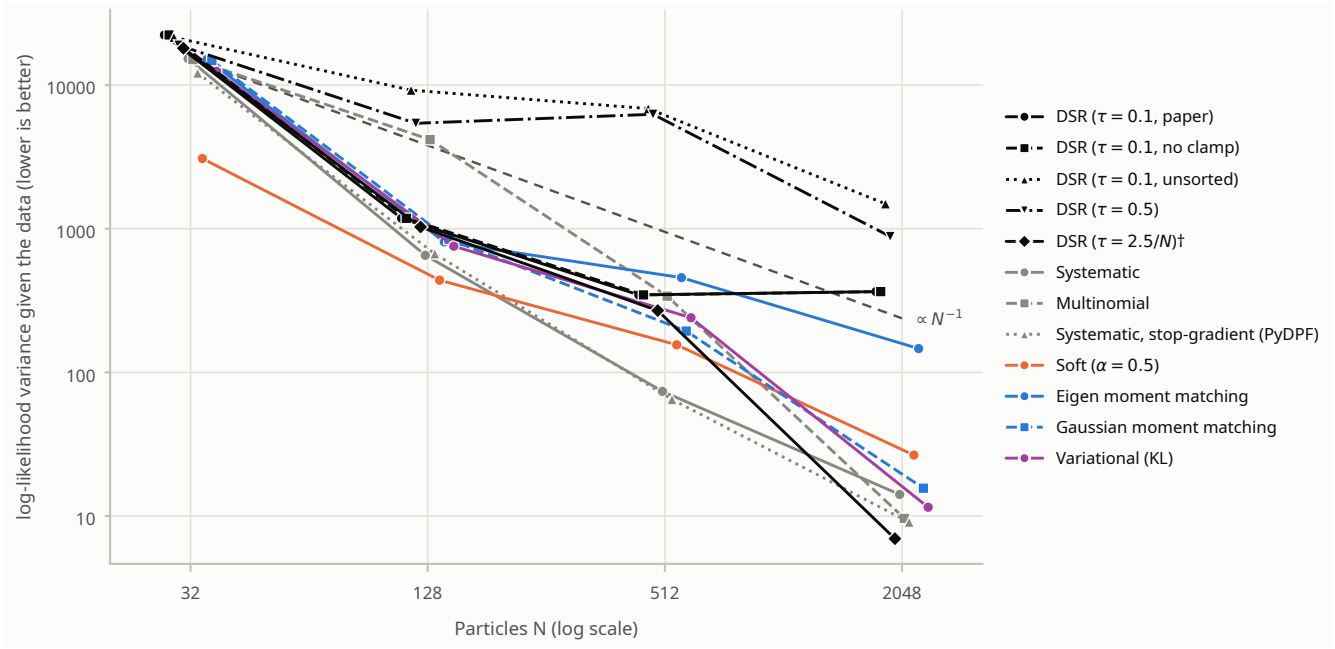


Figure 15: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 16: Stochastic volatility: state RMSE (lower is better). $x_t = 0.95x_{t-1} + 0.2\varepsilon_t$, $y_t = \exp(x_t/2)\eta_t$. 50 steps, resampling when ESS $< 0.5N$; mean $\pm$ 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.425 ± 0.0648	$\mathbf{0.413 \pm 0.0669}$	$\mathbf{0.412 \pm 0.0693}$	$\mathbf{0.411 \pm 0.0691}$
DSR (τ=0.1, no clamp)	0.425 ± 0.0648	$\mathbf{0.413 \pm 0.0669}$	$\mathbf{0.412 \pm 0.0693}$	$\mathbf{0.411 \pm 0.0691}$
DSR (τ=0.1, unsorted)	0.431 ± 0.066	0.423 ± 0.0711	0.418 ± 0.0739	0.418 ± 0.0737
DSR (τ=0.5)	0.431 ± 0.068	0.423 ± 0.071	0.418 ± 0.0733	0.418 ± 0.0734
DSR (τ=2.5/N)†	$\mathbf{0.425 \pm 0.0647}$	0.415 ± 0.0644	0.415 ± 0.0663	0.414 ± 0.066
Systematic	0.428 ± 0.0582	0.417 ± 0.0634	0.415 ± 0.066	0.414 ± 0.0662
Multinomial	0.43 ± 0.0592	0.417 ± 0.0649	0.415 ± 0.0664	0.414 ± 0.0661
Systematic, stop-gradient (PyDPF)	0.427 ± 0.0586	0.416 ± 0.0639	0.414 ± 0.066	0.414 ± 0.0661
Soft (α=0.5)	0.429 ± 0.0603	0.419 ± 0.0635	0.415 ± 0.0664	0.415 ± 0.0664
OT (PyDPF)	0.426 ± 0.0645	0.418 ± 0.0648	–	–
OT (PFJAX)	0.426 ± 0.0649	0.418 ± 0.0648	–	–
OT (FilterFlow)	0.427 ± 0.0647	0.417 ± 0.065	–	–
Eigen moment matching	0.428 ± 0.0603	0.416 ± 0.063	0.415 ± 0.0663	0.414 ± 0.066
Gaussian moment matching	0.426 ± 0.0622	0.416 ± 0.063	0.415 ± 0.0655	0.414 ± 0.0662
Gumbel-softmax	0.427 ± 0.0649	0.415 ± 0.0678	–	–
Variational (KL)	0.426 ± 0.061	0.417 ± 0.0638	0.413 ± 0.0666	0.413 ± 0.067

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

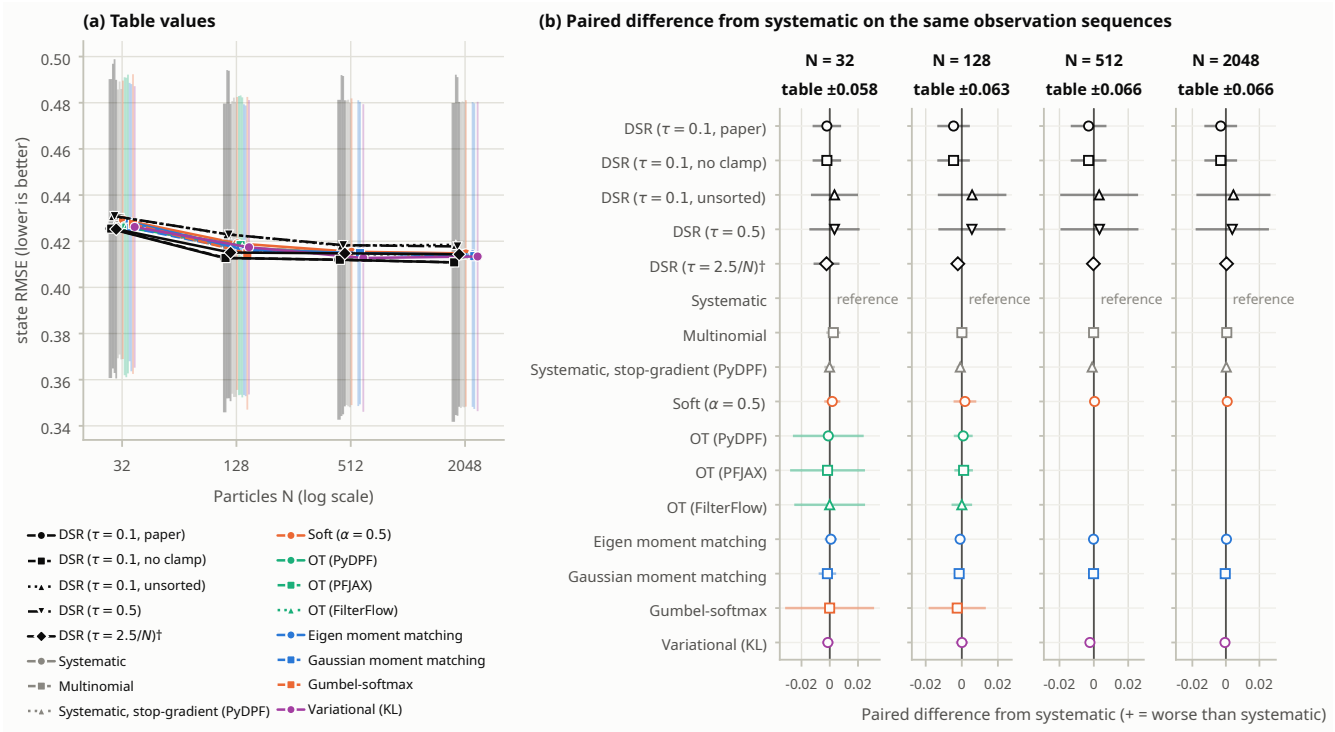


Figure 16: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 17: Stochastic volatility: 95% coverage (closer to 0.95 is better). $x_t = 0.95x_{t-1} + 0.2\varepsilon_t$, $y_t = \exp(x_t/2)\eta_t$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.926 ± 0.0345	**0.947 ± 0.0302**	**0.954 ± 0.0278**	**0.952 ± 0.0318**
DSR (τ=0.1, no clamp)	0.926 ± 0.0345	**0.947 ± 0.0302**	**0.954 ± 0.0278**	**0.952 ± 0.0318**
DSR (τ=0.1, unsorted)	0.886 ± 0.0522	0.896 ± 0.0527	0.905 ± 0.0519	0.904 ± 0.0535
DSR (τ=0.5)	0.88 ± 0.0564	0.893 ± 0.053	0.907 ± 0.0519	0.908 ± 0.053
DSR (τ=2.5/N)†	0.931 ± 0.031	0.962 ± 0.0224	0.975 ± 0.0194	0.976 ± 0.0206
Systematic	0.941 ± 0.025	0.966 ± 0.0184	0.974 ± 0.0189	0.976 ± 0.02
Multinomial	**0.942 ± 0.0276**	0.961 ± 0.0219	0.974 ± 0.0198	0.975 ± 0.0206
Systematic, stop-gradient (PyDPF)	**0.942 ± 0.025**	0.967 ± 0.0197	0.976 ± 0.0185	0.977 ± 0.019
Soft (α=0.5)	0.937 ± 0.0324	0.959 ± 0.0218	0.973 ± 0.0204	0.977 ± 0.0191
OT (PyDPF)	0.94 ± 0.0376	0.97 ± 0.0194	–	–
OT (PFJAX)	0.938 ± 0.0402	0.97 ± 0.0194	–	–
OT (FilterFlow)	0.936 ± 0.042	0.968 ± 0.0205	–	–
Eigen moment matching	0.937 ± 0.0298	0.966 ± 0.0203	0.975 ± 0.0181	0.978 ± 0.0184
Gaussian moment matching	0.939 ± 0.0256	0.965 ± 0.0197	0.977 ± 0.0172	0.978 ± 0.0186
Gumbel-softmax	0.916 ± 0.0343	0.946 ± 0.037	–	–
Variational (KL)	0.938 ± 0.0296	0.96 ± 0.0207	0.972 ± 0.0189	0.973 ± 0.0207

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

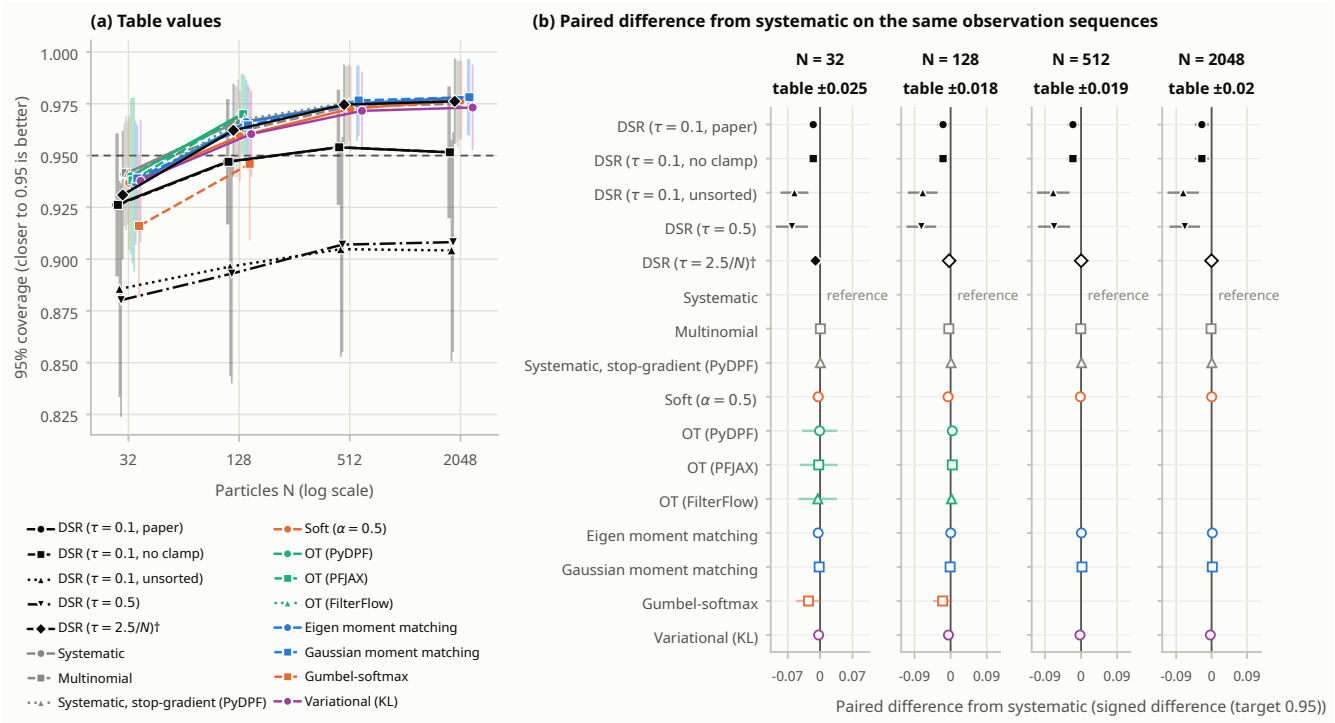


Figure 17: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Dashed line: nominal 0.95. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 18: Stochastic volatility: predictive NLL (lower is better). $x_t = 0.95x_{t-1} + 0.2\varepsilon_t$, $y_t = \exp(x_t/2)\eta_t$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	1.66 ± 0.164	1.66 ± 0.164	1.66 ± 0.162	1.66 ± 0.163
DSR (τ=0.1, no clamp)	1.66 ± 0.164	1.66 ± 0.164	1.66 ± 0.162	1.66 ± 0.163
DSR (τ=0.1, unsorted)	1.67 ± 0.164	1.66 ± 0.163	1.66 ± 0.163	1.66 ± 0.163
DSR (τ=0.5)	1.67 ± 0.164	1.66 ± 0.163	1.66 ± 0.163	1.66 ± 0.163
DSR (τ=2.5/N)†	1.66 ± 0.164	1.66 ± 0.163	1.66 ± 0.162	1.66 ± 0.162
Systematic	1.66 ± 0.163	1.66 ± 0.163	1.66 ± 0.162	1.66 ± 0.162
Multinomial	1.66 ± 0.163	1.66 ± 0.163	1.66 ± 0.162	1.66 ± 0.162
Systematic, stop-gradient (PyDPF)	1.66 ± 0.164	1.66 ± 0.163	1.66 ± 0.162	1.66 ± 0.163
Soft (α=0.5)	1.66 ± 0.164	1.66 ± 0.163	1.66 ± 0.162	1.66 ± 0.162
OT (PyDPF)	1.66 ± 0.162	1.66 ± 0.163	–	–
OT (PFJAX)	$\mathbf{1.66 \pm 0.162}$	1.66 ± 0.163	–	–
OT (FilterFlow)	1.66 ± 0.162	1.66 ± 0.163	–	–
Eigen moment matching	1.66 ± 0.164	1.66 ± 0.163	1.66 ± 0.162	1.66 ± 0.163
Gaussian moment matching	1.66 ± 0.163	$\mathbf{1.66 \pm 0.163}$	1.66 ± 0.162	1.66 ± 0.163
Gumbel-softmax	1.66 ± 0.161	1.66 ± 0.164	–	–
Variational (KL)	1.66 ± 0.163	1.66 ± 0.163	$\mathbf{1.66 \pm 0.162}$	$\mathbf{1.66 \pm 0.162}$

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

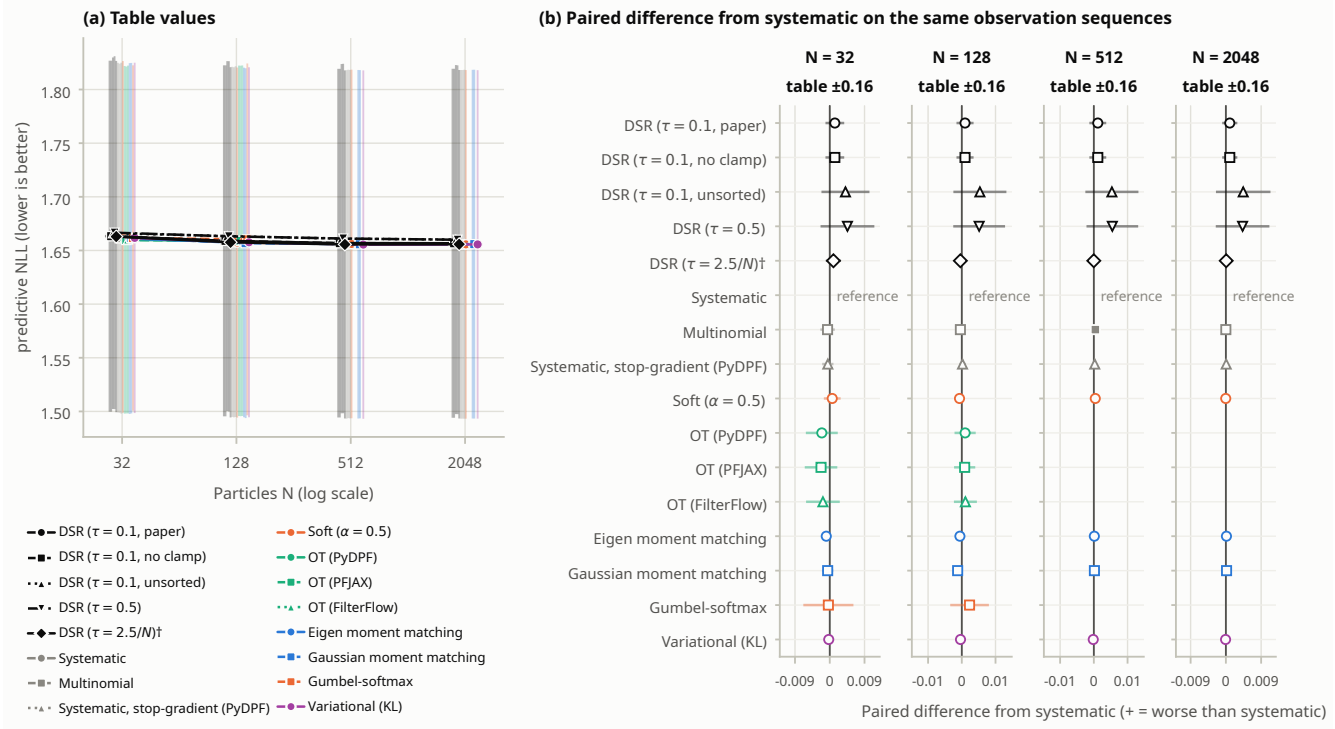


Figure 18: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 19: Stochastic volatility: log-likelihood variance given the data (lower is better). $x_t = 0.95x_{t-1} + 0.2\varepsilon_t$, $y_t = \exp(x_t/2)\eta_t$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.564	0.0958	0.0203	0.0102
DSR (τ=0.1, no clamp)	0.564	0.0958	0.0203	0.0102
DSR (τ=0.1, unsorted)	0.731	0.216	0.0783	0.0342
DSR (τ=0.5)	0.733	0.208	0.0801	0.0342
DSR (τ=2.5/N)†	0.536	0.0849	0.0159	**0.00442**
Systematic	0.557	0.0829	0.018	0.00529
Multinomial	0.429	0.107	0.0193	0.00506
Systematic, stop-gradient (PyDPF)	0.538	0.0806	0.0201	0.00534
Soft (α=0.5)	0.558	0.115	0.0258	0.00876
Eigen moment matching	**0.409**	**0.0746**	**0.0139**	0.00474
Gaussian moment matching	0.547	0.0786	0.0197	0.00579
Variational (KL)	0.499	0.0931	0.0149	0.00492

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

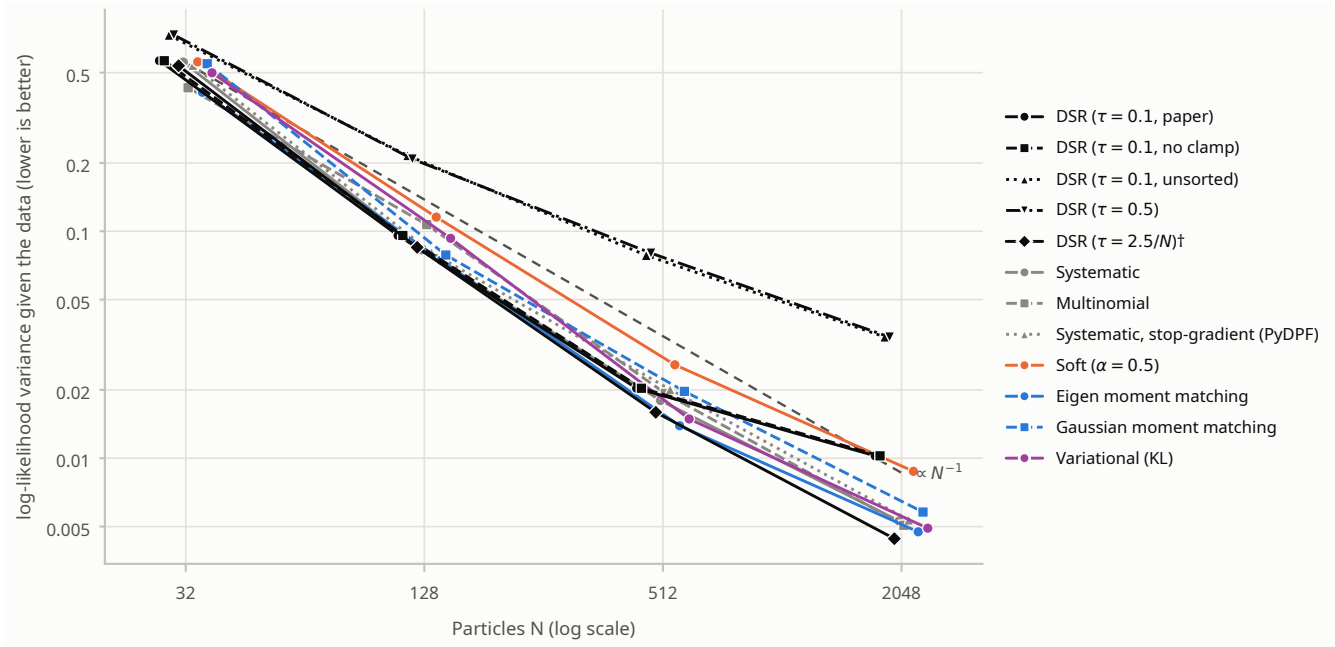


Figure 19: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 20: Bearing-only tracking: position RMSE (lower is better). 2-D constant velocity, bearing noise 0.05 rad, observations hidden at steps 21–30. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	5.12 ± 2.38	4.83 ± 2.73	4.87 ± 2.83	5.19 ± 2.91
DSR (τ=0.1, no clamp)	5.12 ± 2.38	4.83 ± 2.73	4.87 ± 2.83	5.19 ± 2.91
DSR (τ=0.1, unsorted)	4.83 ± 2.19	4.95 ± 2.25	4.84 ± 2.22	4.91 ± 2.27
DSR (τ=0.5)	5.14 ± 2.23	4.81 ± 2.23	4.87 ± 2.28	4.9 ± 2.33
DSR (τ=2.5/N)†	5.11 ± 2.26	5.04 ± 2.33	4.81 ± 2.15	4.42 ± 1.83
Systematic	5.08 ± 1.78	4.55 ± 1.66	4.13 ± 1.68	4.14 ± 1.68
Multinomial	5.38 ± 1.82	4.79 ± 1.63	4.21 ± 1.74	4.16 ± 1.59
Systematic, stop-gradient (PyDPF)	5.5 ± 1.76	4.76 ± 1.75	$\mathbf{4.11 \pm 1.73}$	$\mathbf{4.11 \pm 1.66}$
Soft (α=0.5)	5.62 ± 1.61	4.8 ± 1.71	4.28 ± 1.77	4.2 ± 1.64
OT (PyDPF)	5.31 ± 2.12	5.65 ± 2.28	–	–
OT (PFJAX)	4.97 ± 2.06	5.56 ± 2.29	–	–
OT (FilterFlow)	5.22 ± 2.08	5.3 ± 1.73	–	–
Eigen moment matching	5.12 ± 1.79	4.51 ± 1.71	4.21 ± 1.6	4.24 ± 1.59
Gaussian moment matching	5.23 ± 1.95	$\mathbf{4.51 \pm 1.74}$	4.36 ± 1.63	4.2 ± 1.58
Gumbel-softmax	$\mathbf{4.45 \pm 2.63}$	5.19 ± 2.52	–	–
Variational (KL)	5.19 ± 1.75	4.66 ± 1.63	4.2 ± 1.7	4.27 ± 1.71

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

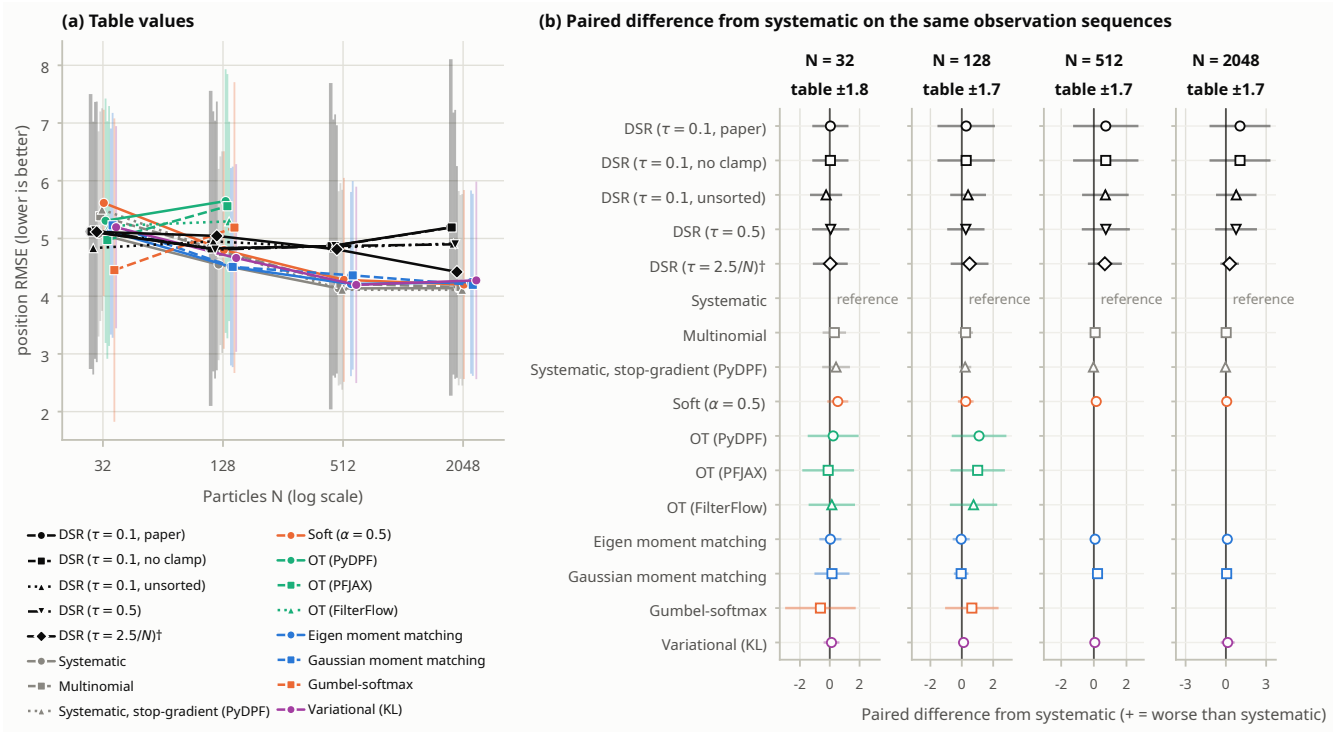


Figure 20: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 21: Bearing-only tracking: RMSE after the observation gap (lower is better). 2-D constant velocity, bearing noise 0.05 rad, observations hidden at steps 21–30. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	9.69 ± 4.53	9.25 ± 5.3	9.31 ± 5.57	9.97 ± 5.76
DSR (τ=0.1, no clamp)	9.69 ± 4.53	9.25 ± 5.3	9.31 ± 5.57	9.97 ± 5.76
DSR (τ=0.1, unsorted)	9.15 ± 4.11	9.52 ± 4.27	9.33 ± 4.2	9.46 ± 4.3
DSR (τ=0.5)	9.77 ± 4.17	9.22 ± 4.24	9.38 ± 4.34	9.44 ± 4.43
DSR (τ=2.5/N)†	9.66 ± 4.25	9.53 ± 4.38	9.03 ± 4.18	8.14 ± 3.46
Systematic	9.54 ± 3.21	8.44 ± 3.08	7.61 ± 3.14	7.58 ± 3.12
Multinomial	10.1 ± 3.37	8.92 ± 2.93	7.76 ± 3.25	7.65 ± 2.94
Systematic, stop-gradient (PyDPF)	10.4 ± 3.29	8.96 ± 3.33	$\mathbf{7.57 \pm 3.19}$	$\mathbf{7.54 \pm 3.04}$
Soft (α=0.5)	10.6 ± 2.94	9.01 ± 3.17	7.92 ± 3.27	7.71 ± 3.04
OT (PyDPF)	9.98 ± 4.01	10.9 ± 4.49	–	–
OT (PFJAX)	9.19 ± 3.9	10.7 ± 4.47	–	–
OT (FilterFlow)	9.96 ± 3.91	10.2 ± 3.24	–	–
Eigen moment matching	9.59 ± 3.36	$\mathbf{8.28 \pm 3.27}$	7.59 ± 3.01	7.69 ± 3
Gaussian moment matching	9.74 ± 3.66	8.3 ± 3.21	7.98 ± 3.1	7.62 ± 2.96
Gumbel-softmax	$\mathbf{8.26 \pm 5.27}$	9.89 ± 4.76	–	–
Variational (KL)	9.81 ± 3.15	8.76 ± 3.03	7.73 ± 3.17	7.9 ± 3.25

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

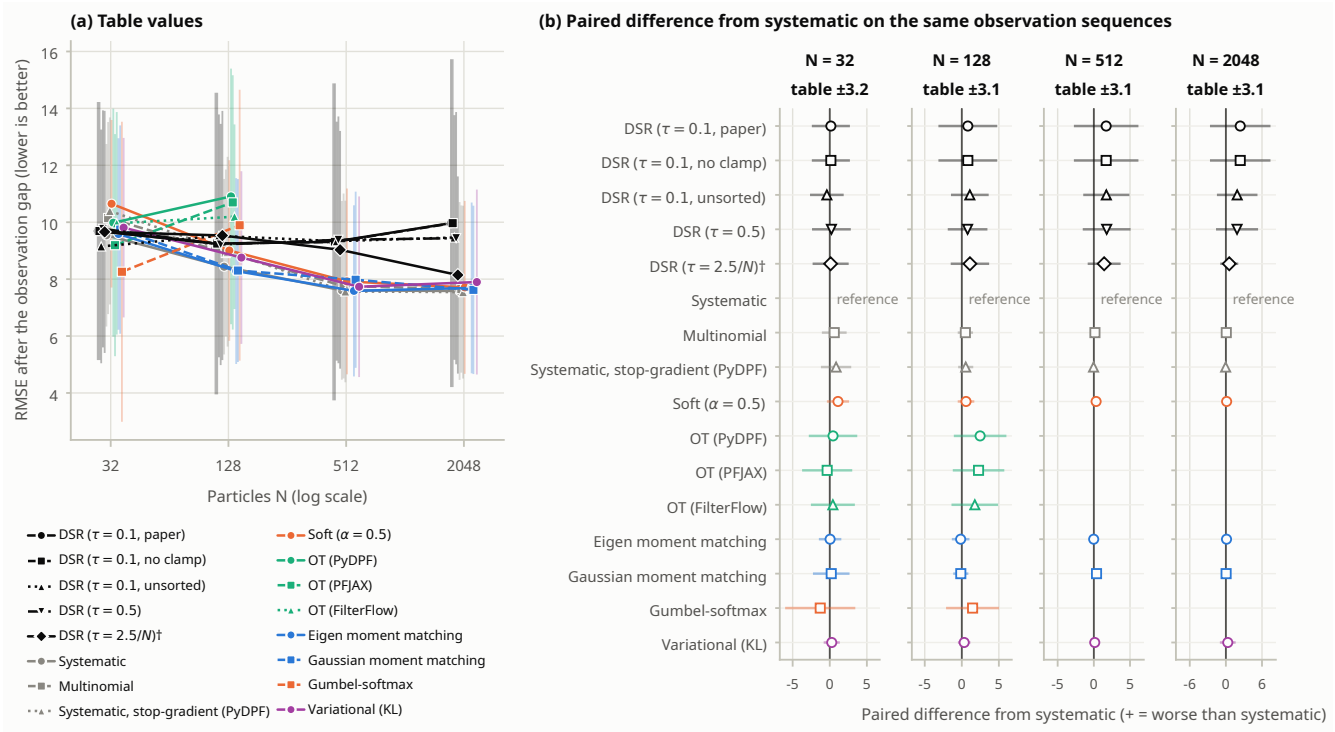


Figure 21: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 22: Bearing-only tracking: 95% coverage (closer to 0.95 is better). 2-D constant velocity, bearing noise 0.05 rad, observations hidden at steps 21–30. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.0692 ± 0.041	0.0836 ± 0.062	0.0874 ± 0.0684	0.0572 ± 0.0502
DSR (τ=0.1, no clamp)	0.0692 ± 0.041	0.0836 ± 0.062	0.0874 ± 0.0684	0.0572 ± 0.0502
DSR (τ=0.1, unsorted)	0.0422 ± 0.0264	0.0202 ± 0.00833	0.0204 ± 0.00991	0.0174 ± 0.00688
DSR (τ=0.5)	0.02 ± 0.0117	0.0234 ± 0.0157	0.0216 ± 0.0114	0.02 ± 0.0104
DSR (τ=2.5/N)†	0.0762 ± 0.0429	0.299 ± 0.163	0.494 ± 0.222	0.682 ± 0.218
Systematic	0.265 ± 0.136	0.636 ± 0.206	0.826 ± 0.18	**0.904 ± 0.117**
Multinomial	0.196 ± 0.121	0.54 ± 0.21	0.809 ± 0.187	0.901 ± 0.119
Systematic, stop-gradient (PyDPF)	0.21 ± 0.137	0.649 ± 0.194	0.85 ± 0.179	0.904 ± 0.115
Soft (α=0.5)	0.179 ± 0.102	0.519 ± 0.196	0.802 ± 0.193	0.891 ± 0.126
OT (PyDPF)	0.09 ± 0.104	0.04 ± 0.0261	–	–
OT (PFJAX)	0.092 ± 0.102	0.042 ± 0.0256	–	–
OT (FilterFlow)	0.054 ± 0.0618	0.052 ± 0.0655	–	–
Eigen moment matching	**0.51 ± 0.187**	**0.778 ± 0.198**	**0.87 ± 0.166**	0.901 ± 0.128
Gaussian moment matching	0.362 ± 0.179	0.724 ± 0.191	0.85 ± 0.174	0.902 ± 0.118
Gumbel-softmax	0.196 ± 0.202	0.216 ± 0.176	–	–
Variational (KL)	0.246 ± 0.134	0.633 ± 0.191	0.83 ± 0.176	0.873 ± 0.125

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

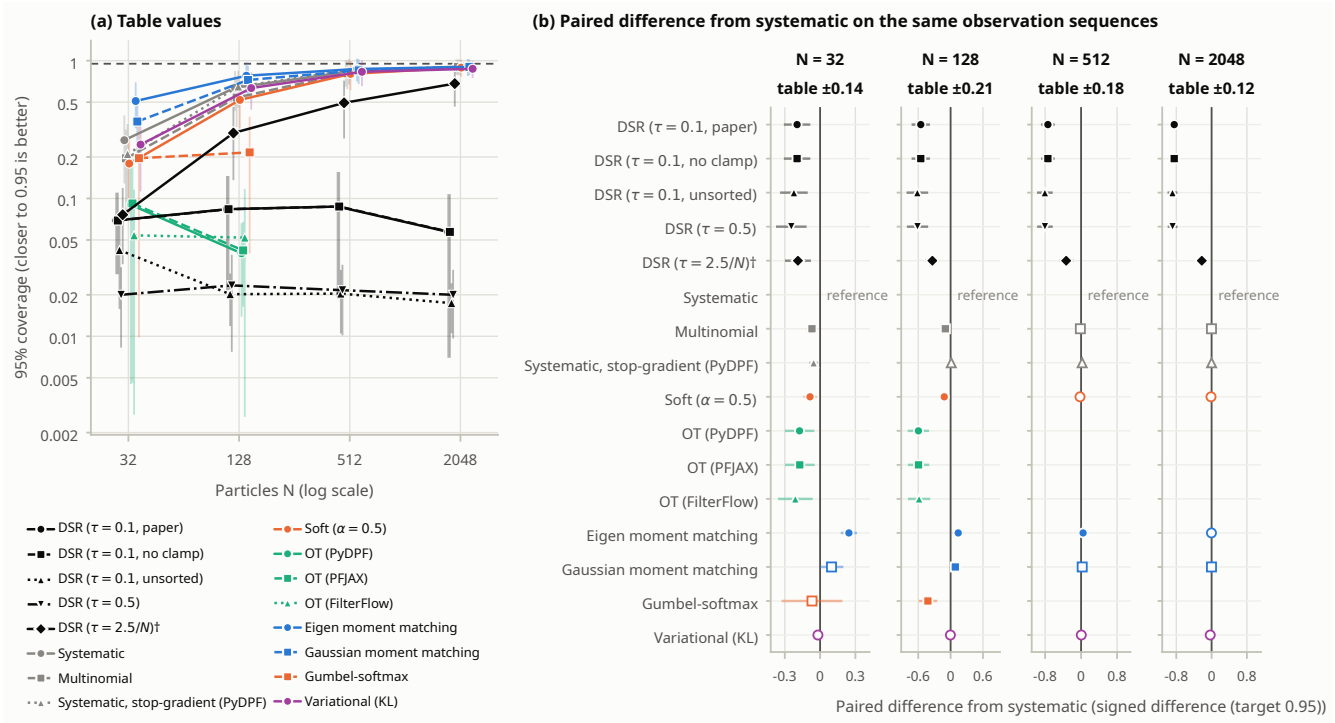


Figure 22: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. Dashed line: nominal 0.95. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 23: Bearing-only tracking: predictive NLL (lower is better). 2-D constant velocity, bearing noise 0.05 rad, observations hidden at steps 21–30. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	-0.528 ± 0.367	-0.638 ± 0.379	-0.612 ± 0.463	-0.435 ± 0.575
DSR (τ=0.1, no clamp)	-0.528 ± 0.367	-0.638 ± 0.379	-0.612 ± 0.463	-0.435 ± 0.575
DSR (τ=0.1, unsorted)	-0.385 ± 0.407	-0.536 ± 0.335	-0.565 ± 0.315	-0.646 ± 0.229
DSR (τ=0.5)	0.104 ± 0.443	-0.454 ± 0.273	-0.545 ± 0.284	-0.611 ± 0.243
DSR (τ=2.5/N)†	-0.613 ± 0.352	-1.16 ± 0.187	-1.27 ± 0.14	-1.31 ± 0.0977
Systematic	-1.2 ± 0.13	-1.31 ± 0.0827	-1.32 ± 0.078	-1.33 ± 0.0779
Multinomial	-1.09 ± 0.119	-1.3 ± 0.0922	$\mathbf{-1.32 \pm 0.0785}$	-1.33 ± 0.0783
Systematic, stop-gradient (PyDPF)	-1.2 ± 0.0929	-1.31 ± 0.0819	-1.32 ± 0.0778	-1.33 ± 0.0781
Soft (α=0.5)	-1 ± 0.181	-1.29 ± 0.0965	-1.32 ± 0.0787	$\mathbf{-1.33 \pm 0.0775}$
OT (PyDPF)	-0.865 ± 0.386	-1.14 ± 0.12	–	–
OT (PFJAX)	-0.905 ± 0.366	-1.14 ± 0.118	–	–
OT (FilterFlow)	-1.08 ± 0.135	-0.797 ± 0.483	–	–
Eigen moment matching	-1.27 ± 0.09	$\mathbf{-1.32 \pm 0.0786}$	-1.32 ± 0.0788	-1.32 ± 0.0785
Gaussian moment matching	$\mathbf{-1.27 \pm 0.0978}$	-1.32 ± 0.0813	-1.32 ± 0.0788	-1.33 ± 0.0783
Gumbel-softmax	-1.21 ± 0.13	-1.23 ± 0.164	–	–
Variational (KL)	-1.2 ± 0.115	-1.31 ± 0.0852	-1.32 ± 0.079	-1.32 ± 0.0788

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

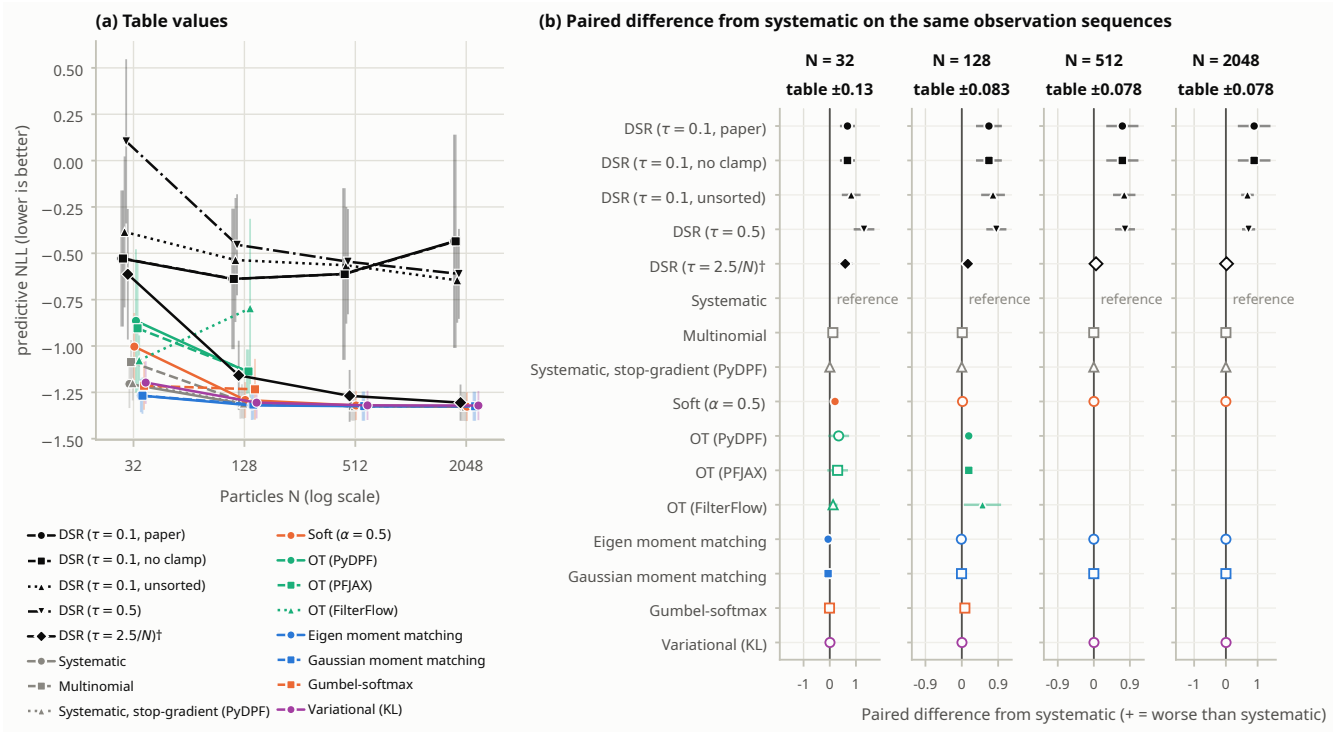


Figure 23: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 24: Bearing-only tracking: log-likelihood variance given the data (lower is better). 2-D constant velocity, bearing noise 0.05 rad, observations hidden at steps 21–30. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	$2.36e+03$	547	302	197
DSR (τ=0.1, no clamp)	$2.36e+03$	547	302	197
DSR (τ=0.1, unsorted)	$3.86e+03$	562	338	261
DSR (τ=0.5)	$4.85e+03$	415	251	203
DSR (τ=2.5/N)†	$1.98e+03$	92.7	31	4.97
Systematic	128	4.74	0.449	0.0906
Multinomial	742	19.1	0.451	0.108
Systematic, stop-gradient (PyDPF)	132	4.65	0.271	0.058
Soft (α=0.5)	$1.11e+03$	9.43	0.543	0.114
Eigen moment matching	106	**0.586**	**0.0826**	**0.0213**
Gaussian moment matching	**14.2**	1.99	0.263	0.0345
Variational (KL)	120	7.52	0.688	0.265

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

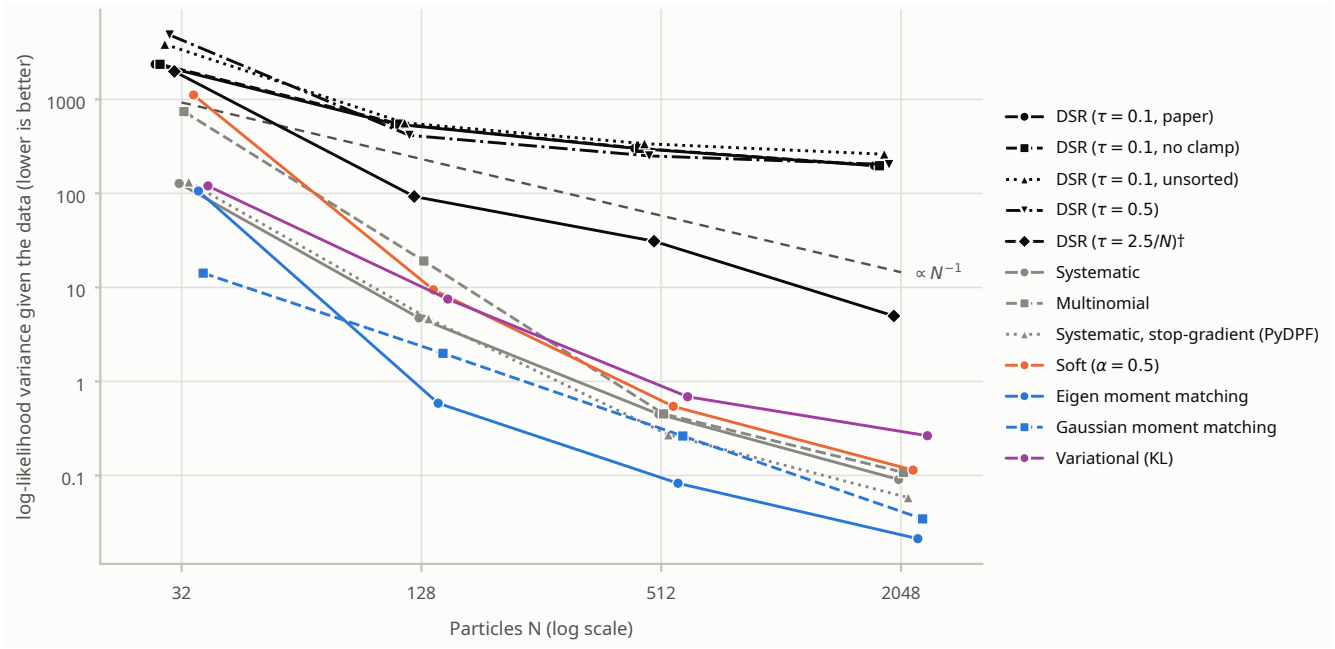


Figure 24: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Stochastic Lorenz models (all methods)

Table 25: Stochastic Lorenz-63: state RMSE (lower is better). $x_{t+1} = F(x_t)x_t + e_t$, F the 5th-order Taylor series of $\exp(A(x_t)\,0.02)$, $e_t \sim \mathcal{N}(0, 0.1I_3)$; 8×8 camera observation (64 pixels) at SMNR 10 dB, as in the DSR paper; $x_0 \sim \mathcal{N}(0, I_3)$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.708 ± 0.209	0.542 ± 0.0917	0.491 ± 0.0404	0.486 ± 0.0408
DSR (τ=0.1, no clamp)	0.708 ± 0.209	0.542 ± 0.0917	0.491 ± 0.0404	0.486 ± 0.0408
DSR (τ=0.1, unsorted)	0.853 ± 0.42	0.619 ± 0.212	0.538 ± 0.0874	0.514 ± 0.0569
DSR (τ=0.5)	1.22 ± 0.705	0.792 ± 0.375	0.548 ± 0.0997	0.52 ± 0.0638
DSR (τ=2.5/N)†	0.64 ± 0.168	0.479 ± 0.036	0.446 ± 0.0301	0.433 ± 0.0304
Multinomial (particles)	0.549 ± 0.0538	0.469 ± 0.0298	0.44 ± 0.0319	0.43 ± 0.0301
Systematic (particles)	0.563 ± 0.0576	0.456 ± 0.032	0.437 ± 0.0293	0.431 ± 0.0297
Stratified (particles)	0.534 ± 0.0459	0.457 ± 0.0317	0.439 ± 0.0318	0.429 ± 0.0297
Residual (particles)	0.609 ± 0.133	0.457 ± 0.0286	$\mathbf{0.435 \pm 0.0316}$	0.429 ± 0.029
SSP (particles)	0.585 ± 0.102	0.46 ± 0.0356	0.438 ± 0.0279	0.43 ± 0.0291
Multinomial (Stone Soup)	0.61 ± 0.118	0.464 ± 0.0309	0.438 ± 0.0325	0.431 ± 0.0287
Systematic (Stone Soup)	0.548 ± 0.0539	0.465 ± 0.0347	0.44 ± 0.0336	$\mathbf{0.429 \pm 0.0284}$
Stratified (Stone Soup)	0.543 ± 0.0545	0.461 ± 0.0274	0.437 ± 0.0281	0.431 ± 0.0291
Residual (Stone Soup)	0.54 ± 0.0483	0.465 ± 0.0328	0.436 ± 0.0294	0.431 ± 0.0295
Multinomial (Cuthbert)	0.606 ± 0.139	0.459 ± 0.028	0.436 ± 0.03	0.431 ± 0.0292
Systematic (Cuthbert)	0.541 ± 0.045	0.461 ± 0.0311	0.437 ± 0.0301	0.43 ± 0.0297
Systematic, stop-gradient (Cuthbert)	0.541 ± 0.045	0.461 ± 0.0311	0.437 ± 0.0301	0.43 ± 0.0297
Multinomial (PFJAX)	0.569 ± 0.074	0.457 ± 0.035	0.439 ± 0.0283	0.43 ± 0.03
Gaussian moment matching, MVN (PFJAX)	$\mathbf{0.531 \pm 0.0422}$	0.456 ± 0.0327	0.44 ± 0.031	0.431 ± 0.0302
OT (PFJAX)	0.55 ± 0.0543	0.468 ± 0.0495	–	–
Multinomial (PyDPF)	0.666 ± 0.161	0.461 ± 0.0323	0.438 ± 0.032	0.431 ± 0.0287
Systematic (PyDPF)	0.584 ± 0.0955	0.464 ± 0.0326	0.439 ± 0.0305	0.431 ± 0.0301
Systematic, stop-gradient (PyDPF)	0.584 ± 0.0955	0.464 ± 0.0326	0.439 ± 0.0305	0.431 ± 0.0301
Soft (α=0.5) (PyDPF)	0.616 ± 0.0947	0.467 ± 0.0374	0.44 ± 0.0319	0.43 ± 0.0302
OT (PyDPF)	0.542 ± 0.0572	0.477 ± 0.0375	–	–
Multinomial (torchfilter)	0.666 ± 0.161	0.461 ± 0.0323	0.438 ± 0.032	0.431 ± 0.0287
Soft (α=0.5) (torchfilter)	0.616 ± 0.0947	0.467 ± 0.0374	0.44 ± 0.0319	0.43 ± 0.0302
Multinomial (FilterFlow)	0.61 ± 0.152	0.462 ± 0.0354	0.439 ± 0.0299	0.431 ± 0.0296
Systematic (FilterFlow)	0.55 ± 0.0618	0.458 ± 0.039	0.438 ± 0.0344	0.429 ± 0.0293
Stratified (FilterFlow)	0.555 ± 0.0663	0.458 ± 0.0325	0.438 ± 0.0315	0.43 ± 0.0296
OT (FilterFlow)	0.595 ± 0.0447	0.491 ± 0.0529	–	–
Covariance-corrected OT (FilterFlow)	0.556 ± 0.055	0.473 ± 0.0368	–	–
Multinomial (Resamplings.jl)	0.598 ± 0.085	0.466 ± 0.029	0.437 ± 0.0301	0.429 ± 0.0273
Systematic (Resamplings.jl)	0.546 ± 0.057	0.457 ± 0.0322	0.436 ± 0.0322	0.431 ± 0.029
Stratified (Resamplings.jl)	0.553 ± 0.062	0.462 ± 0.035	0.438 ± 0.0294	0.43 ± 0.0294
Residual (Resamplings.jl)	0.543 ± 0.0519	0.464 ± 0.0319	0.437 ± 0.0292	0.43 ± 0.028
SSP (Resamplings.jl)	0.538 ± 0.0508	0.457 ± 0.0309	0.437 ± 0.032	0.431 ± 0.0297
Killing (Resamplings.jl)	0.562 ± 0.056	0.466 ± 0.0342	0.438 ± 0.0294	0.43 ± 0.0292
Eigen moment matching	0.533 ± 0.0492	$\mathbf{0.456 \pm 0.0319}$	0.437 ± 0.0302	0.43 ± 0.0289
Gaussian moment matching	0.591 ± 0.105	0.46 ± 0.0329	0.439 ± 0.0315	0.43 ± 0.0302
Gumbel-softmax	0.756 ± 0.46	0.484 ± 0.0639	–	–
Variational (KL)	0.546 ± 0.0595	0.461 ± 0.0324	0.438 ± 0.0299	0.429 ± 0.0288
Variational (TV)	0.577 ± 0.0864	0.46 ± 0.0357	0.435 ± 0.0304	0.432 ± 0.0312

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

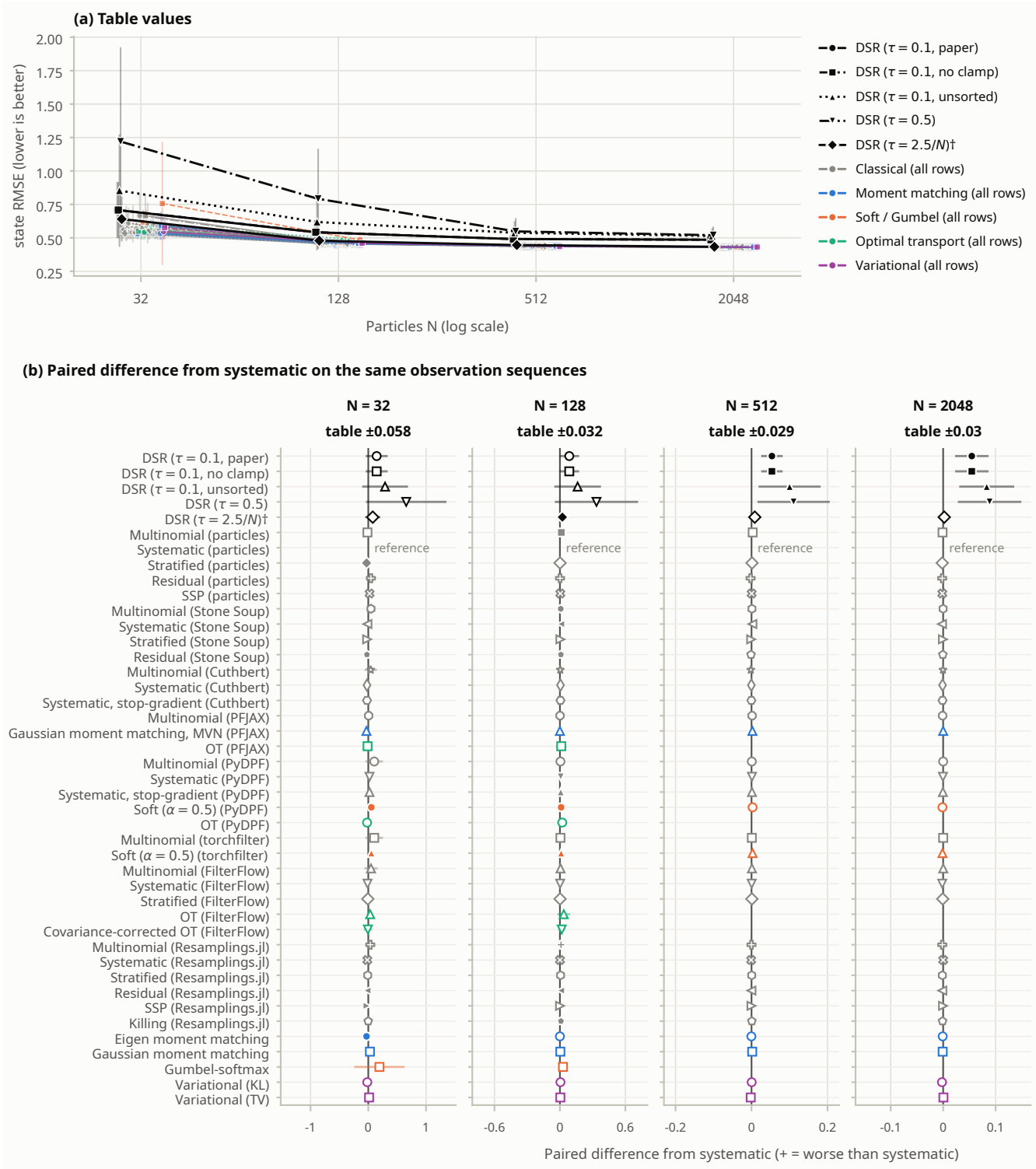


Figure 25: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 26: Stochastic Lorenz-63: 95% coverage (closer to 0.95 is better). $x_{t+1} = F(x_t)x_t + e_t$, F the 5th-order Taylor series of $\exp(A(x_t)\,0.02)$, $e_t \sim \mathcal{N}(0, 0.1I_3)$; 8×8 camera observation (64 pixels) at SMNR 10 dB, as in the DSR paper; $x_0 \sim \mathcal{N}(0, I_3)$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.397 ± 0.0632	0.579 ± 0.0734	0.655 ± 0.0765	0.682 ± 0.0707
DSR (τ=0.1, no clamp)	0.397 ± 0.0632	0.579 ± 0.0734	0.655 ± 0.0765	0.682 ± 0.0707
DSR (τ=0.1, unsorted)	0.378 ± 0.0755	0.516 ± 0.0843	0.576 ± 0.0896	0.597 ± 0.0933
DSR (τ=0.5)	0.332 ± 0.0725	0.489 ± 0.0938	0.563 ± 0.0941	0.593 ± 0.0967
DSR (τ=2.5/N)†	0.418 ± 0.0685	0.723 ± 0.0576	0.877 ± 0.0343	0.932 ± 0.0297
Multinomial (particles)	0.581 ± 0.0664	0.833 ± 0.0444	0.928 ± 0.0297	0.95 ± 0.0245
Systematic (particles)	0.58 ± 0.0676	0.854 ± 0.0422	0.933 ± 0.0283	0.95 ± 0.0246
Stratified (particles)	0.597 ± 0.0716	0.856 ± 0.0382	0.926 ± 0.0283	0.95 ± 0.0239
Residual (particles)	0.575 ± 0.0766	0.843 ± 0.041	0.931 ± 0.0254	0.95 ± 0.024
SSP (particles)	0.575 ± 0.0842	0.852 ± 0.0372	0.93 ± 0.0269	0.952 ± 0.0225
Multinomial (Stone Soup)	0.558 ± 0.0661	0.84 ± 0.0433	0.927 ± 0.0266	0.952 ± 0.0231
Systematic (Stone Soup)	0.583 ± 0.0712	0.846 ± 0.0512	0.925 ± 0.0314	0.953 ± 0.0218
Stratified (Stone Soup)	0.576 ± 0.071	0.852 ± 0.0431	0.927 ± 0.0246	0.95 ± 0.0248
Residual (Stone Soup)	0.591 ± 0.0594	0.846 ± 0.048	$\mathbf{0.935 \pm 0.0269}$	0.952 ± 0.023
Multinomial (Cuthbert)	0.571 ± 0.0739	0.842 ± 0.0399	0.932 ± 0.0253	0.948 ± 0.0259
Systematic (Cuthbert)	0.591 ± 0.0625	0.847 ± 0.042	0.933 ± 0.0301	0.951 ± 0.0234
Systematic, stop-gradient (Cuthbert)	0.591 ± 0.0625	0.847 ± 0.042	0.933 ± 0.0301	0.951 ± 0.0234
Multinomial (PFJAX)	0.56 ± 0.0742	0.846 ± 0.0394	0.929 ± 0.0272	0.95 ± 0.0253
Gaussian moment matching, MVN (PFJAX)	$\mathbf{0.631 \pm 0.0665}$	$\mathbf{0.862 \pm 0.0391}$	0.932 ± 0.0287	0.948 ± 0.0257
OT (PFJAX)	0.532 ± 0.0577	0.804 ± 0.061	–	–
Multinomial (PyDPF)	0.544 ± 0.073	0.841 ± 0.0444	0.932 ± 0.0284	0.949 ± 0.024
Systematic (PyDPF)	0.568 ± 0.0718	0.849 ± 0.0434	0.928 ± 0.029	0.95 ± 0.0234
Systematic, stop-gradient (PyDPF)	0.568 ± 0.0718	0.849 ± 0.0434	0.928 ± 0.029	0.95 ± 0.0234
Soft (α=0.5) (PyDPF)	0.473 ± 0.0717	0.814 ± 0.0563	0.922 ± 0.0333	$\mathbf{0.95 \pm 0.0243}$
OT (PyDPF)	0.506 ± 0.058	0.772 ± 0.0577	–	–
Multinomial (torchfilter)	0.544 ± 0.073	0.841 ± 0.0444	0.932 ± 0.0284	0.949 ± 0.024
Soft (α=0.5) (torchfilter)	0.473 ± 0.0717	0.814 ± 0.0563	0.922 ± 0.0333	$\mathbf{0.95 \pm 0.0243}$
Multinomial (FilterFlow)	0.566 ± 0.0736	0.837 ± 0.0454	0.93 ± 0.0275	0.947 ± 0.0253
Systematic (FilterFlow)	0.583 ± 0.0804	0.846 ± 0.0505	0.926 ± 0.0335	0.951 ± 0.0246
Stratified (FilterFlow)	0.595 ± 0.0689	0.849 ± 0.0448	0.926 ± 0.03	0.948 ± 0.0245
OT (FilterFlow)	0.484 ± 0.0713	0.734 ± 0.0748	–	–
Covariance-corrected OT (FilterFlow)	0.558 ± 0.074	0.814 ± 0.0651	–	–
Multinomial (Resamplings.jl)	0.553 ± 0.0622	0.847 ± 0.0416	0.932 ± 0.0265	0.95 ± 0.0209
Systematic (Resamplings.jl)	0.584 ± 0.0755	0.851 ± 0.0453	0.924 ± 0.0297	0.949 ± 0.0253
Stratified (Resamplings.jl)	0.579 ± 0.0668	0.847 ± 0.0446	0.932 ± 0.0274	0.95 ± 0.0245
Residual (Resamplings.jl)	0.593 ± 0.0704	0.842 ± 0.0422	0.93 ± 0.0289	0.95 ± 0.0231
SSP (Resamplings.jl)	0.596 ± 0.0677	0.852 ± 0.0453	0.929 ± 0.0298	0.947 ± 0.027
Killing (Resamplings.jl)	0.562 ± 0.0696	0.838 ± 0.0461	0.929 ± 0.0264	0.947 ± 0.0238
Eigen moment matching	0.612 ± 0.0674	0.858 ± 0.033	0.932 ± 0.0272	0.949 ± 0.0242
Gaussian moment matching	0.568 ± 0.07	0.847 ± 0.0386	0.929 ± 0.0257	0.947 ± 0.0258
Gumbel-softmax	0.464 ± 0.127	0.702 ± 0.0913	–	–
Variational (KL)	0.588 ± 0.0827	0.845 ± 0.0432	0.927 ± 0.0278	0.949 ± 0.0225
Variational (TV)	0.586 ± 0.0704	0.851 ± 0.0419	0.927 ± 0.027	0.945 ± 0.0266

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

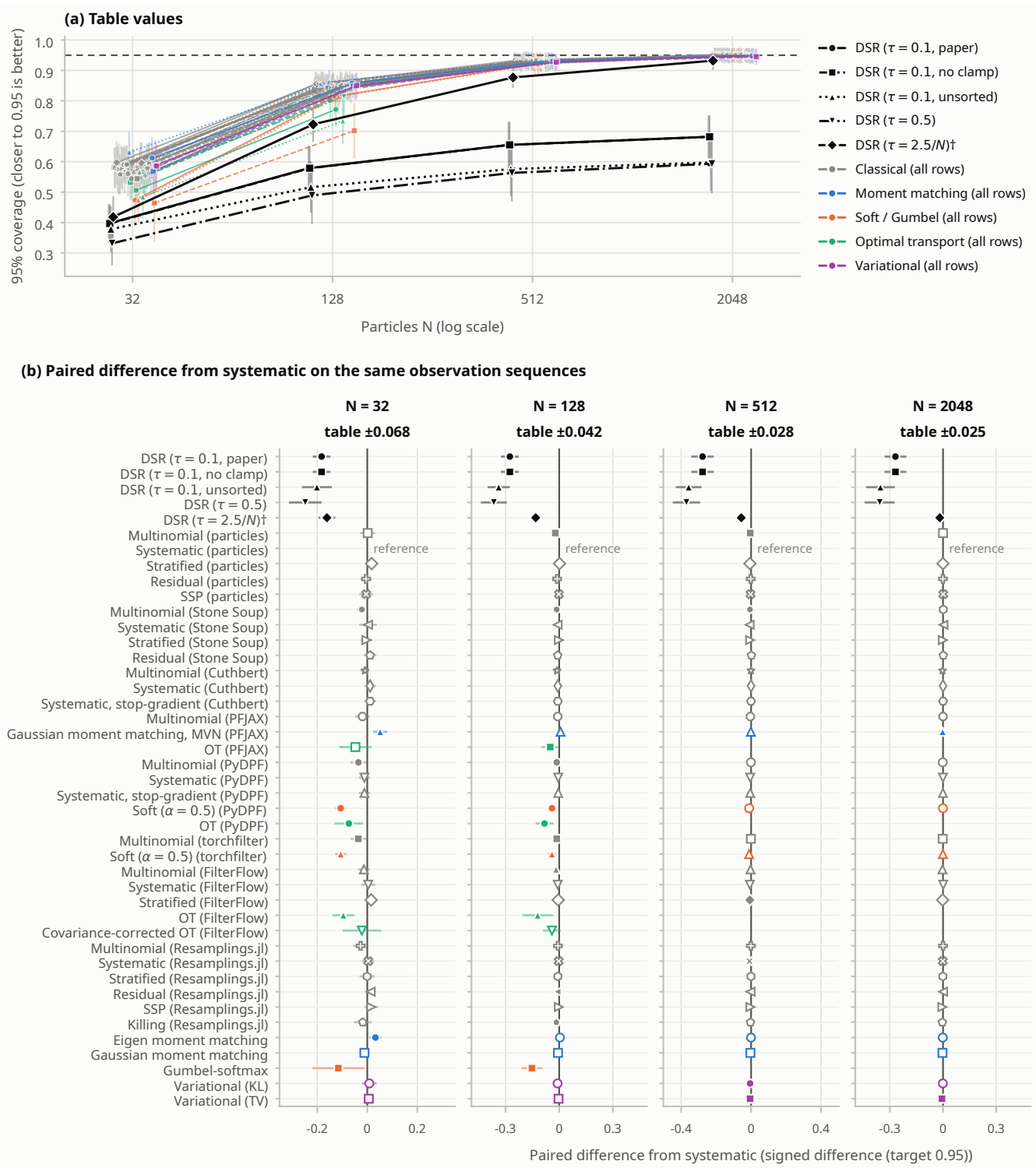


Figure 26: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Dashed line: nominal 0.95. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 27: Stochastic Lorenz-63: predictive NLL (lower is better). $x_{t+1} = F(x_t)x_t + e_t$, F the 5th-order Taylor series of $\exp(A(x_t)\,0.02)$, $e_t \sim \mathcal{N}(0, 0.1I_3)$; 8×8 camera observation (64 pixels) at SMNR 10 dB, as in the DSR paper; $x_0 \sim \mathcal{N}(0, I_3)$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	35.3 ± 4.62	31.8 ± 1	31.4 ± 0.579	31.4 ± 0.582
DSR (τ=0.1, no clamp)	35.3 ± 4.62	31.8 ± 1	31.4 ± 0.579	31.4 ± 0.582
DSR (τ=0.1, unsorted)	38.4 ± 10.3	33.4 ± 4.11	31.8 ± 0.86	31.6 ± 0.578
DSR (τ=0.5)	46 ± 14.8	36.6 ± 6.87	31.9 ± 0.952	31.6 ± 0.587
DSR (τ=2.5/N)†	33.7 ± 3.89	31.3 ± 0.574	31.2 ± 0.566	31.2 ± 0.565
Multinomial (particles)	31.6 ± 0.72	31.2 ± 0.572	31.2 ± 0.567	31.2 ± 0.562
Systematic (particles)	31.8 ± 0.756	31.2 ± 0.575	31.2 ± 0.564	31.2 ± 0.563
Stratified (particles)	31.6 ± 0.675	31.2 ± 0.57	31.2 ± 0.564	31.2 ± 0.563
Residual (particles)	33.1 ± 2.6	31.2 ± 0.576	31.2 ± 0.565	31.2 ± 0.563
SSP (particles)	32.4 ± 1.69	31.2 ± 0.571	31.2 ± 0.565	31.2 ± 0.562
Multinomial (Stone Soup)	32.7 ± 1.99	31.2 ± 0.576	31.2 ± 0.565	31.2 ± 0.562
Systematic (Stone Soup)	31.6 ± 0.681	31.2 ± 0.573	31.2 ± 0.564	31.2 ± 0.563
Stratified (Stone Soup)	31.6 ± 0.681	31.2 ± 0.575	31.2 ± 0.562	31.2 ± 0.564
Residual (Stone Soup)	31.6 ± 0.666	31.2 ± 0.578	31.2 ± 0.563	31.2 ± 0.563
Multinomial (Cuthbert)	33.3 ± 3.68	31.2 ± 0.574	$\mathbf{31.2 \pm 0.564}$	31.2 ± 0.562
Systematic (Cuthbert)	31.6 ± 0.682	31.2 ± 0.576	31.2 ± 0.564	31.2 ± 0.563
Systematic, stop-gradient (Cuthbert)	31.6 ± 0.682	31.2 ± 0.576	31.2 ± 0.564	31.2 ± 0.563
Multinomial (PFJAX)	31.7 ± 0.883	31.2 ± 0.58	31.2 ± 0.566	31.2 ± 0.563
Gaussian moment matching, MVN (PFJAX)	$\mathbf{31.5 \pm 0.612}$	31.2 ± 0.571	31.2 ± 0.565	31.2 ± 0.564
OT (PFJAX)	31.5 ± 0.562	31.3 ± 0.577	–	–
Multinomial (PyDPF)	34.1 ± 3.73	31.2 ± 0.572	31.2 ± 0.565	31.2 ± 0.562
Systematic (PyDPF)	32.2 ± 1.31	31.2 ± 0.57	31.2 ± 0.569	31.2 ± 0.562
Systematic, stop-gradient (PyDPF)	32.2 ± 1.31	31.2 ± 0.57	31.2 ± 0.569	31.2 ± 0.562
Soft (α=0.5) (PyDPF)	32.2 ± 1.16	31.2 ± 0.577	31.2 ± 0.564	31.2 ± 0.564
OT (PyDPF)	31.5 ± 0.602	31.3 ± 0.596	–	–
Multinomial (torchfilter)	34.1 ± 3.73	31.2 ± 0.572	31.2 ± 0.565	31.2 ± 0.562
Soft (α=0.5) (torchfilter)	32.2 ± 1.16	31.2 ± 0.577	31.2 ± 0.564	31.2 ± 0.564
Multinomial (FilterFlow)	33.1 ± 3.01	31.2 ± 0.573	31.2 ± 0.567	31.2 ± 0.563
Systematic (FilterFlow)	31.7 ± 0.774	31.2 ± 0.579	31.2 ± 0.565	31.2 ± 0.562
Stratified (FilterFlow)	31.8 ± 0.78	31.2 ± 0.575	31.2 ± 0.564	31.2 ± 0.564
OT (FilterFlow)	31.7 ± 0.64	31.4 ± 0.576	–	–
Covariance-corrected OT (FilterFlow)	31.5 ± 0.622	31.3 ± 0.585	–	–
Multinomial (Resamplings.jl)	32.2 ± 1.48	31.2 ± 0.568	31.2 ± 0.566	31.2 ± 0.562
Systematic (Resamplings.jl)	31.6 ± 0.744	31.2 ± 0.575	31.2 ± 0.567	31.2 ± 0.562
Stratified (Resamplings.jl)	31.7 ± 0.844	31.2 ± 0.576	31.2 ± 0.564	31.2 ± 0.564
Residual (Resamplings.jl)	31.6 ± 0.644	31.2 ± 0.574	31.2 ± 0.564	31.2 ± 0.563
SSP (Resamplings.jl)	31.6 ± 0.641	31.2 ± 0.572	31.2 ± 0.565	31.2 ± 0.563
Killing (Resamplings.jl)	31.7 ± 0.775	31.2 ± 0.569	31.2 ± 0.566	31.2 ± 0.564
Eigen moment matching	31.5 ± 0.664	$\mathbf{31.2 \pm 0.57}$	31.2 ± 0.565	$\mathbf{31.2 \pm 0.563}$
Gaussian moment matching	32.3 ± 1.4	31.2 ± 0.578	31.2 ± 0.565	31.2 ± 0.563
Gumbel-softmax	35.4 ± 8.44	31.3 ± 0.579	–	–
Variational (KL)	31.6 ± 0.673	31.2 ± 0.57	31.2 ± 0.566	31.2 ± 0.564
Variational (TV)	32.5 ± 2.04	31.2 ± 0.574	31.2 ± 0.567	31.2 ± 0.563

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

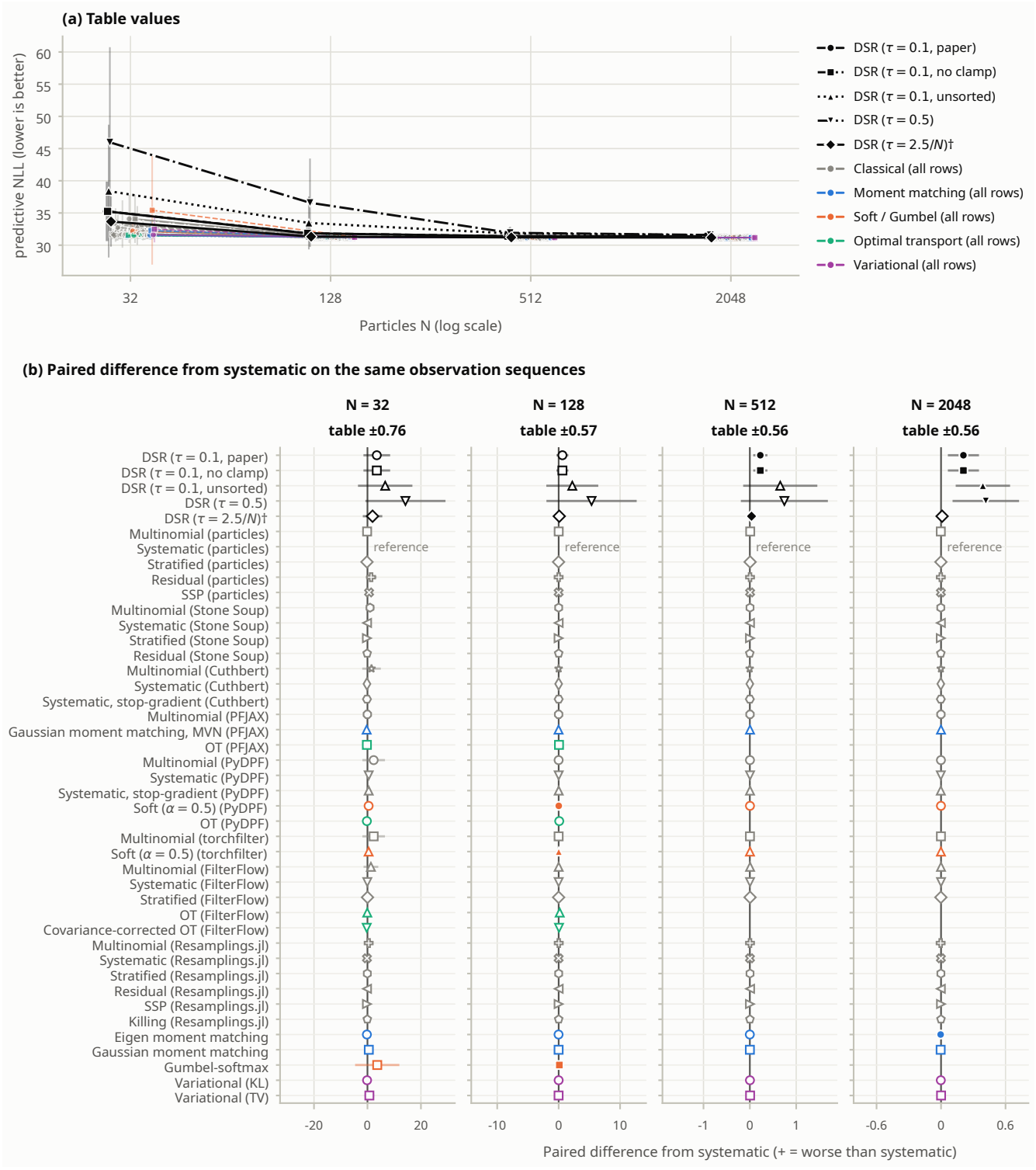


Figure 27: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 28: Stochastic Lorenz-63: log-likelihood variance given the data (lower is better). $x_{t+1} = F(x_t)x_t + e_t$, F the 5th-order Taylor series of $\exp(A(x_t)\,0.02)$, $e_t \sim \mathcal{N}(0, 0.1I_3)$; 8×8 camera observation (64 pixels) at SMNR 10 dB, as in the DSR paper; $x_0 \sim \mathcal{N}(0, I_3)$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	8.46×10^5	$7.86e+03$	47.8	33.6
DSR (τ=0.1, no clamp)	8.46×10^5	$7.86e+03$	47.8	33.6
DSR (τ=0.1, unsorted)	1.04×10^6	3.74×10^5	$6.62e+03$	123
DSR (τ=0.5)	3.76×10^6	2.1×10^6	$7.66e+03$	271
DSR (τ=2.5/N)†	5.95×10^5	40.4	2.07	0.512
Multinomial (particles)	$1.34e+03$	10.3	2.38	0.384
Systematic (particles)	$3.37e+03$	8.85	1.52	0.597
Stratified (particles)	$1.38e+03$	9.89	2.18	0.316
Residual (particles)	1.94×10^5	11.7	1.7	0.381
SSP (particles)	7.28×10^4	39.4	1.8	0.422
Multinomial (Stone Soup)	1.25×10^5	12.1	1.74	0.398
Systematic (Stone Soup)	$1.86e+03$	14.4	1.48	0.322
Stratified (Stone Soup)	912	9.28	1.9	0.337
Residual (Stone Soup)	$2.18e+03$	14.1	1.87	0.449
Multinomial (Cuthbert)	6.63×10^5	9.23	2.15	0.493
Systematic (Cuthbert)	$1.19e+03$	8.75	1.66	0.353
Systematic, stop-gradient (Cuthbert)	$1.19e+03$	8.75	1.66	0.353
Multinomial (PFJAX)	$3.51e+03$	11.4	1.52	0.481
Gaussian moment matching, MVN (PFJAX)	**147**	9.31	1.6	0.381
Multinomial (PyDPF)	9.51×10^5	8.31	1.36	0.489
Systematic (PyDPF)	2.95×10^4	15.4	1.66	0.429
Systematic, stop-gradient (PyDPF)	2.95×10^4	15.4	1.66	0.429
Soft (α=0.5) (PyDPF)	1.65×10^4	14.7	2.13	0.5
Multinomial (torchfilter)	9.51×10^5	8.31	1.36	0.489
Soft (α=0.5) (torchfilter)	1.65×10^4	14.7	2.13	0.5
Multinomial (FilterFlow)	1.74×10^5	10.9	1.93	0.537
Systematic (FilterFlow)	$6.08e+03$	9	1.82	0.334
Stratified (FilterFlow)	$4.44e+03$	9.99	1.6	**0.263**
Multinomial (Resamplings.jl)	6.62×10^4	40.8	1.83	0.539
Systematic (Resamplings.jl)	$1.15e+03$	11.9	1.85	0.33
Stratified (Resamplings.jl)	$4.63e+03$	9.89	1.78	0.357
Residual (Resamplings.jl)	$1.75e+03$	11.1	1.69	0.289
SSP (Resamplings.jl)	$1.56e+03$	9.24	1.55	0.316
Killing (Resamplings.jl)	$3.2e+03$	10.3	2.29	0.484
Eigen moment matching	$1.88e+03$	7.76	**1.31**	0.428
Gaussian moment matching	4.38×10^4	8.54	1.57	0.364
Variational (KL)	$1.47e+03$	18.7	1.37	0.343
Variational (TV)	1.98×10^5	**6.78**	1.57	0.452

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

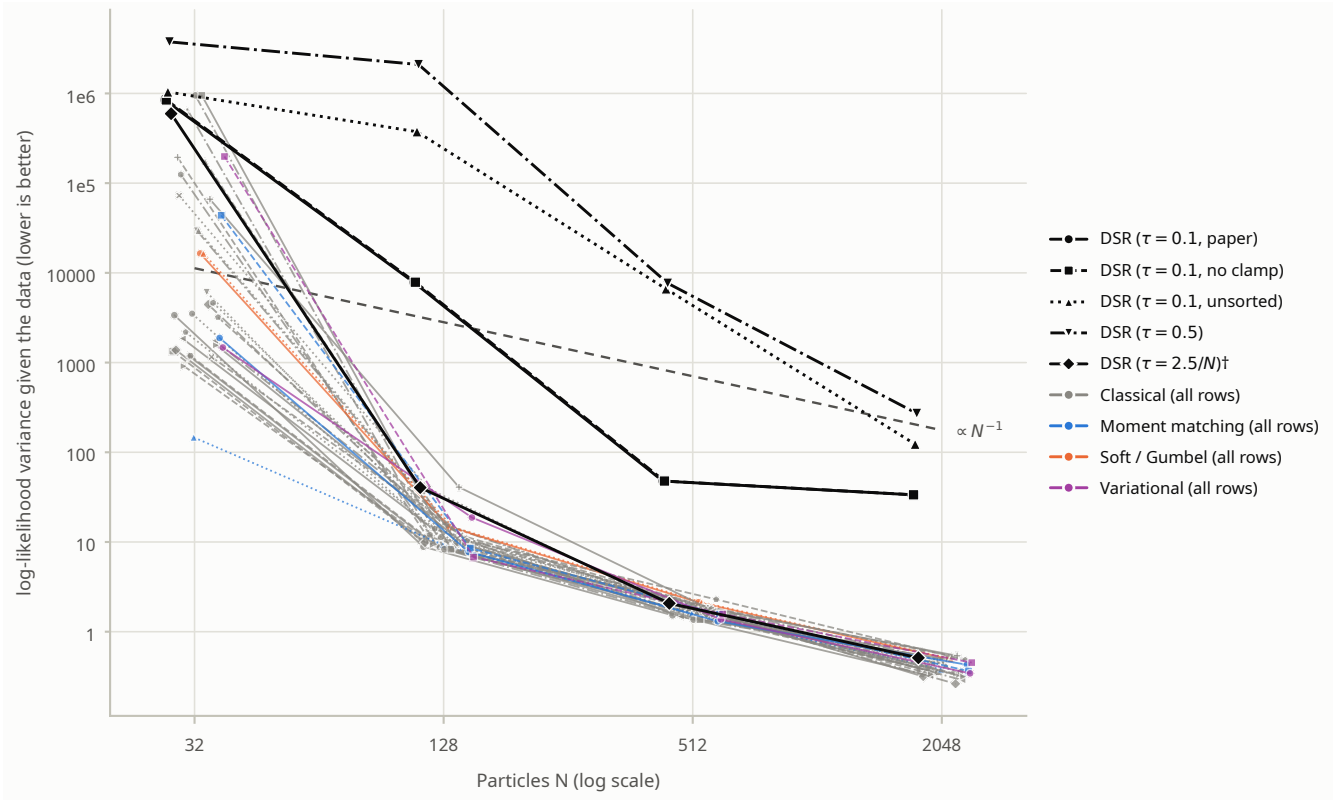


Figure 28: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 29: Stochastic Lorenz-63: mean ESS/N (higher is better). $x_{t+1} = F(x_t)x_t + e_t$, F the 5th-order Taylor series of $\exp(A(x_t)\,0.02)$, $e_t \sim \mathcal{N}(0, 0.1I_3)$; 8×8 camera observation (64 pixels) at SMNR 10 dB, as in the DSR paper; $x_0 \sim \mathcal{N}(0, I_3)$. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.33 ± 0.0277	0.325 ± 0.0327	$\mathbf{0.327 \pm 0.0291}$	0.326 ± 0.029
DSR (τ=0.1, no clamp)	0.33 ± 0.0277	0.325 ± 0.0327	$\mathbf{0.327 \pm 0.0291}$	0.326 ± 0.029
DSR (τ=0.1, unsorted)	0.324 ± 0.0373	$\mathbf{0.326 \pm 0.036}$	0.327 ± 0.0335	$\mathbf{0.327 \pm 0.0313}$
DSR (τ=0.5)	0.32 ± 0.0425	0.323 ± 0.039	0.325 ± 0.0335	0.326 ± 0.0323
DSR (τ=2.5/N)†	$\mathbf{0.332 \pm 0.0316}$	0.32 ± 0.0274	0.312 ± 0.0251	0.306 ± 0.0251
Multinomial (particles)	0.318 ± 0.0258	0.304 ± 0.0239	0.3 ± 0.0238	0.298 ± 0.025
Systematic (particles)	0.317 ± 0.026	0.304 ± 0.0247	0.3 ± 0.0241	0.298 ± 0.0252
Stratified (particles)	0.319 ± 0.0253	0.306 ± 0.0242	0.3 ± 0.0238	0.298 ± 0.025
Residual (particles)	0.315 ± 0.0274	0.306 ± 0.0242	0.301 ± 0.0247	0.298 ± 0.0251
SSP (particles)	0.315 ± 0.0261	0.305 ± 0.0247	0.3 ± 0.0246	0.298 ± 0.0249
Multinomial (Stone Soup)	0.316 ± 0.0266	0.305 ± 0.0237	0.3 ± 0.0248	0.298 ± 0.0253
Systematic (Stone Soup)	0.318 ± 0.025	0.305 ± 0.0243	0.3 ± 0.0245	0.298 ± 0.0251
Stratified (Stone Soup)	0.32 ± 0.0254	0.305 ± 0.0237	0.3 ± 0.0239	0.298 ± 0.025
Residual (Stone Soup)	0.317 ± 0.0241	0.305 ± 0.0245	0.3 ± 0.0238	0.298 ± 0.025
Multinomial (Cuthbert)	0.316 ± 0.0269	0.306 ± 0.0248	0.301 ± 0.0245	0.298 ± 0.025
Systematic (Cuthbert)	0.317 ± 0.0259	0.305 ± 0.0241	0.3 ± 0.0238	0.298 ± 0.0247
Systematic, stop-gradient (Cuthbert)	0.317 ± 0.0259	0.305 ± 0.0241	0.3 ± 0.0238	0.298 ± 0.0247
Multinomial (PFJAX)	0.317 ± 0.0263	0.306 ± 0.0245	0.301 ± 0.0249	0.298 ± 0.0251
Gaussian moment matching, MVN (PFJAX)	0.312 ± 0.0249	0.304 ± 0.0243	0.3 ± 0.0248	0.298 ± 0.025
OT (PFJAX)	0.326 ± 0.028	0.317 ± 0.0287	–	–
Multinomial (PyDPF)	0.315 ± 0.0231	0.306 ± 0.0254	0.3 ± 0.0245	0.298 ± 0.0247
Systematic (PyDPF)	0.314 ± 0.026	0.305 ± 0.025	0.3 ± 0.0246	0.298 ± 0.0252
Systematic, stop-gradient (PyDPF)	0.314 ± 0.026	0.305 ± 0.025	0.3 ± 0.0246	0.298 ± 0.0252
Soft (α=0.5) (PyDPF)	0.232 ± 0.0241	0.219 ± 0.0256	0.213 ± 0.0257	0.211 ± 0.0256
OT (PyDPF)	0.326 ± 0.0279	0.312 ± 0.0305	–	–
Multinomial (torchfilter)	0.315 ± 0.0231	0.306 ± 0.0254	0.3 ± 0.0245	0.298 ± 0.0247
Soft (α=0.5) (torchfilter)	0.232 ± 0.0241	0.219 ± 0.0256	0.213 ± 0.0257	0.211 ± 0.0256
Multinomial (FilterFlow)	0.316 ± 0.0282	0.306 ± 0.0238	0.301 ± 0.0244	0.298 ± 0.0249
Systematic (FilterFlow)	0.316 ± 0.0252	0.305 ± 0.0249	0.3 ± 0.0243	0.298 ± 0.0251
Stratified (FilterFlow)	0.318 ± 0.0254	0.307 ± 0.0241	0.3 ± 0.0235	0.298 ± 0.0251
OT (FilterFlow)	0.321 ± 0.0279	0.316 ± 0.0281	–	–
Covariance-corrected OT (FilterFlow)	0.324 ± 0.0268	0.299 ± 0.026	–	–
Multinomial (Resamplings.jl)	0.317 ± 0.0263	0.304 ± 0.0244	0.301 ± 0.0248	0.298 ± 0.0249
Systematic (Resamplings.jl)	0.32 ± 0.0265	0.306 ± 0.0248	0.3 ± 0.0243	0.298 ± 0.0249
Stratified (Resamplings.jl)	0.317 ± 0.0257	0.305 ± 0.0248	0.3 ± 0.0243	0.298 ± 0.025
Residual (Resamplings.jl)	0.318 ± 0.0243	0.305 ± 0.0246	0.3 ± 0.0243	0.298 ± 0.025
SSP (Resamplings.jl)	0.318 ± 0.0256	0.306 ± 0.0243	0.301 ± 0.0246	0.298 ± 0.0251
Killing (Resamplings.jl)	0.317 ± 0.0263	0.305 ± 0.0241	0.301 ± 0.024	0.298 ± 0.0252
Eigen moment matching	0.319 ± 0.0266	0.306 ± 0.0252	0.3 ± 0.0246	0.298 ± 0.0252
Gaussian moment matching	0.317 ± 0.0283	0.306 ± 0.0246	0.301 ± 0.0249	0.298 ± 0.0255
Gumbel-softmax	0.321 ± 0.0421	0.317 ± 0.025	–	–
Variational (KL)	0.318 ± 0.026	0.307 ± 0.0244	0.301 ± 0.0244	0.3 ± 0.0256
Variational (TV)	0.318 ± 0.0258	0.308 ± 0.0246	0.303 ± 0.0249	0.3 ± 0.0259

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

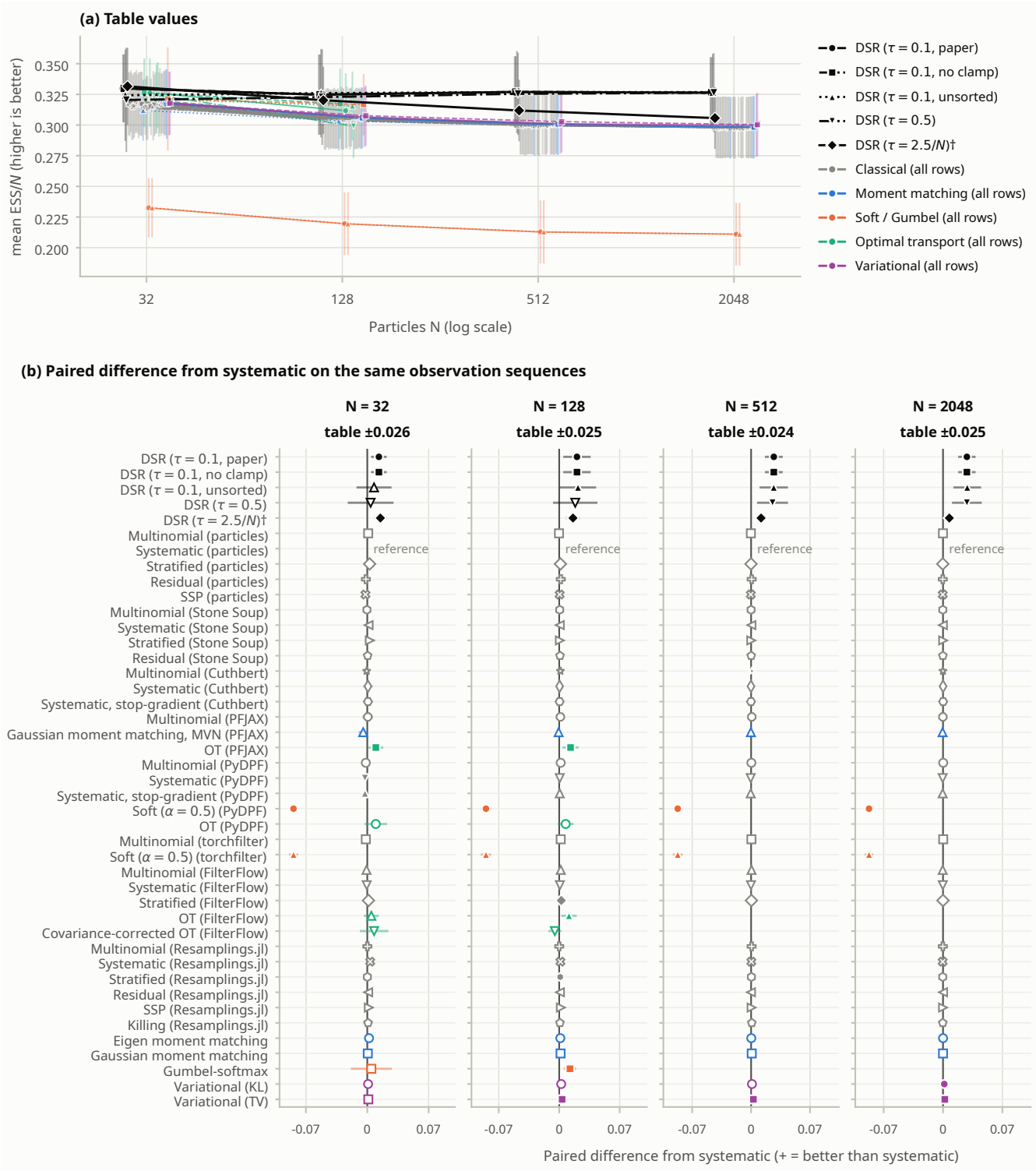


Figure 29: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 30: Stochastic Lorenz-96: state RMSE (lower is better). d=10, F=8, RK4 over Δt=0.05 per step plus $\mathcal{N}(0, 0.1I)$; $y_t = x_t + \mathcal{N}(0, I)$; x_0 from the climatological Gaussian. 50 steps, resampling when ESS $< 0.5N$; mean $\pm$ 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel). Numerical failures, excluded from the means: Gaussian moment matching, MVN (PFJAX) 9% at N=32; Gaussian moment matching, MVN (PFJAX) 3% at N=128.

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	3.09 ± 0.606	2.46 ± 0.589	2.11 ± 0.589	1.9 ± 0.575
DSR (τ=0.1, no clamp)	3.09 ± 0.606	2.46 ± 0.589	2.11 ± 0.589	1.9 ± 0.575
DSR (τ=0.1, unsorted)	3.07 ± 0.599	2.47 ± 0.606	2.13 ± 0.581	1.9 ± 0.572
DSR (τ=0.5)	3.2 ± 0.566	2.65 ± 0.586	2.25 ± 0.577	1.97 ± 0.523
DSR (τ=2.5/N)[†]	3.02 ± 0.613	2.17 ± 0.616	1.45 ± 0.467	1.08 ± 0.301
Multinomial (particles)	2.8 ± 0.634	2.06 ± 0.6	1.41 ± 0.449	1.09 ± 0.308
Systematic (particles)	2.76 ± 0.624	2.09 ± 0.668	1.45 ± 0.466	1.07 ± 0.296
Stratified (particles)	2.81 ± 0.599	2.07 ± 0.658	1.43 ± 0.466	1.08 ± 0.308
Residual (particles)	2.83 ± 0.641	2.06 ± 0.662	1.45 ± 0.477	1.11 ± 0.337
SSP (particles)	2.77 ± 0.643	2.02 ± 0.628	1.41 ± 0.443	1.1 ± 0.317
Multinomial (Stone Soup)	2.8 ± 0.616	2.06 ± 0.618	1.45 ± 0.498	1.11 ± 0.327
Systematic (Stone Soup)	2.74 ± 0.634	2.05 ± 0.619	1.44 ± 0.467	1.09 ± 0.302
Stratified (Stone Soup)	2.76 ± 0.66	2.06 ± 0.627	1.43 ± 0.478	1.08 ± 0.293
Residual (Stone Soup)	2.79 ± 0.631	2.07 ± 0.648	1.46 ± 0.484	1.06 ± 0.287
Multinomial (Cuthbert)	2.87 ± 0.571	1.99 ± 0.62	1.4 ± 0.437	1.09 ± 0.306
Systematic (Cuthbert)	2.79 ± 0.6	2.05 ± 0.652	1.39 ± 0.438	1.12 ± 0.34
Systematic, stop-gradient (Cuthbert)	2.79 ± 0.6	2.05 ± 0.652	1.39 ± 0.438	1.12 ± 0.34
Multinomial (PFJAX)	2.86 ± 0.608	1.99 ± 0.637	1.43 ± 0.459	1.08 ± 0.31
Gaussian moment matching, MVN (PFJAX)	$\mathbf{2.28 \pm 0.732}$	$\mathbf{1.52 \pm 0.521}$	$\mathbf{0.934 \pm 0.235}$	$\mathbf{0.735 \pm 0.115}$
OT (PFJAX)	3.27 ± 0.681	2.28 ± 0.935	–	–
Multinomial (PyDPF)	2.82 ± 0.633	2.03 ± 0.598	1.48 ± 0.5	1.1 ± 0.312
Systematic (PyDPF)	2.73 ± 0.636	2.05 ± 0.653	1.41 ± 0.442	1.06 ± 0.303
Systematic, stop-gradient (PyDPF)	2.73 ± 0.636	2.05 ± 0.653	1.41 ± 0.442	1.06 ± 0.303
Soft (α=0.5) (PyDPF)	2.96 ± 0.641	2.14 ± 0.693	1.48 ± 0.484	1.07 ± 0.294
OT (PyDPF)	3.11 ± 0.666	2.44 ± 1.01	–	–
Multinomial (torchfilter)	2.82 ± 0.633	2.03 ± 0.598	1.48 ± 0.5	1.1 ± 0.312
Soft (α=0.5) (torchfilter)	2.96 ± 0.641	2.14 ± 0.693	1.48 ± 0.484	1.07 ± 0.294
Multinomial (FilterFlow)	2.83 ± 0.641	2.05 ± 0.607	1.46 ± 0.495	1.13 ± 0.347
Systematic (FilterFlow)	2.78 ± 0.61	2.05 ± 0.665	1.4 ± 0.439	1.08 ± 0.307
Stratified (FilterFlow)	2.74 ± 0.593	2.03 ± 0.654	1.43 ± 0.457	1.06 ± 0.289
OT (FilterFlow)	3.36 ± 0.661	2.52 ± 0.985	–	–
Covariance-corrected OT (FilterFlow)	2.91 ± 0.791	2.19 ± 0.93	–	–
Multinomial (Resamplings.jl)	2.8 ± 0.636	2.03 ± 0.621	1.46 ± 0.506	1.09 ± 0.309
Systematic (Resamplings.jl)	2.82 ± 0.608	2.02 ± 0.608	1.46 ± 0.49	1.11 ± 0.322
Stratified (Resamplings.jl)	2.77 ± 0.655	2.03 ± 0.631	1.39 ± 0.429	1.12 ± 0.338
Residual (Resamplings.jl)	2.82 ± 0.635	2.11 ± 0.663	1.43 ± 0.478	1.09 ± 0.313
SSP (Resamplings.jl)	2.78 ± 0.65	2.06 ± 0.657	1.39 ± 0.429	1.07 ± 0.297
Killing (Resamplings.jl)	2.79 ± 0.619	2.01 ± 0.595	1.43 ± 0.471	1.07 ± 0.309
Eigen moment matching	2.62 ± 0.695	1.75 ± 0.595	1.1 ± 0.323	0.849 ± 0.204
Gaussian moment matching	2.55 ± 0.643	1.8 ± 0.593	1.15 ± 0.341	0.865 ± 0.225
Gumbel-softmax	3.15 ± 0.758	2.44 ± 1.05	–	–
Variational (KL)	2.77 ± 0.63	2.05 ± 0.63	1.41 ± 0.454	1.09 ± 0.311
Variational (TV)	2.79 ± 0.638	2.02 ± 0.636	1.44 ± 0.464	1.11 ± 0.322

[†]Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

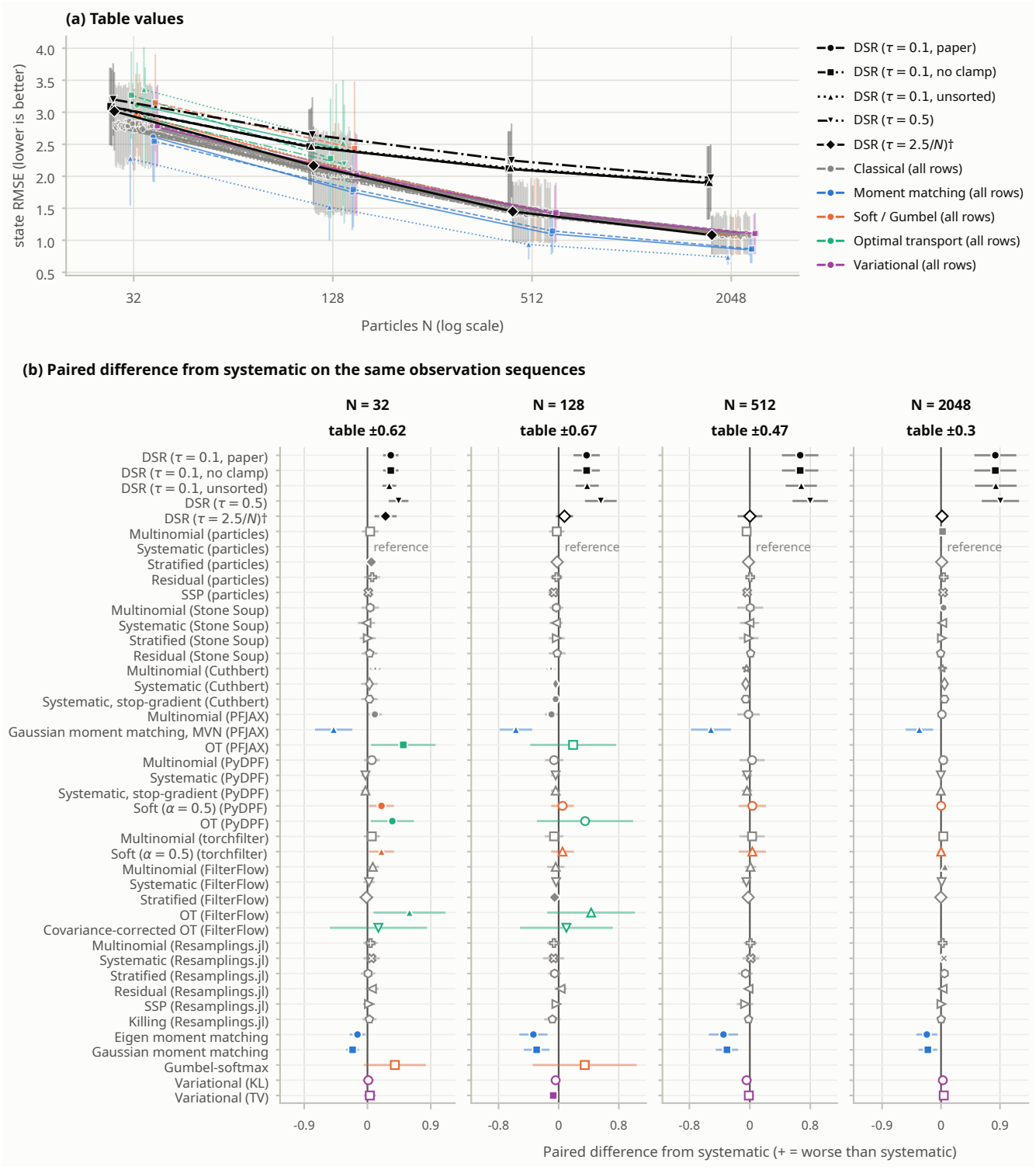


Figure 30: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 31: Stochastic Lorenz-96: 95% coverage (closer to 0.95 is better). d=10, F=8, RK4 over Δt=0.05 per step plus $\mathcal{N}(0, 0.1I)$; $y_t = x_t + \mathcal{N}(0, I)$; x_0 from the climatological Gaussian. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel). Numerical failures, excluded from the means: Gaussian moment matching, MVN (PFJAX) 9% at N=32; Gaussian moment matching, MVN (PFJAX) 3% at N=128.

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0	0.0024 ± 0.00222	0.004 ± 0.00427	0.0102 ± 0.0117
DSR (τ=0.1, no clamp)	0	0.0024 ± 0.00222	0.004 ± 0.00427	0.0102 ± 0.0117
DSR (τ=0.1, unsorted)	0	0.0016 ± 0.00232	0.0024 ± 0.00176	0.006 ± 0.00691
DSR (τ=0.5)	0	$4 \times 10^{-4} \pm 6.03 \times 10^{-4}$	0.001 ± 0.00122	0.0054 ± 0.00611
DSR (τ=2.5/N)†	$2 \times 10^{-4} \pm 4.52 \times 10^{-4}$	0.008 ± 0.0058	0.0828 ± 0.0384	0.272 ± 0.0761
Multinomial (particles)	0.001 ± 0.00139	0.0634 ± 0.0351	0.308 ± 0.09	0.544 ± 0.1
Systematic (particles)	0.0016 ± 0.00162	0.0788 ± 0.0458	0.303 ± 0.0933	0.563 ± 0.0973
Stratified (particles)	0.001 ± 0.00122	0.0808 ± 0.0423	0.312 ± 0.0928	0.558 ± 0.0938
Residual (particles)	0.0018 ± 0.00228	0.0792 ± 0.0431	0.31 ± 0.106	0.548 ± 0.109
SSP (particles)	0.0012 ± 0.00138	0.079 ± 0.0421	0.318 ± 0.0947	0.552 ± 0.101
Multinomial (Stone Soup)	0.0016 ± 0.00148	0.0708 ± 0.0399	0.299 ± 0.0987	0.539 ± 0.0998
Systematic (Stone Soup)	0.0012 ± 0.00154	0.0706 ± 0.0367	0.314 ± 0.0968	0.552 ± 0.0984
Stratified (Stone Soup)	0.0018 ± 0.00228	0.0696 ± 0.0352	0.312 ± 0.106	0.546 ± 0.0962
Residual (Stone Soup)	$4 \times 10^{-4} \pm 9.05 \times 10^{-4}$	0.0728 ± 0.048	0.312 ± 0.106	0.559 ± 0.0873
Multinomial (Cuthbert)	$6 \times 10^{-4} \pm 9.66 \times 10^{-4}$	0.0742 ± 0.0399	0.304 ± 0.0842	0.544 ± 0.103
Systematic (Cuthbert)	0.0012 ± 0.00154	0.075 ± 0.0439	0.328 ± 0.105	0.541 ± 0.1
Systematic, stop-gradient (Cuthbert)	0.0012 ± 0.00154	0.075 ± 0.0439	0.328 ± 0.105	0.541 ± 0.1
Multinomial (PFJAX)	0.0012 ± 0.00121	0.0702 ± 0.0439	0.293 ± 0.0868	0.555 ± 0.101
Gaussian moment matching, MVN (PFJAX)	0.00507 ± 0.00451	$\mathbf{0.182 \pm 0.0745}$	$\mathbf{0.552 \pm 0.0763}$	$\mathbf{0.746 \pm 0.056}$
OT (PFJAX)	0	0.034 ± 0.0536	–	–
Multinomial (PyDPF)	0.0012 ± 0.00138	0.0776 ± 0.0489	0.296 ± 0.0951	0.538 ± 0.101
Systematic (PyDPF)	0.001 ± 0.00101	0.0852 ± 0.05	0.321 ± 0.0999	0.56 ± 0.0936
Systematic, stop-gradient (PyDPF)	0.001 ± 0.00101	0.0852 ± 0.05	0.321 ± 0.0999	0.56 ± 0.0936
Soft (α=0.5) (PyDPF)	0.0012 ± 0.00154	0.035 ± 0.0248	0.249 ± 0.092	0.519 ± 0.093
OT (PyDPF)	0	0.004 ± 0.00905	–	–
Multinomial (torchfilter)	0.0012 ± 0.00138	0.0776 ± 0.0489	0.296 ± 0.0951	0.538 ± 0.101
Soft (α=0.5) (torchfilter)	0.0012 ± 0.00154	0.035 ± 0.0248	0.249 ± 0.092	0.519 ± 0.093
Multinomial (FilterFlow)	$4 \times 10^{-4} \pm 6.03 \times 10^{-4}$	0.0724 ± 0.0405	0.314 ± 0.103	0.537 ± 0.102
Systematic (FilterFlow)	0.0012 ± 0.00181	0.0836 ± 0.0436	0.322 ± 0.0999	0.556 ± 0.0999
Stratified (FilterFlow)	0.001 ± 0.00101	0.0786 ± 0.0468	0.321 ± 0.102	0.554 ± 0.0933
OT (FilterFlow)	0	0.002 ± 0.00452	–	–
Covariance-corrected OT (FilterFlow)	0	0.018 ± 0.0207	–	–
Multinomial (Resamplings.jl)	$4 \times 10^{-4} \pm 6.03 \times 10^{-4}$	0.0698 ± 0.041	0.297 ± 0.0997	0.541 ± 0.0986
Systematic (Resamplings.jl)	$6 \times 10^{-4} \pm 6.91 \times 10^{-4}$	0.0758 ± 0.0355	0.317 ± 0.105	0.549 ± 0.1
Stratified (Resamplings.jl)	$8 \times 10^{-4} \pm 0.00138$	0.077 ± 0.0458	0.323 ± 0.0953	0.546 ± 0.103
Residual (Resamplings.jl)	0.0012 ± 0.00121	0.0638 ± 0.0347	0.313 ± 0.106	0.543 ± 0.107
SSP (Resamplings.jl)	0.0014 ± 0.00152	0.0692 ± 0.0369	0.319 ± 0.0899	0.551 ± 0.098
Killing (Resamplings.jl)	0.0012 ± 0.00181	0.0672 ± 0.0372	0.311 ± 0.093	0.545 ± 0.104
Eigen moment matching	$\mathbf{0.0052 \pm 0.00596}$	0.156 ± 0.0651	0.514 ± 0.0838	0.713 ± 0.0662
Gaussian moment matching	0.0032 ± 0.00325	0.147 ± 0.0743	0.48 ± 0.0947	0.706 ± 0.071
Gumbel-softmax	0.002 ± 0.00452	0.016 ± 0.0131	–	–
Variational (KL)	0.0022 ± 0.00207	0.0704 ± 0.042	0.323 ± 0.0877	0.543 ± 0.0978
Variational (TV)	0.0024 ± 0.003	0.0796 ± 0.0416	0.307 ± 0.101	0.543 ± 0.0979

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

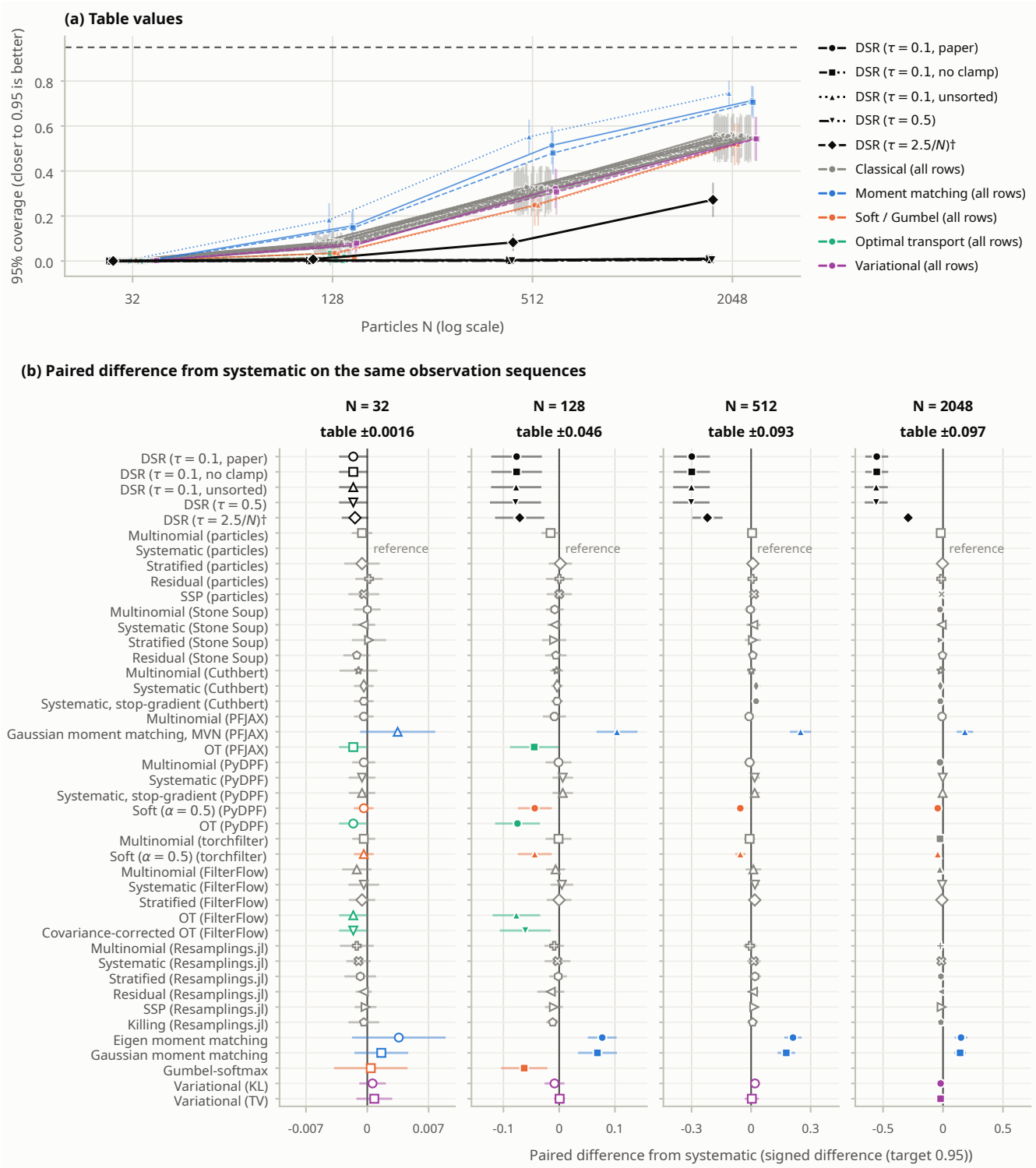


Figure 31: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Dashed line: nominal 0.95. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 32: Stochastic Lorenz-96: predictive NLL (lower is better). d=10, F=8, RK4 over Δt=0.05 per step plus $\mathcal{N}(0, 0.1I)$; $y_t = x_t + \mathcal{N}(0, I)$; x_0 from the climatological Gaussian. 50 steps, resampling when ESS $< 0.5N$; mean $\pm$ 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel). Numerical failures, excluded from the means: Gaussian moment matching, MVN (PFJAX) 9% at N=32; Gaussian moment matching, MVN (PFJAX) 3% at N=128.

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	69.2 ± 18.8	51.5 ± 15.1	42.7 ± 14.2	37.8 ± 13.1
DSR (τ=0.1, no clamp)	69.2 ± 18.8	51.5 ± 15.1	42.7 ± 14.2	37.8 ± 13.1
DSR (τ=0.1, unsorted)	68 ± 18.9	52 ± 15.7	43 ± 14.3	38 ± 13.3
DSR (τ=0.5)	72.1 ± 18.1	56.5 ± 16.1	45.8 ± 14.7	39 ± 12
DSR (τ=2.5/N)†	67.1 ± 18.6	45 ± 14.7	29.4 ± 9.22	22.8 ± 4.71
Multinomial (particles)	60.9 ± 18.1	43.1 ± 13.7	29.3 ± 9.33	23.1 ± 5.09
Systematic (particles)	60 ± 17.8	43.8 ± 15.5	30.1 ± 9.76	22.7 ± 4.72
Stratified (particles)	61.2 ± 17.3	43.6 ± 15.5	29.8 ± 9.76	22.8 ± 4.97
Residual (particles)	62 ± 18.8	43.3 ± 15.3	30 ± 9.71	23.6 ± 5.73
SSP (particles)	60.2 ± 18.5	42.3 ± 14.4	29 ± 8.73	23.2 ± 5.16
Multinomial (Stone Soup)	60.6 ± 17.7	43.2 ± 14.4	30.1 ± 10.4	23.2 ± 5.26
Systematic (Stone Soup)	59.5 ± 18.1	42.9 ± 14.3	29.8 ± 9.51	22.9 ± 4.95
Stratified (Stone Soup)	60.2 ± 18.8	43.3 ± 14.6	29.7 ± 9.66	22.8 ± 4.8
Residual (Stone Soup)	60.5 ± 18.1	43.4 ± 14.8	30.4 ± 10.1	22.5 ± 4.55
Multinomial (Cuthbert)	62.8 ± 16.5	41.3 ± 14.2	28.8 ± 8.63	22.9 ± 4.87
Systematic (Cuthbert)	60.4 ± 17.2	42.7 ± 15.2	28.5 ± 8.44	23.7 ± 5.86
Systematic, stop-gradient (Cuthbert)	60.4 ± 17.2	42.7 ± 15.2	28.5 ± 8.44	23.7 ± 5.86
Multinomial (PFJAX)	62.8 ± 17.7	41.3 ± 14.6	29.4 ± 9.19	22.9 ± 5.17
Gaussian moment matching, MVN (PFJAX)	$\mathbf{48.2 \pm 19.7}$	$\mathbf{30.8 \pm 10.3}$	$\mathbf{20.3 \pm 3.29}$	$\mathbf{17.7 \pm 1}$
OT (PFJAX)	73.2 ± 23.2	49.5 ± 23.7	–	–
Multinomial (PyDPF)	61.8 ± 17.9	42.2 ± 13.8	30.6 ± 10.2	23.3 ± 5.17
Systematic (PyDPF)	59.3 ± 18	42.9 ± 15.1	29 ± 8.75	22.6 ± 4.92
Systematic, stop-gradient (PyDPF)	59.3 ± 18	42.9 ± 15.1	29 ± 8.75	22.6 ± 4.92
Soft (α=0.5) (PyDPF)	65.7 ± 19.8	44.9 ± 16.7	30.2 ± 9.83	22.4 ± 4.55
OT (PyDPF)	68.1 ± 22.1	54.7 ± 27.8	–	–
Multinomial (torchfilter)	61.8 ± 17.9	42.2 ± 13.8	30.6 ± 10.2	23.3 ± 5.17
Soft (α=0.5) (torchfilter)	65.7 ± 19.8	44.9 ± 16.7	30.2 ± 9.83	22.4 ± 4.55
Multinomial (FilterFlow)	62.2 ± 18.8	43 ± 14	30.4 ± 10.3	23.9 ± 6
Systematic (FilterFlow)	60.3 ± 17.4	43 ± 15.5	29 ± 8.63	22.9 ± 4.91
Stratified (FilterFlow)	59.2 ± 17.2	42.4 ± 14.9	29.7 ± 9.38	22.6 ± 4.61
OT (FilterFlow)	76.2 ± 23	56.4 ± 26.5	–	–
Covariance-corrected OT (FilterFlow)	64 ± 26.7	47.5 ± 23.3	–	–
Multinomial (Resamplings.jl)	61.2 ± 18.8	42.3 ± 14.3	30.4 ± 10.5	23.1 ± 5.07
Systematic (Resamplings.jl)	61.3 ± 17.6	42.1 ± 13.7	30.4 ± 10.2	23.3 ± 5.2
Stratified (Resamplings.jl)	60.3 ± 19.1	42.3 ± 14.6	28.7 ± 8.49	23.9 ± 5.94
Residual (Resamplings.jl)	61.6 ± 18.5	44.9 ± 15.8	29.7 ± 9.85	23.1 ± 5.1
SSP (Resamplings.jl)	60.3 ± 19	43.2 ± 15.2	28.3 ± 8.36	22.7 ± 4.66
Killing (Resamplings.jl)	60.6 ± 18.1	41.2 ± 13.3	29.5 ± 9.47	22.6 ± 4.71
Eigen moment matching	57 ± 19.2	36 ± 12.4	23.2 ± 5.15	19.4 ± 2.66
Gaussian moment matching	54 ± 17.6	37 ± 12.2	24.3 ± 5.85	19.9 ± 3.34
Gumbel-softmax	70.5 ± 25.5	55.1 ± 29.4	–	–
Variational (KL)	60.2 ± 18.3	42.7 ± 14.5	29 ± 9.26	23.2 ± 5.15
Variational (TV)	60.7 ± 18.7	41.9 ± 14.6	29.9 ± 9.69	23.5 ± 5.48

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

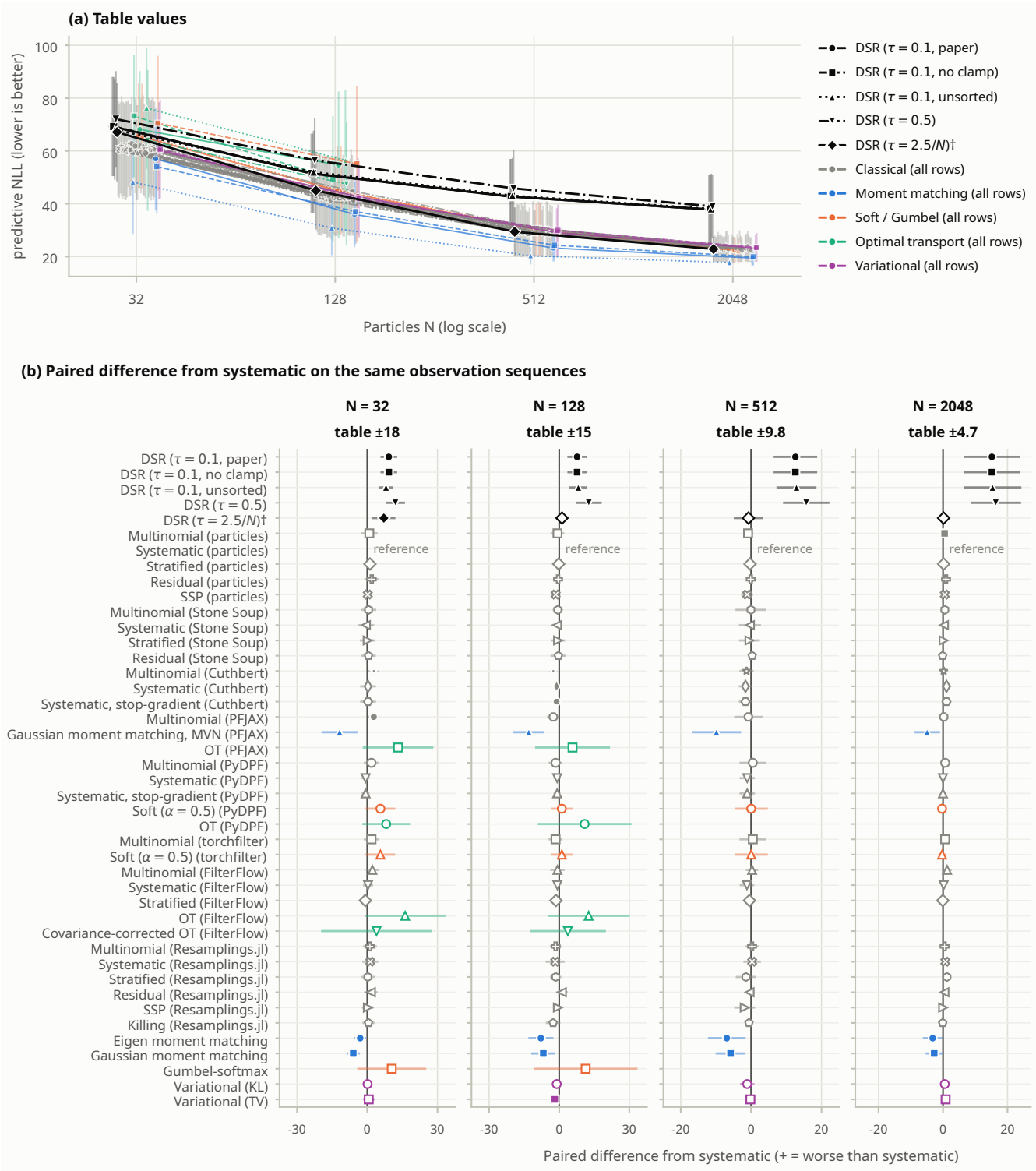


Figure 32: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Table 33: Stochastic Lorenz-96: log-likelihood variance given the data (lower is better). d=10, F=8, RK4 over Δt=0.05 per step plus $\mathcal{N}(0, 0.1I)$; $y_t = x_t + \mathcal{N}(0, I)$; x_0 from the climatological Gaussian. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel). Numerical failures, excluded from the means: Gaussian moment matching, MVN (PFJAX) 9% at N=32; Gaussian moment matching, MVN (PFJAX) 3% at N=128.

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	1.38×10^6	8.57×10^5	5.2×10^5	3.38×10^5
DSR (τ=0.1, no clamp)	1.38×10^6	8.57×10^5	5.2×10^5	3.38×10^5
DSR (τ=0.1, unsorted)	1.12×10^6	1.01×10^6	5.4×10^5	4.11×10^5
DSR (τ=0.5)	1.29×10^6	9.67×10^5	5.83×10^5	4.08×10^5
DSR (τ=2.5/N)†	1.37×10^6	1.01×10^6	5.22×10^5	2.45×10^5
Multinomial (particles)	1.39×10^6	1.07×10^6	7.75×10^5	2.81×10^5
Systematic (particles)	1.48×10^6	9.24×10^5	6.93×10^5	2.4×10^5
Stratified (particles)	1.36×10^6	1.12×10^6	7.22×10^5	2.42×10^5
Residual (particles)	1.4×10^6	1.05×10^6	6.19×10^5	2.96×10^5
SSP (particles)	1.35×10^6	1.07×10^6	5.53×10^5	2.76×10^5
Multinomial (Stone Soup)	1.29×10^6	1.2×10^6	5.76×10^5	2.18×10^5
Systematic (Stone Soup)	1.32×10^6	1.07×10^6	6.49×10^5	2.54×10^5
Stratified (Stone Soup)	1.21×10^6	1.13×10^6	6.38×10^5	2.61×10^5
Residual (Stone Soup)	1.28×10^6	1.13×10^6	7.26×10^5	2.74×10^5
Multinomial (Cuthbert)	1.44×10^6	1.05×10^6	6.05×10^5	2.01×10^5
Systematic (Cuthbert)	1.32×10^6	9.35×10^5	5.44×10^5	3.04×10^5
Systematic, stop-gradient (Cuthbert)	1.32×10^6	9.35×10^5	5.44×10^5	3.04×10^5
Multinomial (PFJAX)	1.51×10^6	9.1×10^5	5.82×10^5	2.27×10^5
Gaussian moment matching, MVN (PFJAX)	$\mathbf{8.17 \times 10^5}$	$\mathbf{6.89 \times 10^5}$	$\mathbf{1 \times 10^5}$	$\mathbf{5.18e+03}$
Multinomial (PyDPF)	1.53×10^6	1.05×10^6	6.47×10^5	2.83×10^5
Systematic (PyDPF)	1.43×10^6	9.29×10^5	6.27×10^5	2.24×10^5
Systematic, stop-gradient (PyDPF)	1.43×10^6	9.29×10^5	6.27×10^5	2.24×10^5
Soft (α=0.5) (PyDPF)	1.55×10^6	9.24×10^5	5.59×10^5	1.73×10^5
Multinomial (torchfilter)	1.53×10^6	1.05×10^6	6.47×10^5	2.83×10^5
Soft (α=0.5) (torchfilter)	1.55×10^6	9.24×10^5	5.59×10^5	1.73×10^5
Multinomial (FilterFlow)	1.47×10^6	1.25×10^6	6.77×10^5	3.16×10^5
Systematic (FilterFlow)	1.41×10^6	9.4×10^5	6.08×10^5	2.55×10^5
Stratified (FilterFlow)	1.45×10^6	9.25×10^5	7.03×10^5	2.57×10^5
Multinomial (Resamplings.jl)	1.34×10^6	1.05×10^6	6.82×10^5	2.94×10^5
Systematic (Resamplings.jl)	1.38×10^6	1.12×10^6	7.45×10^5	2.43×10^5
Stratified (Resamplings.jl)	1.31×10^6	9.49×10^5	6.1×10^5	3.64×10^5
Residual (Resamplings.jl)	1.31×10^6	1.17×10^6	6.28×10^5	2.52×10^5
SSP (Resamplings.jl)	1.15×10^6	1.07×10^6	4.94×10^5	2.15×10^5
Killing (Resamplings.jl)	1.33×10^6	8.9×10^5	6.45×10^5	1.86×10^5
Eigen moment matching	1.41×10^6	7.78×10^5	2.75×10^5	8.27×10^4
Gaussian moment matching	1.17×10^6	8.78×10^5	4.08×10^5	1.59×10^5
Variational (KL)	1.39×10^6	1.05×10^6	6.09×10^5	2.87×10^5
Variational (TV)	1.23×10^6	9.23×10^5	7.81×10^5	2.98×10^5

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

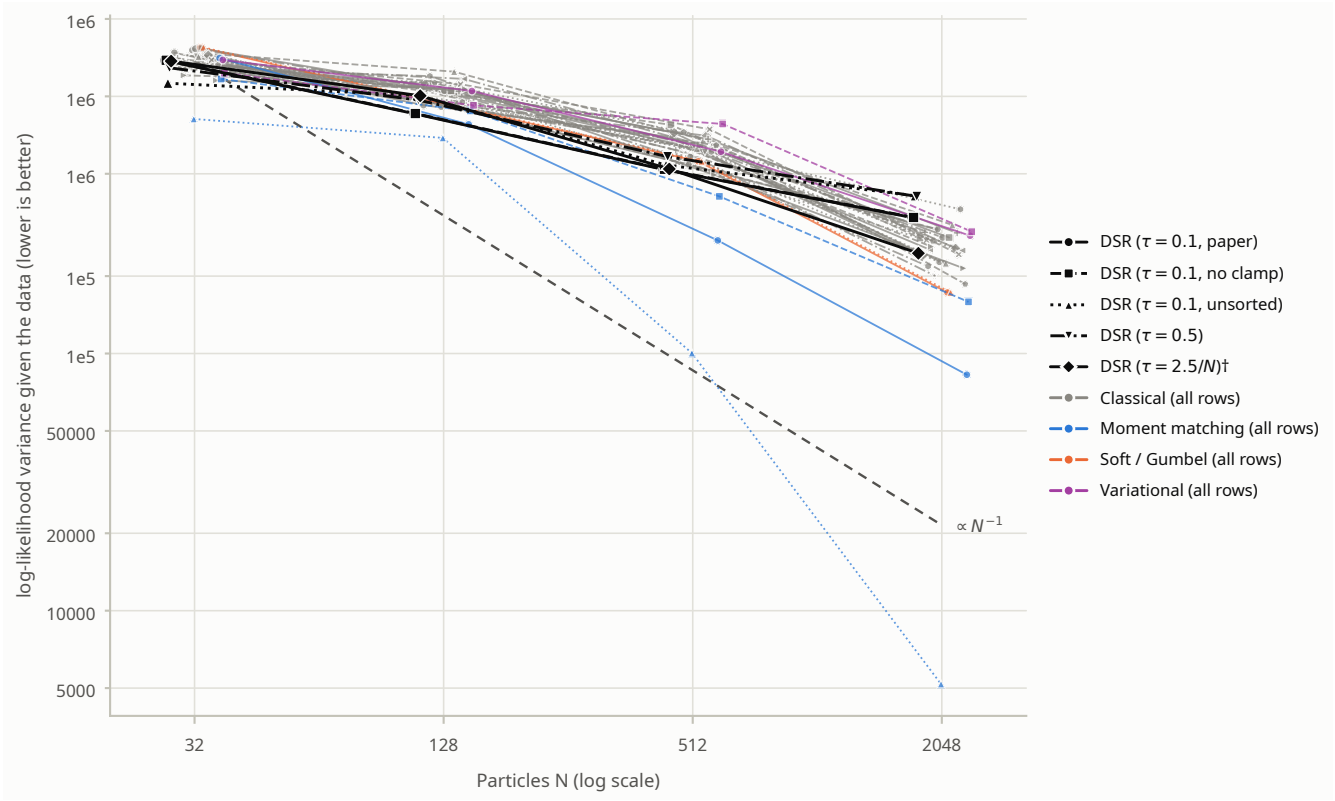


Figure 33: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 34: Stochastic Lorenz-96: mean ESS/N (higher is better). d=10, F=8, RK4 over Δt=0.05 per step plus $\mathcal{N}(0, 0.1I)$; $y_t = x_t + \mathcal{N}(0, I)$; x_0 from the climatological Gaussian. 50 steps, resampling when ESS $< 0.5N$; mean ± 95% CI over 10 observation sequences (100 filter runs; 10 for OT and Gumbel). Numerical failures, excluded from the means: Gaussian moment matching, MVN (PFJAX) 9% at N=32; Gaussian moment matching, MVN (PFJAX) 3% at N=128.

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.135 ± 0.033	0.118 ± 0.0409	0.115 ± 0.0467	0.124 ± 0.0474
DSR (τ=0.1, no clamp)	0.135 ± 0.033	0.118 ± 0.0409	0.115 ± 0.0467	0.124 ± 0.0474
DSR (τ=0.1, unsorted)	0.132 ± 0.0317	0.119 ± 0.0417	0.114 ± 0.0438	0.126 ± 0.0469
DSR (τ=0.5)	0.129 ± 0.031	0.107 ± 0.0369	0.107 ± 0.0432	0.115 ± 0.0431
DSR (τ=2.5/N)†	0.138 ± 0.0346	$\mathbf{0.131 \pm 0.0406}$	$\mathbf{0.15 \pm 0.0369}$	$\mathbf{0.152 \pm 0.0253}$
Multinomial (particles)	0.133 ± 0.0286	0.107 ± 0.0284	0.106 ± 0.0203	0.102 ± 0.015
Systematic (particles)	0.136 ± 0.0306	0.106 ± 0.0312	0.102 ± 0.021	0.102 ± 0.0145
Stratified (particles)	0.133 ± 0.0274	0.108 ± 0.0292	0.104 ± 0.0208	0.103 ± 0.0153
Residual (particles)	0.133 ± 0.0285	0.11 ± 0.0297	0.102 ± 0.022	0.101 ± 0.0156
SSP (particles)	0.136 ± 0.0296	0.11 ± 0.0288	0.105 ± 0.0207	0.101 ± 0.0153
Multinomial (Stone Soup)	0.135 ± 0.0288	0.11 ± 0.0277	0.105 ± 0.0205	0.102 ± 0.0153
Systematic (Stone Soup)	0.135 ± 0.0282	0.107 ± 0.0275	0.104 ± 0.0211	0.102 ± 0.0145
Stratified (Stone Soup)	0.136 ± 0.0304	0.108 ± 0.0274	0.105 ± 0.0218	0.102 ± 0.0146
Residual (Stone Soup)	0.134 ± 0.0283	0.108 ± 0.0309	0.102 ± 0.0217	0.103 ± 0.0142
Multinomial (Cuthbert)	0.13 ± 0.0259	0.113 ± 0.028	0.107 ± 0.019	0.103 ± 0.0152
Systematic (Cuthbert)	0.133 ± 0.0277	0.108 ± 0.0311	0.107 ± 0.0202	0.101 ± 0.0161
Systematic, stop-gradient (Cuthbert)	0.133 ± 0.0277	0.108 ± 0.0311	0.107 ± 0.0202	0.101 ± 0.0161
Multinomial (PFJAX)	0.134 ± 0.0296	0.113 ± 0.0295	0.105 ± 0.0195	0.102 ± 0.0148
Gaussian moment matching, MVN (PFJAX)	0.137 ± 0.0305	0.118 ± 0.0233	0.113 ± 0.0123	0.108 ± 0.0107
OT (PFJAX)	0.111 ± 0.0341	0.125 ± 0.0534	–	–
Multinomial (PyDPF)	0.135 ± 0.0289	0.11 ± 0.0275	0.102 ± 0.0239	0.103 ± 0.0153
Systematic (PyDPF)	0.136 ± 0.0288	0.11 ± 0.0299	0.105 ± 0.02	0.103 ± 0.0147
Systematic, stop-gradient (PyDPF)	0.136 ± 0.0288	0.11 ± 0.0299	0.105 ± 0.02	0.103 ± 0.0147
Soft (α=0.5) (PyDPF)	0.101 ± 0.0173	0.0747 ± 0.0214	0.068 ± 0.0142	0.0656 ± 0.009
OT (PyDPF)	0.116 ± 0.0339	0.11 ± 0.0521	–	–
Multinomial (torchfilter)	0.135 ± 0.0289	0.11 ± 0.0275	0.102 ± 0.0239	0.103 ± 0.0153
Soft (α=0.5) (torchfilter)	0.101 ± 0.0173	0.0747 ± 0.0214	0.068 ± 0.0142	0.0656 ± 0.009
Multinomial (FilterFlow)	0.133 ± 0.0289	0.11 ± 0.0284	0.106 ± 0.022	0.101 ± 0.0158
Systematic (FilterFlow)	0.133 ± 0.0283	0.109 ± 0.0312	0.105 ± 0.019	0.103 ± 0.0146
Stratified (FilterFlow)	0.134 ± 0.0269	0.108 ± 0.0316	0.104 ± 0.0205	0.103 ± 0.0146
OT (FilterFlow)	0.112 ± 0.0342	0.11 ± 0.0589	–	–
Covariance-corrected OT (FilterFlow)	0.121 ± 0.0295	0.119 ± 0.0413	–	–
Multinomial (Resamplings.jl)	0.135 ± 0.0275	0.112 ± 0.0299	0.104 ± 0.0227	0.102 ± 0.0153
Systematic (Resamplings.jl)	0.13 ± 0.0277	0.111 ± 0.0286	0.103 ± 0.0218	0.101 ± 0.0161
Stratified (Resamplings.jl)	0.134 ± 0.0271	0.108 ± 0.0296	0.105 ± 0.02	0.101 ± 0.0161
Residual (Resamplings.jl)	0.131 ± 0.0266	0.106 ± 0.0299	0.104 ± 0.022	0.102 ± 0.0147
SSP (Resamplings.jl)	0.133 ± 0.0279	0.108 ± 0.0298	0.106 ± 0.0191	0.102 ± 0.014
Killing (Resamplings.jl)	0.131 ± 0.0281	0.111 ± 0.0283	0.106 ± 0.0211	0.103 ± 0.0151
Eigen moment matching	0.137 ± 0.03	0.114 ± 0.0255	0.111 ± 0.0145	0.105 ± 0.0113
Gaussian moment matching	$\mathbf{0.141 \pm 0.0287}$	0.113 ± 0.0275	0.109 ± 0.0151	0.106 ± 0.012
Gumbel-softmax	0.118 ± 0.0346	0.115 ± 0.0512	–	–
Variational (KL)	0.136 ± 0.0291	0.108 ± 0.0302	0.109 ± 0.0199	0.104 ± 0.0159
Variational (TV)	0.135 ± 0.0283	0.112 ± 0.0305	0.107 ± 0.0211	0.104 ± 0.0155

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

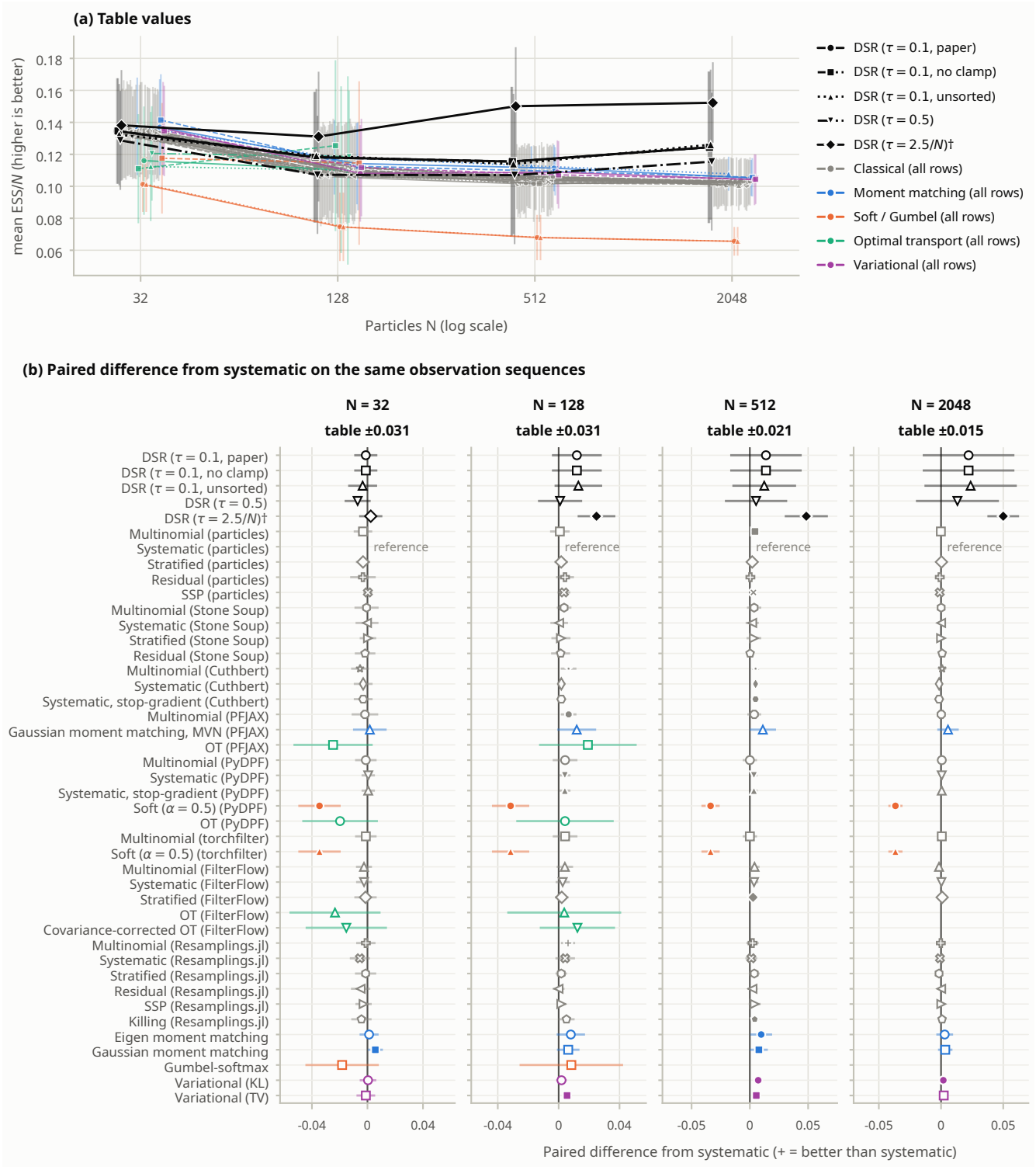


Figure 34: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. **(b)** Each method minus systematic resampling, computed per observation sequence and then averaged (mean ± paired 95% t interval over the 10 sequences). Filled markers: the interval excludes 0. The table's ± for systematic, printed under each N, is dominated by how hard each sequence is, which every method shares; pairing removes that, so differences invisible in (a) can be resolved here.

Parameter learning (stochastic volatility)

Table 35: Stochastic volatility, parameter learning: observation-scale fitting and likelihood score. N=128; $y_t = s\exp(x_t/2)\eta_t$ with true scale $s^\star$=1; 50 steps, resampling when ESS $< 0.5N$. $\hat{s}$ maximizes the PF log-likelihood over the grid $\{0.6, 0.8, 1.0, 1.2, 1.4\}$ with common random numbers; $\hat{s}_{\text{grid}}$ is the maximizer for a 1001-point numerical grid filter. The score is a central difference in $\log s$ (δ=0.01) at s=1, compared with the grid-filter score. Mean ± 95% CI over 10 observation sequences.

Method	$\lvert\hat{s} - s^\star\rvert$	$\lvert\hat{s} - \hat{s}_{\text{grid}}\rvert$	Score error	Score RMSE
DSR (τ=0.1, paper)	0.24 ± 0.0905	0.08 ± 0.0739	1.32 ± 5.17	6.98
Systematic (particles)	0.22 ± 0.125	$\mathbf{0.02 \pm 0.0452}$	1.67 ± 6.51	8.79
Systematic (Stone Soup)	0.24 ± 0.113	0.08 ± 0.0739	1.13 ± 7.64	10.2
Systematic (Resamplings.jl)	0.28 ± 0.0739	0.12 ± 0.0739	-2.87 ± 5.14	7.39
Systematic, stop-gradient (Cuthbert)	0.22 ± 0.125	$\mathbf{0.02 \pm 0.0452}$	-4.49 ± 4.5	7.47
Soft (α=0.5, torchfilter)	$\mathbf{0.2 \pm 0.0954}$	0.08 ± 0.0739	-2.01 ± 7.5	10.1
OT (PyDPF)	0.26 ± 0.0966	0.06 ± 0.0691	$\mathbf{-0.322 \pm 0.829}$	**1.15**
OT (PFJAX)	0.26 ± 0.0966	0.06 ± 0.0691	-0.339 ± 0.851	1.18
OT (FilterFlow)	0.28 ± 0.1	0.08 ± 0.0739	-0.385 ± 0.942	1.31
Eigen moment matching	0.26 ± 0.0966	0.06 ± 0.0691	-0.804 ± 1.47	2.11
Gaussian moment matching	0.22 ± 0.106	0.06 ± 0.0691	-0.673 ± 3.93	5.25
Gumbel-softmax	0.26 ± 0.0966	0.1 ± 0.0754	0.602 ± 3.44	4.61
Variational (KL)	0.24 ± 0.113	0.04 ± 0.0603	-1.55 ± 4.81	6.57

Lower is better for the two parameter errors and the score RMSE; closer to 0 is better for the mean score error. Parameter estimates lie on a 0.2-spaced grid, so errors are multiples of 0.2 per dataset. Only the paper-default DSR (τ=0.1, Morton sorting) was run in this stage; baselines are the representative method of each other repository. Bold marks the best mean per column, not a statistically significant difference.

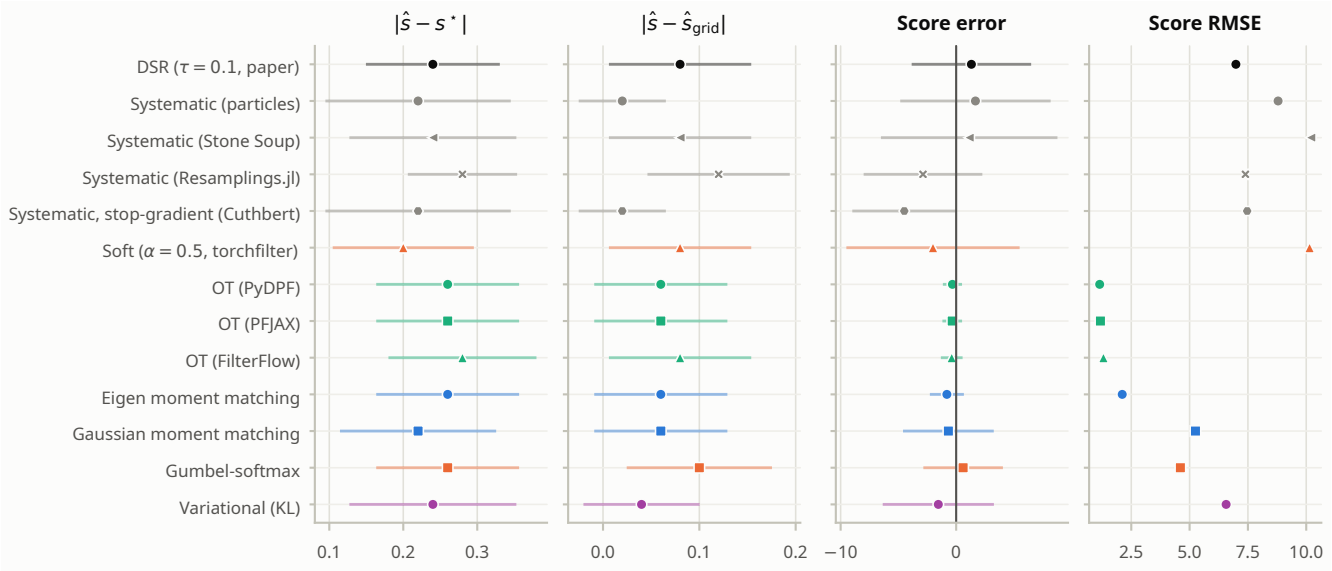


Figure 35: Plot of Table **??**: mean ± 95% CI over 10 observation sequences, one panel per column. Rows are in table order. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family.

Fixed weighted clouds (resampling only)

Table 36: Gaussian cloud: sliced Wasserstein-1 (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	$0.383 \pm 7.88 \times 10^{-4}$	$0.331 \pm 1.09 \times 10^{-4}$	$0.355 \pm 1.15 \times 10^{-6}$	$0.345 \pm 1.18 \times 10^{-5}$
DSR (τ=0.1, no clamp)	$0.383 \pm 7.88 \times 10^{-4}$	$0.331 \pm 1.09 \times 10^{-4}$	$0.355 \pm 1.15 \times 10^{-6}$	$0.345 \pm 1.18 \times 10^{-5}$
DSR (τ=0.1, unsorted)	$0.642 \pm 9.17 \times 10^{-5}$	$0.666 \pm 3.1 \times 10^{-5}$	$0.792 \pm 1.73 \times 10^{-6}$	0.774
DSR (τ=0.5)	$0.743 \pm 3.81 \times 10^{-5}$	$0.637 \pm 4.47 \times 10^{-6}$	$0.747 \pm 9.68 \times 10^{-7}$	0.713
DSR (τ=2.5/N)†	$0.326 \pm 8.05 \times 10^{-4}$	$0.14 \pm 5.73 \times 10^{-5}$	$0.072 \pm 2.69 \times 10^{-5}$	$0.031 \pm 3.24 \times 10^{-6}$
Systematic	0.0951 ± 0.00479	$\mathbf{0.0432 \pm 0.00271}$	$0.0226 \pm 6.38 \times 10^{-4}$	$0.0107 \pm 3.72 \times 10^{-4}$
Multinomial	0.195 ± 0.0113	0.0957 ± 0.00491	0.0552 ± 0.00276	0.0288 ± 0.00137
Systematic, stop-gradient (PyDPF)	$\mathbf{0.0901 \pm 0.00377}$	0.045 ± 0.00256	$\mathbf{0.0219 \pm 6.25 \times 10^{-4}}$	$\mathbf{0.0104 \pm 3.66 \times 10^{-4}}$
Soft (α=0.5)	0.2 ± 0.0117	0.0921 ± 0.00393	0.0578 ± 0.00302	0.0276 ± 0.00128
OT (PyDPF)	0.118	0.0546	0.0481	0.0403
OT (PFJAX)	0.119	0.0553	0.0478	0.0403
OT (FilterFlow)	0.153	0.0771	0.0812	0.0756
Eigen moment matching	0.186	0.064	0.0518	0.0247
Gaussian moment matching	0.285 ± 0.0111	0.121 ± 0.00497	0.0743 ± 0.00229	0.0338 ± 0.00124
Gumbel-softmax	0.291 ± 0.0277	0.23 ± 0.00949	0.278 ± 0.0321	0.264 ± 0.0156
Variational (KL)	0.151	0.124	0.118	0.116

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

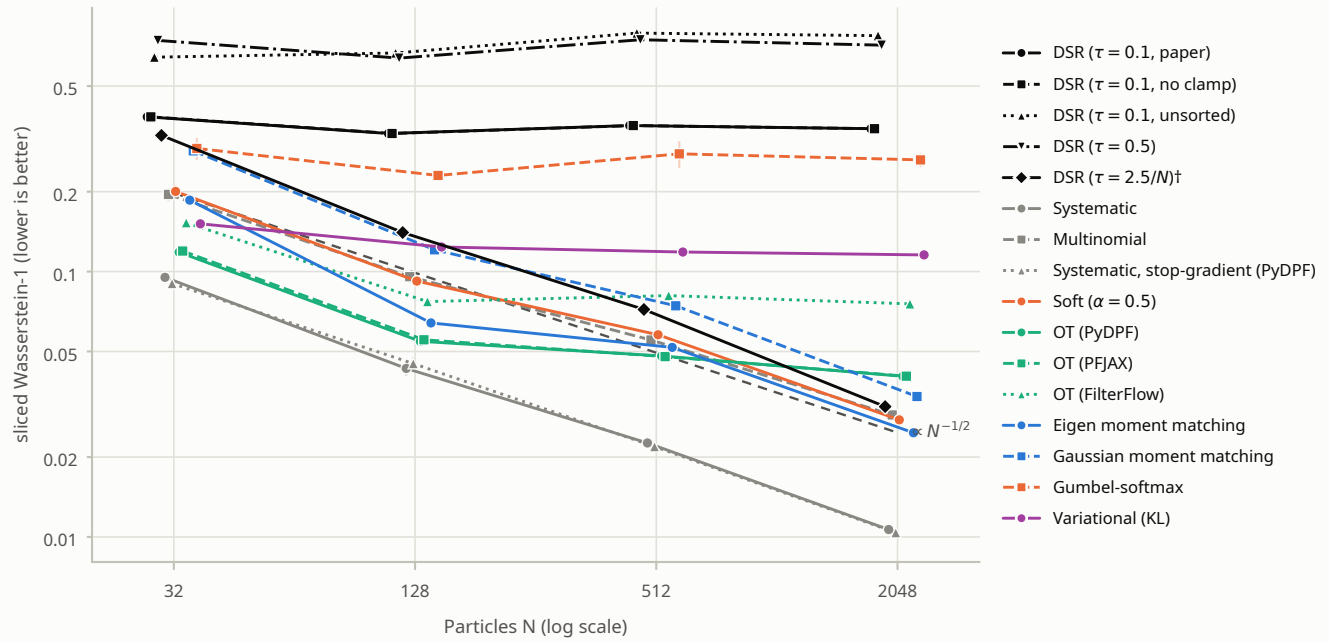


Figure 36: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 37: Gaussian cloud: Gaussian-kernel MMD^2 (random features) (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	$0.0792 \pm 1.64 \times 10^{-4}$	$0.0834 \pm 5.07 \times 10^{-5}$	$0.0695 \pm 7.43 \times 10^{-6}$	$0.0705 \pm 2.53 \times 10^{-6}$
DSR (τ=0.1, no clamp)	$0.0792 \pm 1.64 \times 10^{-4}$	$0.0834 \pm 5.07 \times 10^{-5}$	$0.0695 \pm 7.43 \times 10^{-6}$	$0.0705 \pm 2.53 \times 10^{-6}$
DSR (τ=0.1, unsorted)	$0.166 \pm 9.98 \times 10^{-5}$	$0.153 \pm 2.85 \times 10^{-7}$	$0.143 \pm 3.23 \times 10^{-7}$	0.149
DSR (τ=0.5)	$0.181 \pm 3.15 \times 10^{-5}$	$0.152 \pm 1.07 \times 10^{-5}$	$0.141 \pm 4.23 \times 10^{-6}$	$0.146 \pm 1.04 \times 10^{-6}$
DSR (τ=2.5/N)†	$0.0562 \pm 2.06 \times 10^{-4}$	$0.0116 \pm 1.21 \times 10^{-5}$	$0.00134 \pm 6.69 \times 10^{-7}$	$1.84 \times 10^{-4} \pm 9.25 \times 10^{-8}$
Systematic	$0.00292 \pm 2.02 \times 10^{-4}$	$7.23 \times 10^{-4} \pm 1.31 \times 10^{-4}$	$1.75 \times 10^{-4} \pm 1.82 \times 10^{-5}$	$3.08 \times 10^{-5} \pm 3.15 \times 10^{-6}$
Multinomial	0.0136 ± 0.00216	$0.0039 \pm 5.5 \times 10^{-4}$	$8.84 \times 10^{-4} \pm 1.15 \times 10^{-4}$	$2.32 \times 10^{-4} \pm 2.77 \times 10^{-5}$
Systematic, stop-gradient (PyDPF)	$0.00282 \pm 1.92 \times 10^{-4}$	$8.04 \times 10^{-4} \pm 1.27 \times 10^{-4}$	$\mathbf{1.69 \times 10^{-4} \pm 1.9 \times 10^{-5}}$	$2.62 \times 10^{-5} \pm 2.94 \times 10^{-6}$
Soft (α=0.5)	0.016 ± 0.00263	$0.00368 \pm 4.88 \times 10^{-4}$	$9.87 \times 10^{-4} \pm 1.31 \times 10^{-4}$	$2.18 \times 10^{-4} \pm 2.88 \times 10^{-5}$
OT (PyDPF)	**0.00196**	$\mathbf{6.73 \times 10^{-4}}$	6.04×10^{-4}	5.87×10^{-4}
OT (PFJAX)	0.00205	7.19×10^{-4}	5.94×10^{-4}	5.88×10^{-4}
OT (FilterFlow)	0.00615	0.00239	0.00218	0.00223
Eigen moment matching	0.00863	6.9×10^{-4}	2.46×10^{-4}	$\mathbf{2.07 \times 10^{-5}}$
Gaussian moment matching	0.0223 ± 0.00266	$0.00447 \pm 5.37 \times 10^{-4}$	$0.0015 \pm 1.4 \times 10^{-4}$	$2.21 \times 10^{-4} \pm 2.81 \times 10^{-5}$
Gumbel-softmax	0.0426 ± 0.00986	0.0308 ± 0.00324	0.0276 ± 0.00221	0.0272 ± 0.004
Variational (KL)	0.0104	0.0081	0.00535	0.00582

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

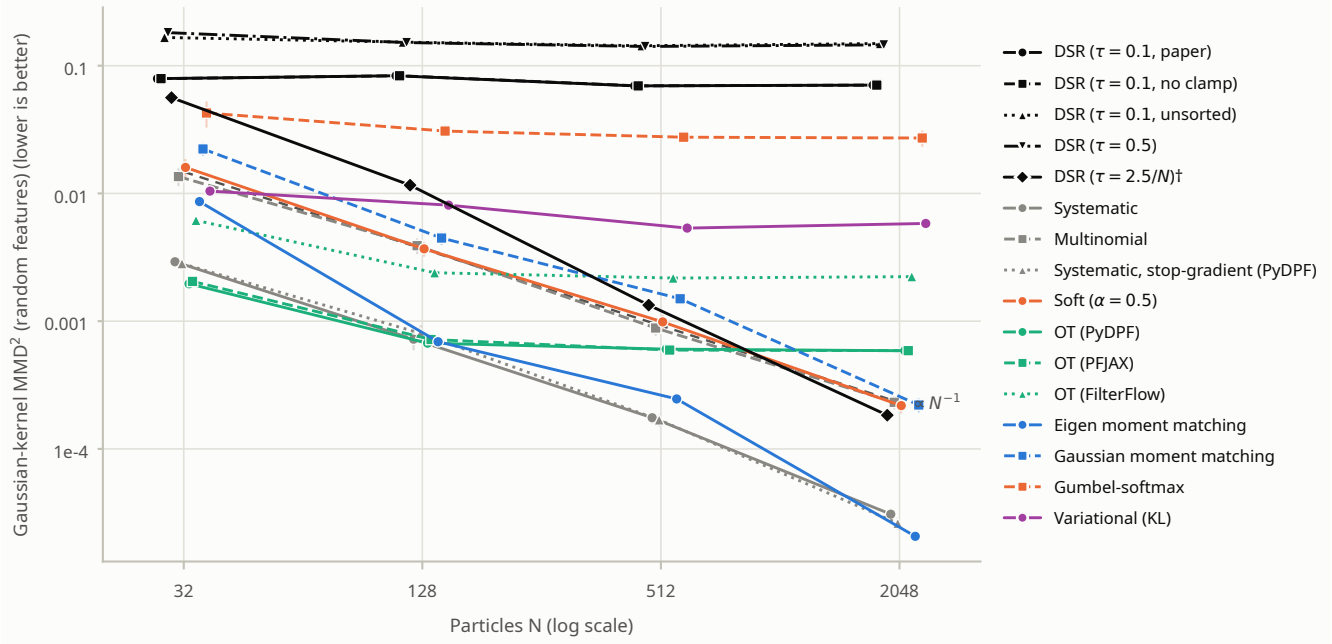


Figure 37: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 38: Gaussian cloud: covariance error (Frobenius) (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	1.05 ± 0.00134	$0.924 \pm 1.07 \times 10^{-4}$	$1.16 \pm 1.88 \times 10^{-5}$	1.07
DSR (τ=0.1, no clamp)	1.05 ± 0.00134	$0.924 \pm 1.07 \times 10^{-4}$	$1.16 \pm 1.88 \times 10^{-5}$	1.07
DSR (τ=0.1, unsorted)	$1.33 \pm 3.3 \times 10^{-5}$	$1.16 \pm 7.95 \times 10^{-6}$	1.54	1.42
DSR (τ=0.5)	$1.39 \pm 3.5 \times 10^{-6}$	1.15	1.53	1.41
DSR (τ=2.5/N)†	0.915 ± 0.00211	$0.47 \pm 1.59 \times 10^{-4}$	$0.277 \pm 4.55 \times 10^{-5}$	$0.124 \pm 1.05 \times 10^{-5}$
Systematic	0.273 ± 0.0136	0.0698 ± 0.00694	0.055 ± 0.00397	0.0253 ± 0.00184
Multinomial	0.343 ± 0.0313	0.15 ± 0.0127	0.101 ± 0.00899	0.0524 ± 0.00372
Systematic, stop-gradient (PyDPF)	0.27 ± 0.0129	0.0725 ± 0.00803	0.0548 ± 0.00357	0.0237 ± 0.00179
Soft (α=0.5)	0.282 ± 0.0248	0.14 ± 0.0119	0.109 ± 0.0102	0.0479 ± 0.00383
OT (PyDPF)	0.216	0.114	0.144	0.136
OT (PFJAX)	0.223	0.119	0.148	0.137
OT (FilterFlow)	0.357	0.21	0.262	0.255
Eigen moment matching	$\mathbf{1.57 \times 10^{-16}}$	$\mathbf{3.54 \times 10^{-16}}$	$\mathbf{1.99 \times 10^{-15}}$	$\mathbf{7.02 \times 10^{-16}}$
Gaussian moment matching	0.378 ± 0.0319	0.167 ± 0.0164	0.107 ± 0.00992	0.0516 ± 0.00415
Gumbel-softmax	0.706 ± 0.091	0.592 ± 0.0287	0.798 ± 0.0376	0.718 ± 0.0476
Variational (KL)	0.0867	0.082	0.119	0.0861

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

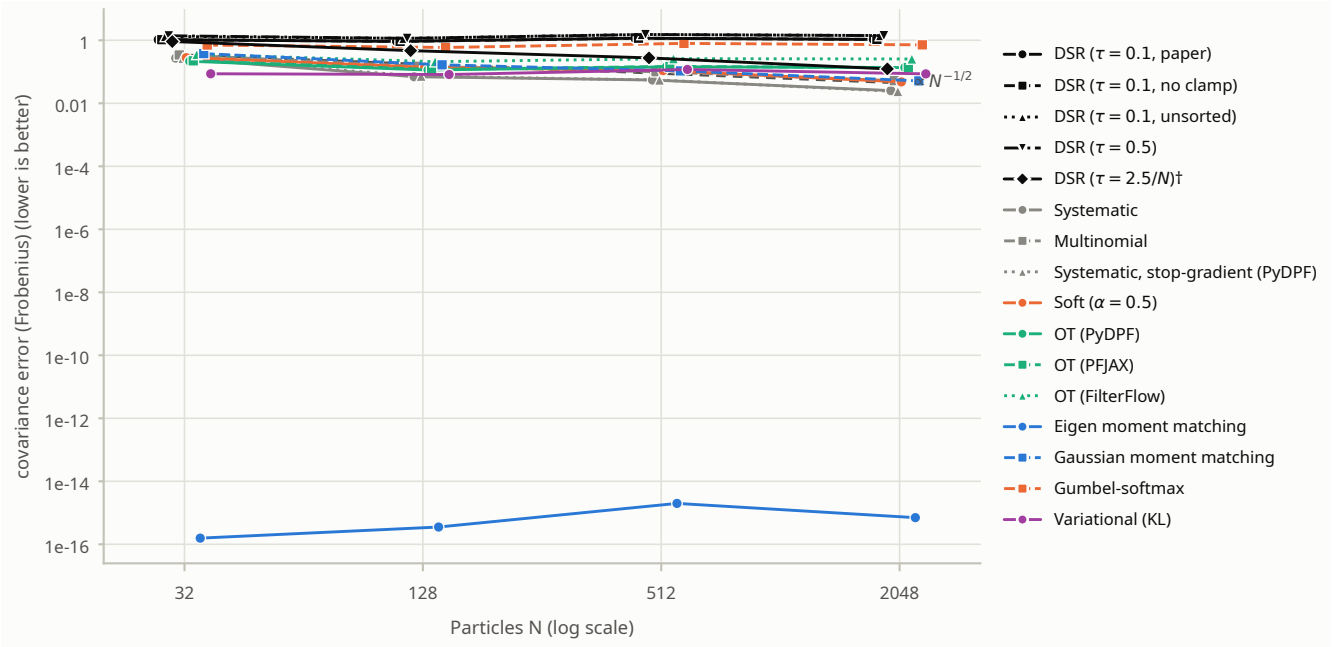


Figure 38: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 39: Gaussian cloud: MSE of $\mathbb{E}[x^2]$ (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.444 ± 0.00282	$0.377 \pm 3.21 \times 10^{-4}$	$0.625 \pm 2 \times 10^{-4}$	$0.525 \pm 4.27 \times 10^{-5}$
DSR (τ=0.1, no clamp)	0.444 ± 0.00282	$0.377 \pm 3.21 \times 10^{-4}$	$0.625 \pm 2 \times 10^{-4}$	$0.525 \pm 4.27 \times 10^{-5}$
DSR (τ=0.1, unsorted)	0.868 ± 0.00109	$0.65 \pm 6.96 \times 10^{-5}$	$1.18 \pm 4.7 \times 10^{-6}$	1.01
DSR (τ=0.5)	$0.884 \pm 8.21 \times 10^{-4}$	$0.639 \pm 9.53 \times 10^{-5}$	$1.15 \pm 4.45 \times 10^{-5}$	$0.976 \pm 1.24 \times 10^{-5}$
DSR (τ=2.5/N)†	0.349 ± 0.00267	$0.0948 \pm 5.75 \times 10^{-5}$	$0.029 \pm 7.26 \times 10^{-5}$	$0.00625 \pm 1.3 \times 10^{-5}$
Systematic	0.00892 ± 0.00145	$0.00147 \pm 3.02 \times 10^{-4}$	$0.00144 \pm 1.84 \times 10^{-4}$	$8.91 \times 10^{-5} \pm 1.36 \times 10^{-5}$
Multinomial	0.0411 ± 0.00804	0.0109 ± 0.00237	0.00508 ± 0.00113	$0.00148 \pm 2.72 \times 10^{-4}$
Systematic, stop-gradient (PyDPF)	0.0106 ± 0.0021	$0.00201 \pm 4.02 \times 10^{-4}$	$0.00147 \pm 1.5 \times 10^{-4}$	$9.14 \times 10^{-5} \pm 1.44 \times 10^{-5}$
Soft (α=0.5)	0.0626 ± 0.0156	0.0111 ± 0.00207	0.00686 ± 0.0013	$0.00149 \pm 3 \times 10^{-4}$
OT (PyDPF)	0.0246	0.00703	0.0108	0.00936
OT (PFJAX)	0.0246	0.00705	0.011	0.00943
OT (FilterFlow)	0.063	0.022	0.0342	0.0326
Eigen moment matching	$\mathbf{7.7 \times 10^{-31}}$	$\mathbf{3.08 \times 10^{-32}}$	$\mathbf{6.71 \times 10^{-30}}$	$\mathbf{2.39 \times 10^{-30}}$
Gaussian moment matching	0.102 ± 0.0247	0.0172 ± 0.00435	0.00584 ± 0.00119	$0.00146 \pm 3.2 \times 10^{-4}$
Gumbel-softmax	0.25 ± 0.0707	0.172 ± 0.0219	0.334 ± 0.0619	0.257 ± 0.0155
Variational (KL)	0.0618	0.0168	0.0316	0.0262

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

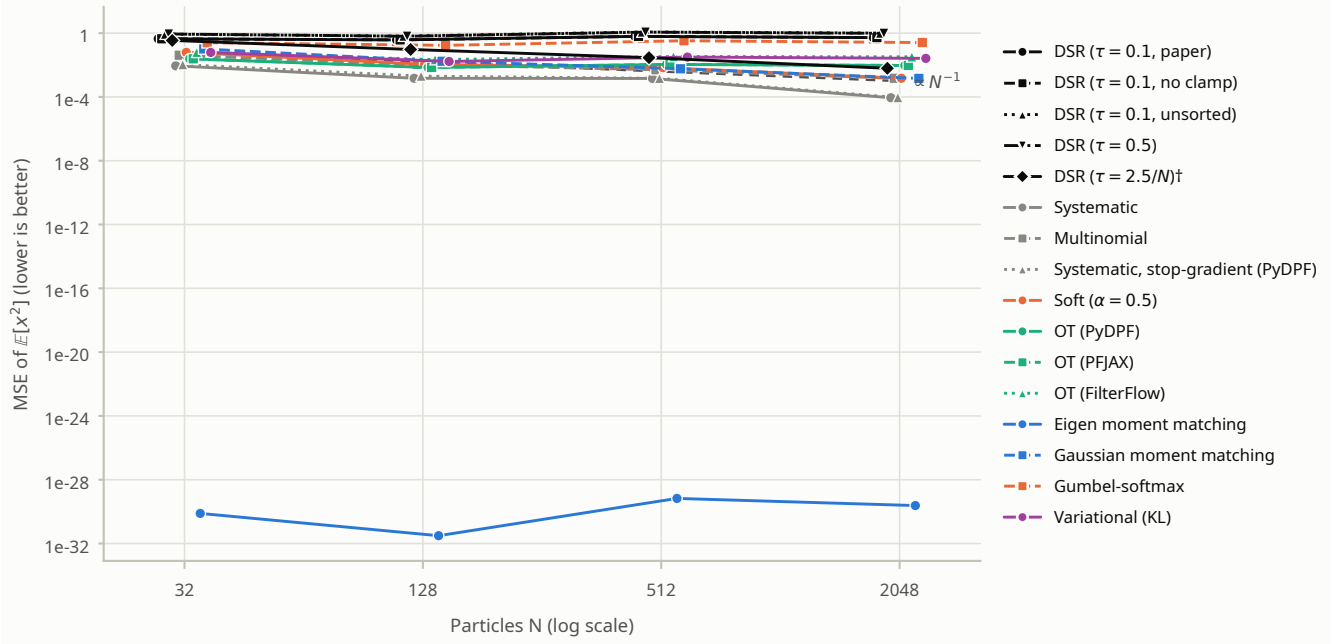


Figure 39: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 40: Bimodal mixture cloud: sliced Wasserstein-1 (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	$0.583 \pm 4.39 \times 10^{-4}$	$0.657 \pm 7.73 \times 10^{-5}$	$0.66 \pm 2.54 \times 10^{-5}$	$0.648 \pm 8.5 \times 10^{-6}$
DSR (τ=0.1, no clamp)	$0.583 \pm 4.39 \times 10^{-4}$	$0.657 \pm 7.73 \times 10^{-5}$	$0.66 \pm 2.54 \times 10^{-5}$	$0.648 \pm 8.5 \times 10^{-6}$
DSR (τ=0.1, unsorted)	$1.82 \pm 6.09 \times 10^{-5}$	$2.03 \pm 3.21 \times 10^{-5}$	$1.99 \pm 5.77 \times 10^{-6}$	2.02
DSR (τ=0.5)	$1.72 \pm 3.26 \times 10^{-4}$	$1.84 \pm 9.28 \times 10^{-5}$	$1.81 \pm 2.29 \times 10^{-5}$	$1.82 \pm 5.67 \times 10^{-6}$
DSR (τ=2.5/N)†	$0.451 \pm 3.85 \times 10^{-4}$	$\mathbf{0.159 \pm 1.89 \times 10^{-4}}$	$\mathbf{0.0739 \pm 2.47 \times 10^{-5}}$	$\mathbf{0.0298 \pm 6.59 \times 10^{-6}}$
Systematic	0.834 ± 0.0724	0.887 ± 0.077	0.865 ± 0.0751	0.868 ± 0.0754
Multinomial	0.398 ± 0.0378	0.195 ± 0.0188	0.104 ± 0.00951	0.0511 ± 0.00467
Systematic, stop-gradient (PyDPF)	0.875 ± 0.0761	0.93 ± 0.0809	0.908 ± 0.0789	0.911 ± 0.0792
Soft (α=0.5)	0.367 ± 0.0309	0.186 ± 0.0175	0.0978 ± 0.00805	0.0492 ± 0.00452
OT (PyDPF)	0.284	0.436	0.374	–
OT (PFJAX)	**0.273**	0.43	0.364	0.369
OT (FilterFlow)	0.466	0.628	0.576	0.571
Eigen moment matching	0.625	0.716	0.688	0.686
Gaussian moment matching	0.847 ± 0.0271	0.86 ± 0.0134	0.793 ± 0.00546	0.786 ± 0.003
Gumbel-softmax	0.728 ± 0.0638	0.772 ± 0.038	0.712 ± 0.0794	0.73 ± 0.0229
Variational (KL)	0.614	0.652	0.636	0.639

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

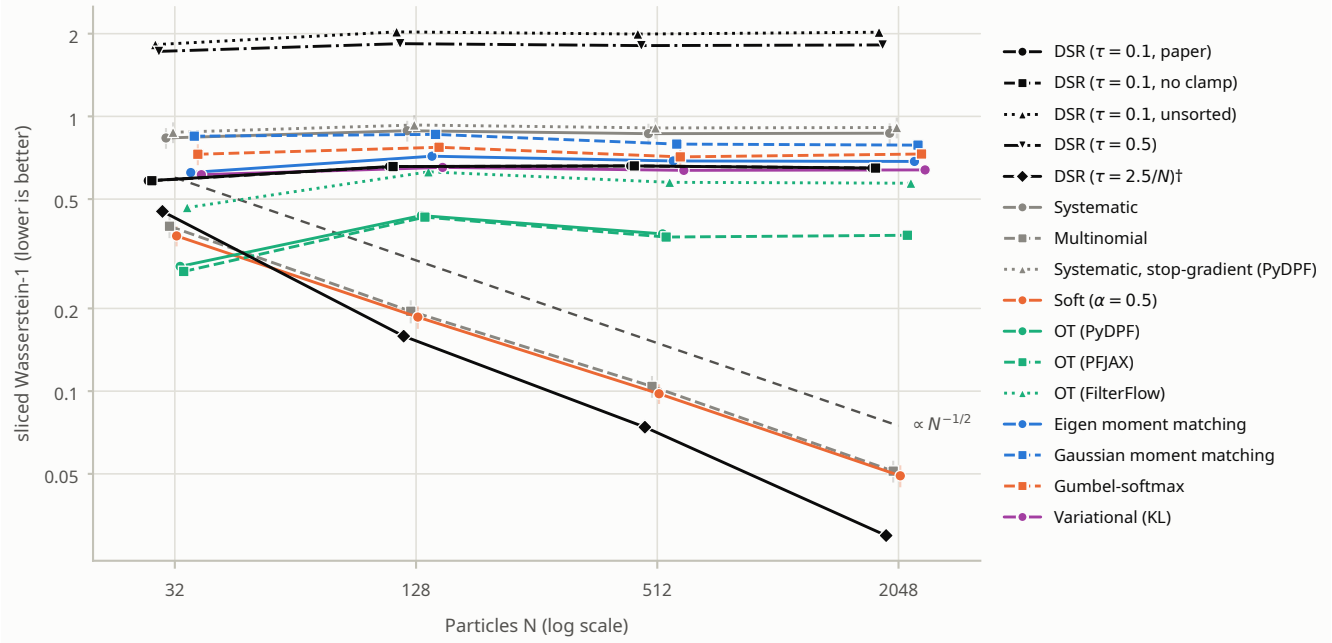


Figure 40: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 41: Bimodal mixture cloud: Gaussian-kernel MMD^2 (random features) (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	$0.0397 \pm 1.07 \times 10^{-4}$	$0.0444 \pm 1.38 \times 10^{-5}$	$0.0456 \pm 3.62 \times 10^{-6}$	$0.0427 \pm 1.05 \times 10^{-6}$
DSR (τ=0.1, no clamp)	$0.0397 \pm 1.07 \times 10^{-4}$	$0.0444 \pm 1.38 \times 10^{-5}$	$0.0456 \pm 3.62 \times 10^{-6}$	$0.0427 \pm 1.05 \times 10^{-6}$
DSR (τ=0.1, unsorted)	$0.227 \pm 1.15 \times 10^{-5}$	$0.228 \pm 7.43 \times 10^{-7}$	$0.219 \pm 2.71 \times 10^{-7}$	0.219
DSR (τ=0.5)	$0.218 \pm 7.77 \times 10^{-5}$	$0.217 \pm 2.39 \times 10^{-5}$	$0.209 \pm 5.48 \times 10^{-6}$	$0.209 \pm 1.29 \times 10^{-6}$
DSR (τ=2.5/N)†	$0.0235 \pm 8.84 \times 10^{-5}$	$\mathbf{0.0025 \pm 7.81 \times 10^{-6}}$	$\mathbf{2.53 \times 10^{-4} \pm 7.04 \times 10^{-7}}$	$\mathbf{2.12 \times 10^{-5} \pm 2.93 \times 10^{-8}}$
Systematic	0.0853 ± 0.0152	0.0875 ± 0.0157	0.0849 ± 0.0152	0.0849 ± 0.0152
Multinomial	0.0177 ± 0.0038	$0.00363 \pm 7.99 \times 10^{-4}$	$0.00101 \pm 2.13 \times 10^{-4}$	$2.42 \times 10^{-4} \pm 5.04 \times 10^{-5}$
Systematic, stop-gradient (PyDPF)	0.0939 ± 0.016	0.0964 ± 0.0164	0.0935 ± 0.016	0.0935 ± 0.016
Soft (α=0.5)	0.0148 ± 0.00263	$0.00326 \pm 7.19 \times 10^{-4}$	$8.64 \times 10^{-4} \pm 1.75 \times 10^{-4}$	$2.21 \times 10^{-4} \pm 5.22 \times 10^{-5}$
OT (PyDPF)	0.00467	0.0123	0.00853	–
OT (PFJAX)	**0.00444**	0.0126	0.0086	0.00927
OT (FilterFlow)	0.0146	0.0249	0.0205	0.021
Eigen moment matching	0.0217	0.0242	0.0226	0.0226
Gaussian moment matching	0.0409 ± 0.00459	0.0325 ± 0.00195	$0.0281 \pm 7.93 \times 10^{-4}$	$0.0273 \pm 4.11 \times 10^{-4}$
Gumbel-softmax	0.0578 ± 0.0128	0.0555 ± 0.00518	0.0458 ± 0.00823	0.0475 ± 0.00456
Variational (KL)	0.0388	0.0398	0.0386	0.0386

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

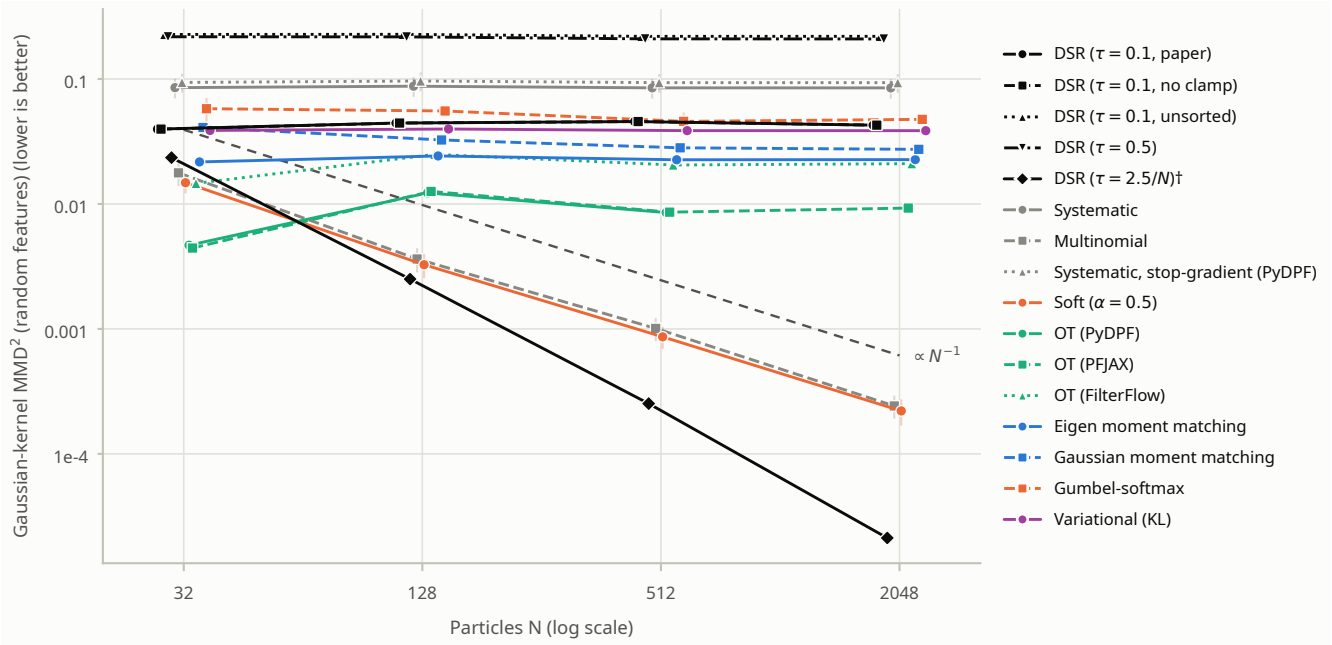


Figure 41: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 42: Bimodal mixture cloud: covariance error (Frobenius) (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	3.72 ± 0.0141	4.09 ± 0.00371	$4.02 \pm 8.46 \times 10^{-4}$	$3.91 \pm 2.25 \times 10^{-4}$
DSR (τ=0.1, no clamp)	3.72 ± 0.0141	4.09 ± 0.00371	$4.02 \pm 8.46 \times 10^{-4}$	$3.91 \pm 2.25 \times 10^{-4}$
DSR (τ=0.1, unsorted)	$8.06 \pm 4.56 \times 10^{-5}$	8.77	8.37	8.42
DSR (τ=0.5)	$7.96 \pm 1.02 \times 10^{-4}$	$8.64 \pm 2.62 \times 10^{-5}$	8.25	8.29
DSR (τ=2.5/N)†	2.9 ± 0.0156	0.998 ± 0.00325	$0.345 \pm 3.73 \times 10^{-4}$	$0.109 \pm 1.16 \times 10^{-4}$
Systematic	2.59 ± 0.609	2.91 ± 0.68	2.77 ± 0.647	2.79 ± 0.651
Multinomial	1.14 ± 0.14	0.559 ± 0.058	0.281 ± 0.031	0.144 ± 0.0148
Systematic, stop-gradient (PyDPF)	2.93 ± 0.64	3.29 ± 0.714	3.14 ± 0.679	3.16 ± 0.684
Soft (α=0.5)	0.997 ± 0.115	0.522 ± 0.0556	0.281 ± 0.0235	0.137 ± 0.0137
OT (PyDPF)	1.13	1.93	1.64	–
OT (PFJAX)	1.25	2.13	1.79	1.84
OT (FilterFlow)	2.09	2.86	2.61	2.62
Eigen moment matching	$\mathbf{5.38 \times 10^{-15}}$	$\mathbf{1.31 \times 10^{-15}}$	$\mathbf{3.73 \times 10^{-15}}$	$\mathbf{3.21 \times 10^{-14}}$
Gaussian moment matching	1.65 ± 0.207	0.93 ± 0.132	0.463 ± 0.0571	0.251 ± 0.0303
Gumbel-softmax	3.97 ± 0.557	4.49 ± 0.33	4.31 ± 0.487	4.24 ± 0.163
Variational (KL)	0.733	0.833	0.801	0.809

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

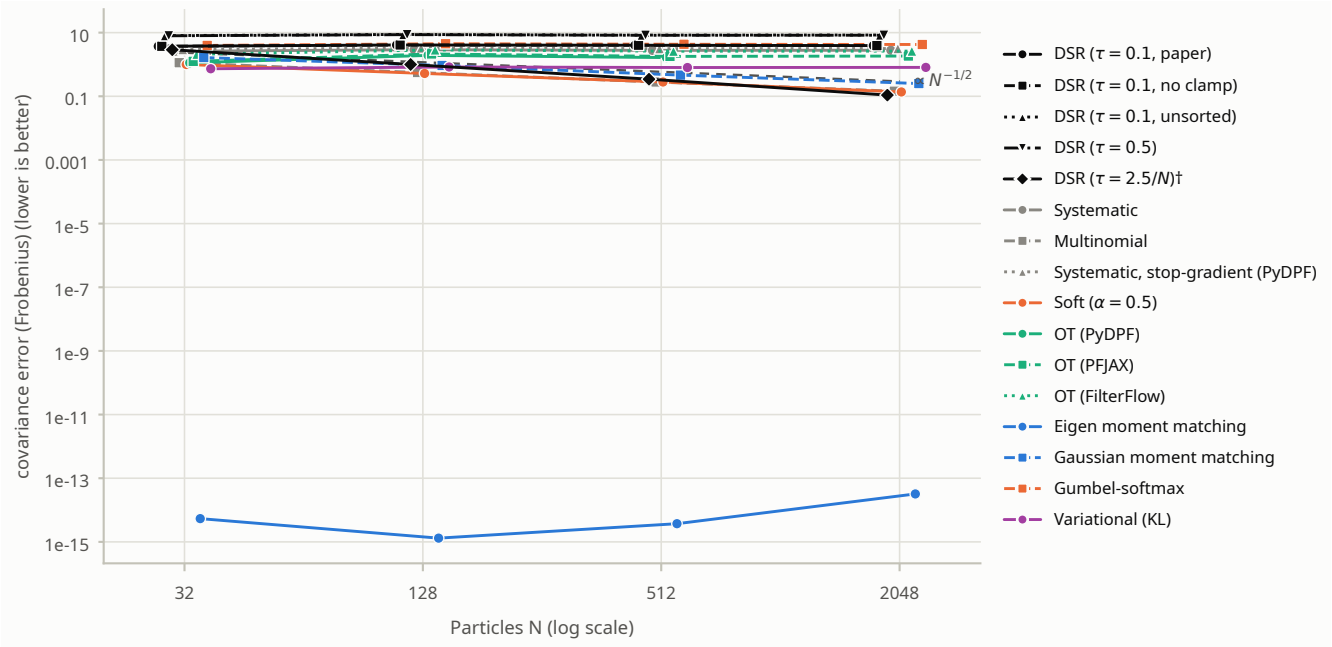


Figure 42: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 43: Bimodal mixture cloud: MSE of $\mathbb{E}[x^2]$ (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	6 ± 0.0119	7.69 ± 0.00441	7.48 ± 0.00103	$7.12 \pm 2.31 \times 10^{-4}$
DSR (τ=0.1, no clamp)	6 ± 0.0119	7.69 ± 0.00441	7.48 ± 0.00103	$7.12 \pm 2.31 \times 10^{-4}$
DSR (τ=0.1, unsorted)	32.1 ± 0.00714	$38.4 \pm 2.52 \times 10^{-4}$	35	35.4
DSR (τ=0.5)	30.3 ± 0.0288	36.4 ± 0.00852	33.3 ± 0.00187	$33.6 \pm 4.83 \times 10^{-4}$
DSR (τ=2.5/N)†	3.56 ± 0.00402	$0.354 \pm 2.75 \times 10^{-4}$	$0.0383 \pm 6.3 \times 10^{-5}$	$0.00444 \pm 4.7 \times 10^{-6}$
Systematic	0.00932 ± 0.00167	$0.00307 \pm 5.49 \times 10^{-4}$	$9.26 \times 10^{-5} \pm 1.66 \times 10^{-5}$	$4.96 \times 10^{-4} \pm 8.86 \times 10^{-5}$
Multinomial	0.136 ± 0.0343	0.0394 ± 0.0104	0.00873 ± 0.00226	$0.00222 \pm 5.4 \times 10^{-4}$
Systematic, stop-gradient (PyDPF)	0.0103 ± 0.00175	$0.00338 \pm 5.77 \times 10^{-4}$	$1.02 \times 10^{-4} \pm 1.74 \times 10^{-5}$	$5.46 \times 10^{-4} \pm 9.31 \times 10^{-5}$
Soft (α=0.5)	0.155 ± 0.0408	0.0319 ± 0.00882	0.00996 ± 0.00237	$0.00197 \pm 3.81 \times 10^{-4}$
OT (PyDPF)	0.697	2.01	1.47	–
OT (PFJAX)	0.763	2.28	1.61	1.69
OT (FilterFlow)	2.14	4.08	3.41	3.44
Eigen moment matching	$\mathbf{1.42 \times 10^{-29}}$	$\mathbf{2.22 \times 10^{-31}}$	$\mathbf{1.6 \times 10^{-30}}$	$\mathbf{1.59 \times 10^{-28}}$
Gaussian moment matching	1.95 ± 0.528	0.816 ± 0.236	0.153 ± 0.0427	0.0449 ± 0.0124
Gumbel-softmax	8.19 ± 1.21	10.6 ± 0.804	8.08 ± 0.99	8.99 ± 0.937
Variational (KL)	0.00424	0.00139	4.21×10^{-5}	2.25×10^{-4}

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

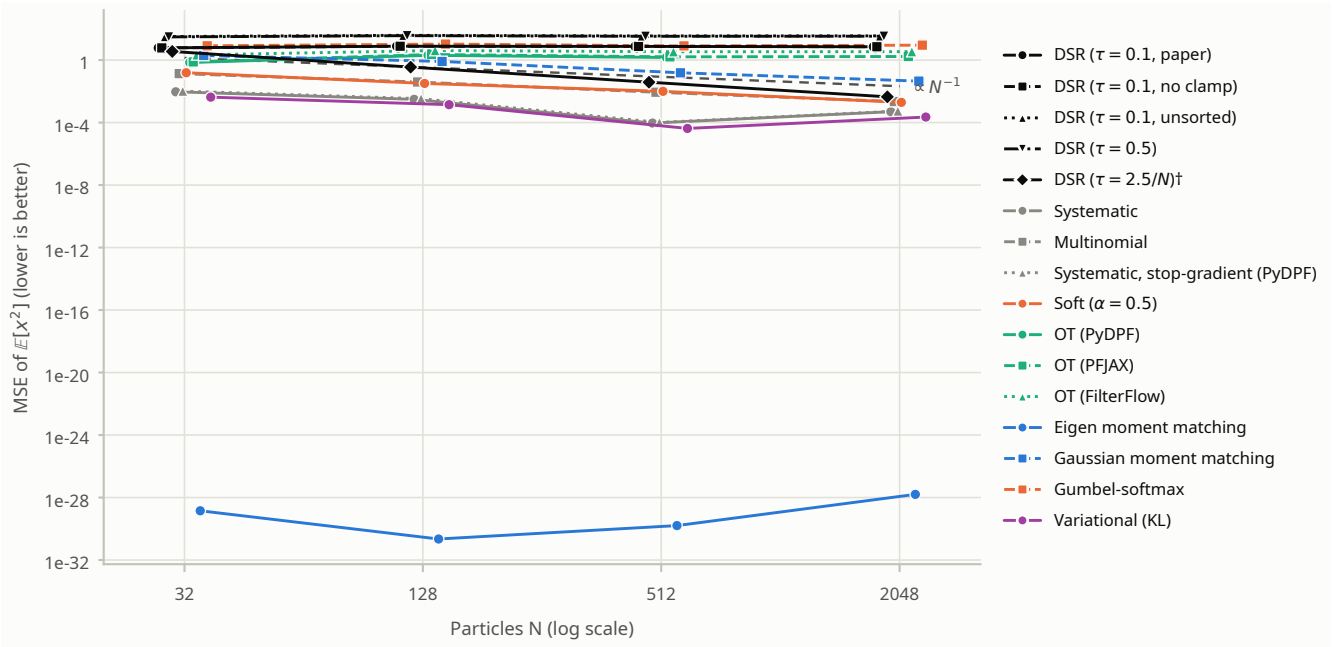


Figure 43: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 44: Bimodal mixture cloud: mode-mass error (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.0104 ± 0.00155	$0.00522 \pm 3.87 \times 10^{-4}$	$0.00617 \pm 7.63 \times 10^{-5}$	$0.00365 \pm 4.58 \times 10^{-5}$
DSR (τ=0.1, no clamp)	0.0104 ± 0.00155	$0.00522 \pm 3.87 \times 10^{-4}$	$0.00617 \pm 7.63 \times 10^{-5}$	$0.00365 \pm 4.58 \times 10^{-5}$
DSR (τ=0.1, unsorted)	0.35	0.35	0.35	0.35
DSR (τ=0.5)	0.35	0.35	0.35	0.35
DSR (τ=2.5/N)†	0.0137 ± 0.00183	$\mathbf{0.0018 \pm 2.66 \times 10^{-4}}$	$\mathbf{5.9 \times 10^{-4} \pm 8.78 \times 10^{-5}}$	$\mathbf{1.81 \times 10^{-4} \pm 3.66 \times 10^{-5}}$
Systematic	0.204 ± 0.0177	0.204 ± 0.0177	0.204 ± 0.0177	0.204 ± 0.0177
Multinomial	0.0786 ± 0.0107	0.0329 ± 0.00501	0.0181 ± 0.00256	0.00869 ± 0.00128
Systematic, stop-gradient (PyDPF)	0.214 ± 0.0186	0.214 ± 0.0186	0.214 ± 0.0186	0.214 ± 0.0186
Soft (α=0.5)	0.0682 ± 0.00952	0.0302 ± 0.0046	0.0161 ± 0.00228	0.008 ± 0.00126
OT (PyDPF)	0.025	0.0484	0.0328	–
OT (PFJAX)	**0.00625**	0.0328	0.0191	0.0255
OT (FilterFlow)	0.025	0.0953	0.0582	0.0655
Eigen moment matching	0.15	0.15	0.15	0.15
Gaussian moment matching	0.071 ± 0.0115	0.0369 ± 0.0058	0.0283 ± 0.00352	0.0262 ± 0.0021
Gumbel-softmax	0.0744 ± 0.0461	0.0431 ± 0.0257	0.057 ± 0.0684	0.0328 ± 0.034
Variational (KL)	0.15	0.15	0.15	0.15

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

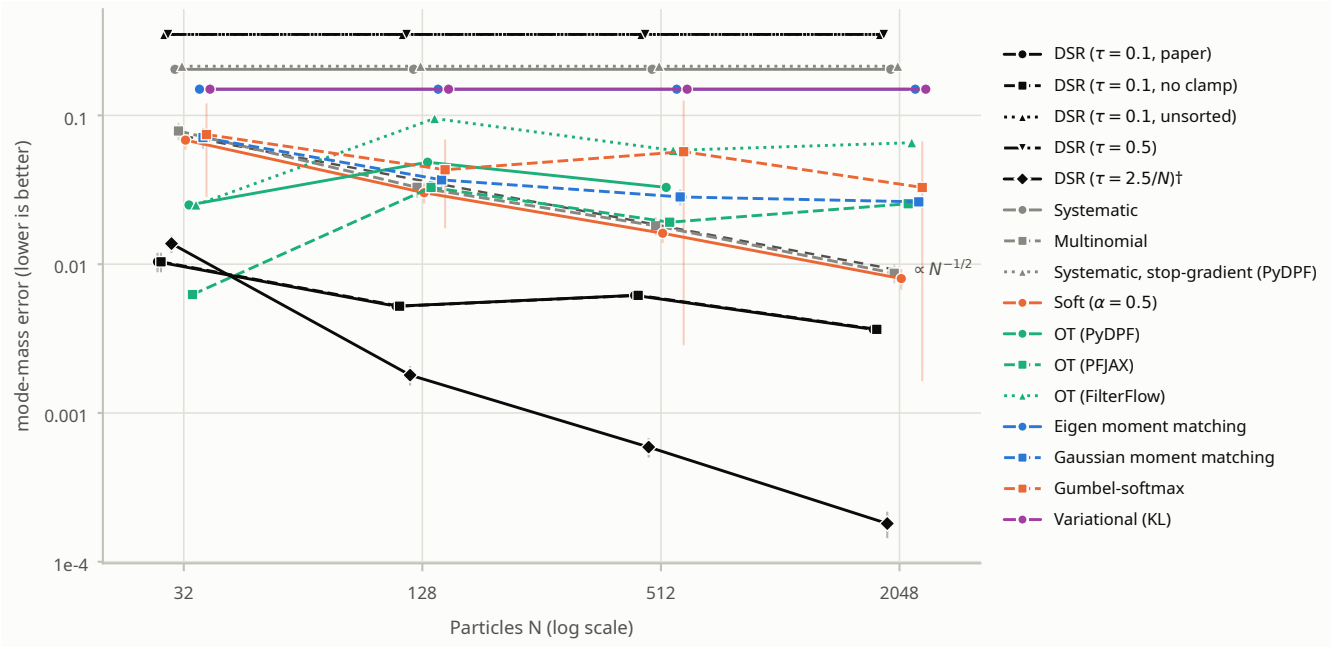


Figure 44: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 45: Bimodal mixture cloud: fraction of draws keeping both modes (each mode $> 1\%$ of the mass) (higher is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	**1**	**1**	**1**	**1**
DSR (τ=0.1, no clamp)	**1**	**1**	**1**	**1**
DSR (τ=0.1, unsorted)	0	0	0	0
DSR (τ=0.5)	0	0	0	0
DSR (τ=2.5/N)†	**1**	**1**	**1**	**1**
Systematic	0.865 ± 0.0443	0.865 ± 0.0443	0.865 ± 0.0443	0.865 ± 0.0443
Multinomial	**1**	**1**	**1**	**1**
Systematic, stop-gradient (PyDPF)	0.84 ± 0.0465	0.84 ± 0.0465	0.84 ± 0.0465	0.84 ± 0.0465
Soft (α=0.5)	**1**	**1**	**1**	**1**
OT (PyDPF)	**1**	**1**	**1**	–
OT (PFJAX)	**1**	**1**	**1**	**1**
OT (FilterFlow)	**1**	**1**	**1**	**1**
Eigen moment matching	**1**	**1**	**1**	**1**
Gaussian moment matching	**1**	**1**	**1**	**1**
Gumbel-softmax	**1**	**1**	**1**	**1**
Variational (KL)	**1**	**1**	**1**	**1**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

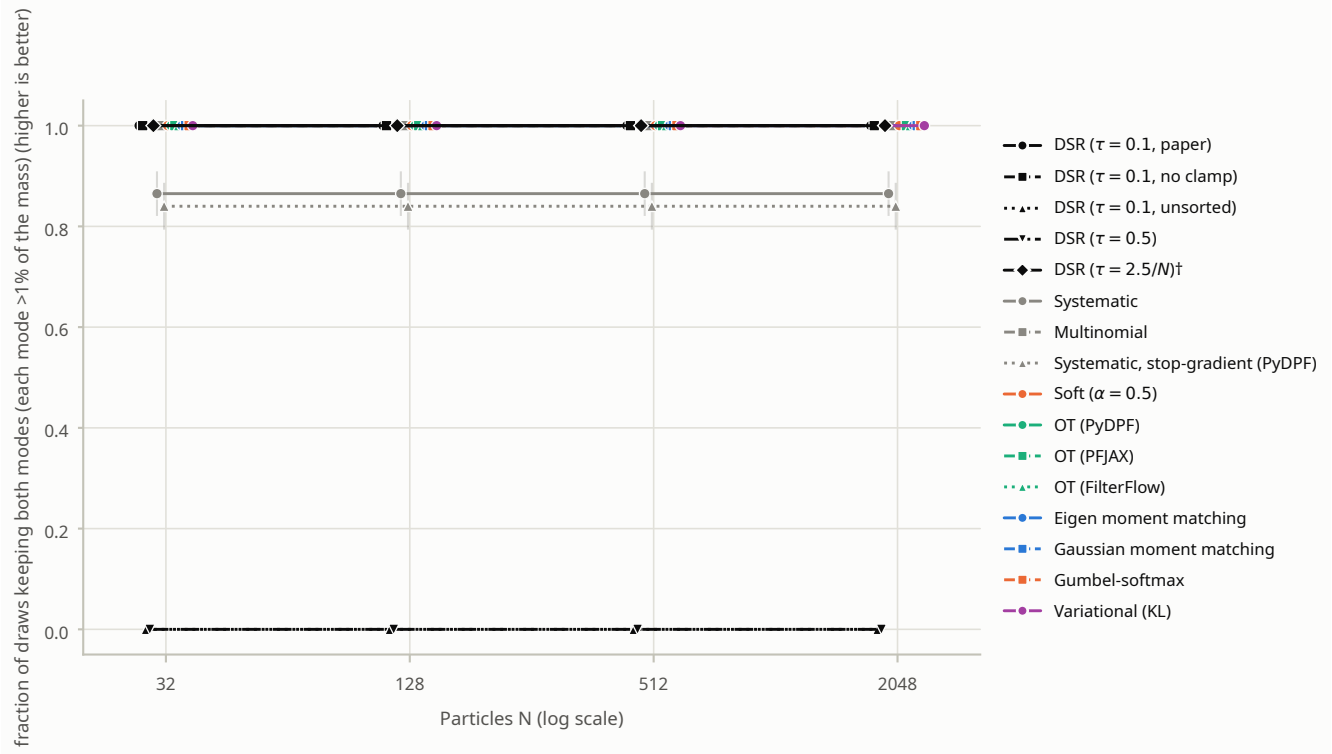


Figure 45: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap.

Table 46: Skewed-weight cloud: sliced Wasserstein-1 (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	$0.409 \pm 7.16 \times 10^{-4}$	$0.295 \pm 1.03 \times 10^{-4}$	$0.333 \pm 1.06 \times 10^{-5}$	$0.34 \pm 3.49 \times 10^{-6}$
DSR (τ=0.1, no clamp)	$0.409 \pm 7.16 \times 10^{-4}$	$0.295 \pm 1.03 \times 10^{-4}$	$0.333 \pm 1.06 \times 10^{-5}$	$0.34 \pm 3.49 \times 10^{-6}$
DSR (τ=0.1, unsorted)	0.455 ± 0.00124	$0.472 \pm 6.7 \times 10^{-6}$	$0.689 \pm 8.7 \times 10^{-6}$	$0.729 \pm 1.24 \times 10^{-6}$
DSR (τ=0.5)	$0.785 \pm 9.91 \times 10^{-5}$	$0.521 \pm 6.58 \times 10^{-7}$	$0.731 \pm 4.29 \times 10^{-6}$	0.696
DSR (τ=2.5/N)†	$0.367 \pm 5.65 \times 10^{-4}$	$0.128 \pm 9.28 \times 10^{-5}$	$0.0709 \pm 2.66 \times 10^{-6}$	$0.0297 \pm 4.4 \times 10^{-6}$
Systematic	0.0587 ± 0.00405	0.0231 ± 0.0016	$0.0155 \pm 5.88 \times 10^{-4}$	$0.00764 \pm 3.46 \times 10^{-4}$
Multinomial	0.182 ± 0.0141	0.0808 ± 0.004	0.0542 ± 0.00287	0.0275 ± 0.00108
Systematic, stop-gradient (PyDPF)	0.0633 ± 0.00383	$\mathbf{0.0218 \pm 0.00153}$	$0.0162 \pm 6.04 \times 10^{-4}$	$0.00757 \pm 2.84 \times 10^{-4}$
Soft (α=0.5)	0.232 ± 0.0205	0.102 ± 0.00551	0.0735 ± 0.00383	0.0333 ± 0.00169
OT (PyDPF)	0.118	0.0687	0.0517	–
OT (PFJAX)	0.112	0.0685	0.0521	0.0422
OT (FilterFlow)	0.165	0.0931	0.0813	0.0769
Eigen moment matching	0.3	0.134	0.0889	0.0407
Gaussian moment matching	0.384 ± 0.0107	0.147 ± 0.00362	0.12 ± 0.00239	0.0462 ± 0.00104
Gumbel-softmax	0.318 ± 0.0232	0.181 ± 0.013	0.27 ± 0.041	0.248 ± 0.00815
Variational (KL)	**0.0586**	0.0238	**0.0102**	**0.00597**

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

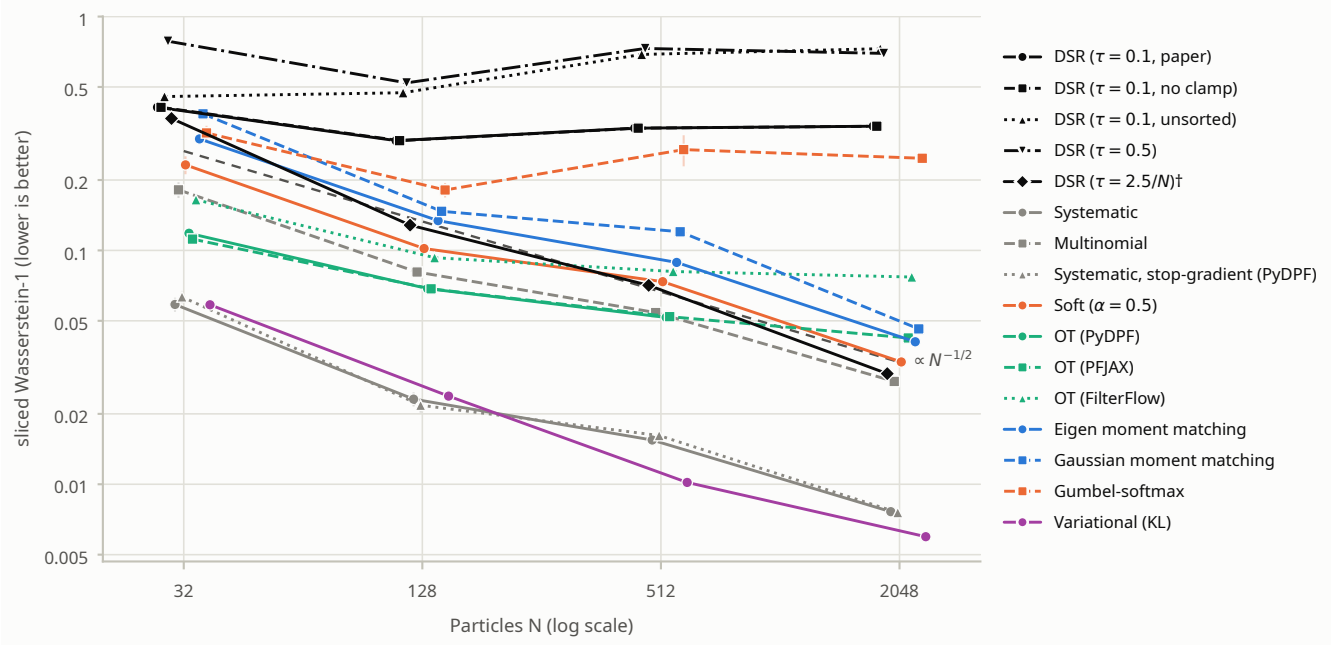


Figure 46: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 47: Skewed-weight cloud: Gaussian-kernel MMD^2 (random features) (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	$0.0585 \pm 3.64 \times 10^{-4}$	$0.0751 \pm 4.26 \times 10^{-5}$	$0.0554 \pm 4.51 \times 10^{-6}$	$0.0722 \pm 1.23 \times 10^{-6}$
DSR (τ=0.1, no clamp)	$0.0585 \pm 3.64 \times 10^{-4}$	$0.0751 \pm 4.26 \times 10^{-5}$	$0.0554 \pm 4.51 \times 10^{-6}$	$0.0722 \pm 1.23 \times 10^{-6}$
DSR (τ=0.1, unsorted)	$0.0648 \pm 3.25 \times 10^{-4}$	$0.112 \pm 1.38 \times 10^{-5}$	$0.114 \pm 1.26 \times 10^{-6}$	0.138
DSR (τ=0.5)	$0.178 \pm 3.07 \times 10^{-5}$	$0.129 \pm 7.71 \times 10^{-6}$	$0.117 \pm 2.36 \times 10^{-6}$	$0.136 \pm 4.76 \times 10^{-7}$
DSR (τ=2.5/N)†	$0.04 \pm 3.39 \times 10^{-4}$	$0.0068 \pm 1.99 \times 10^{-5}$	$8.85 \times 10^{-4} \pm 2.02 \times 10^{-7}$	$1.22 \times 10^{-4} \pm 1.97 \times 10^{-8}$
Systematic	$\mathbf{0.00109 \pm 1.58 \times 10^{-4}}$	$2.81 \times 10^{-4} \pm 4.34 \times 10^{-5}$	$5.77 \times 10^{-5} \pm 6.08 \times 10^{-6}$	$1.86 \times 10^{-5} \pm 2.1 \times 10^{-6}$
Multinomial	0.0141 ± 0.00233	$0.0036 \pm 5.16 \times 10^{-4}$	$7.91 \times 10^{-4} \pm 1.04 \times 10^{-4}$	$2.27 \times 10^{-4} \pm 2.45 \times 10^{-5}$
Systematic, stop-gradient (PyDPF)	$0.00124 \pm 1.47 \times 10^{-4}$	$\mathbf{2.42 \times 10^{-4} \pm 3.94 \times 10^{-5}}$	$6.19 \times 10^{-5} \pm 5.75 \times 10^{-6}$	$1.67 \times 10^{-5} \pm 1.88 \times 10^{-6}$
Soft (α=0.5)	0.024 ± 0.00457	$0.00562 \pm 7.95 \times 10^{-4}$	$0.00151 \pm 1.93 \times 10^{-4}$	$3.42 \times 10^{-4} \pm 4.9 \times 10^{-5}$
OT (PyDPF)	0.00223	5.28×10^{-4}	4.79×10^{-4}	–
OT (PFJAX)	0.00202	5.52×10^{-4}	4.9×10^{-4}	5.36×10^{-4}
OT (FilterFlow)	0.00509	0.00217	0.00193	0.00208
Eigen moment matching	0.00404	0.00685	8.83×10^{-4}	3.13×10^{-4}
Gaussian moment matching	0.0232 ± 0.00272	$0.00845 \pm 8.5 \times 10^{-4}$	$0.00259 \pm 2.21 \times 10^{-4}$	$4.56 \times 10^{-4} \pm 3.73 \times 10^{-5}$
Gumbel-softmax	0.0386 ± 0.00782	0.0215 ± 0.00305	0.0226 ± 0.00341	0.0232 ± 0.00296
Variational (KL)	0.00129	2.7×10^{-4}	$\mathbf{2.41 \times 10^{-5}}$	$\mathbf{1.03 \times 10^{-5}}$

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

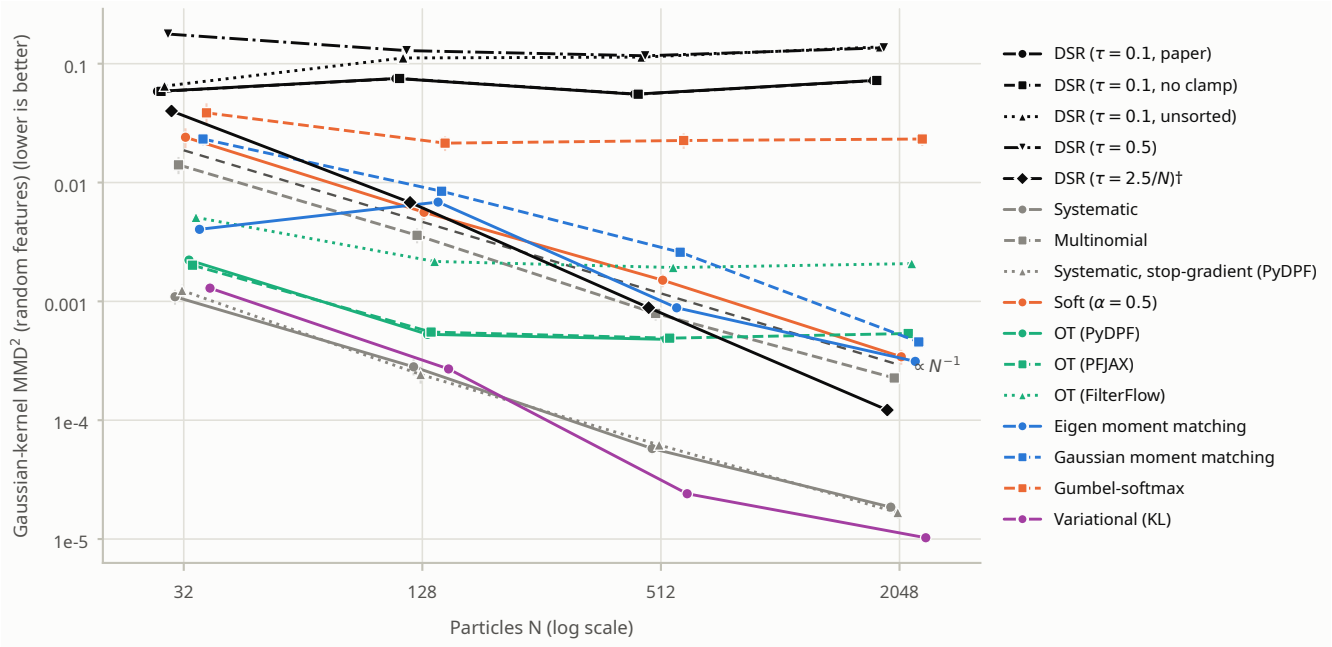


Figure 47: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 48: Skewed-weight cloud: covariance error (Frobenius) (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.994 ± 0.00307	$0.768 \pm 1.5 \times 10^{-5}$	$1.08 \pm 4.66 \times 10^{-5}$	$1.1 \pm 1.81 \times 10^{-6}$
DSR (τ=0.1, no clamp)	0.994 ± 0.00307	$0.768 \pm 1.5 \times 10^{-5}$	$1.08 \pm 4.66 \times 10^{-5}$	$1.1 \pm 1.81 \times 10^{-6}$
DSR (τ=0.1, unsorted)	1.17 ± 0.00151	0.863	1.49	1.39
DSR (τ=0.5)	$1.63 \pm 2.18 \times 10^{-5}$	0.897	1.51	1.38
DSR (τ=2.5/N)†	0.833 ± 0.00357	$0.361 \pm 6.54 \times 10^{-6}$	$0.276 \pm 2.81 \times 10^{-5}$	$0.109 \pm 1.33 \times 10^{-5}$
Systematic	0.0911 ± 0.00551	0.0425 ± 0.00445	0.0347 ± 0.00278	0.0139 ± 0.00104
Multinomial	0.292 ± 0.0317	0.135 ± 0.0122	0.115 ± 0.00965	0.0488 ± 0.00431
Systematic, stop-gradient (PyDPF)	0.0963 ± 0.0062	0.0383 ± 0.00474	0.0353 ± 0.00283	0.0135 ± 0.00118
Soft (α=0.5)	0.392 ± 0.0373	0.17 ± 0.015	0.144 ± 0.0127	0.059 ± 0.00461
OT (PyDPF)	0.185	0.0974	0.132	–
OT (PFJAX)	0.193	0.098	0.133	0.132
OT (FilterFlow)	0.293	0.183	0.254	0.252
Eigen moment matching	$\mathbf{1.12 \times 10^{-15}}$	$\mathbf{8.04 \times 10^{-16}}$	$\mathbf{2.69 \times 10^{-15}}$	$\mathbf{1.13 \times 10^{-15}}$
Gaussian moment matching	0.401 ± 0.0387	0.127 ± 0.0134	0.104 ± 0.00963	0.0503 ± 0.0041
Gumbel-softmax	0.782 ± 0.0964	0.45 ± 0.036	0.767 ± 0.143	0.677 ± 0.0324
Variational (KL)	0.0725	0.0184	0.02	0.00708

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

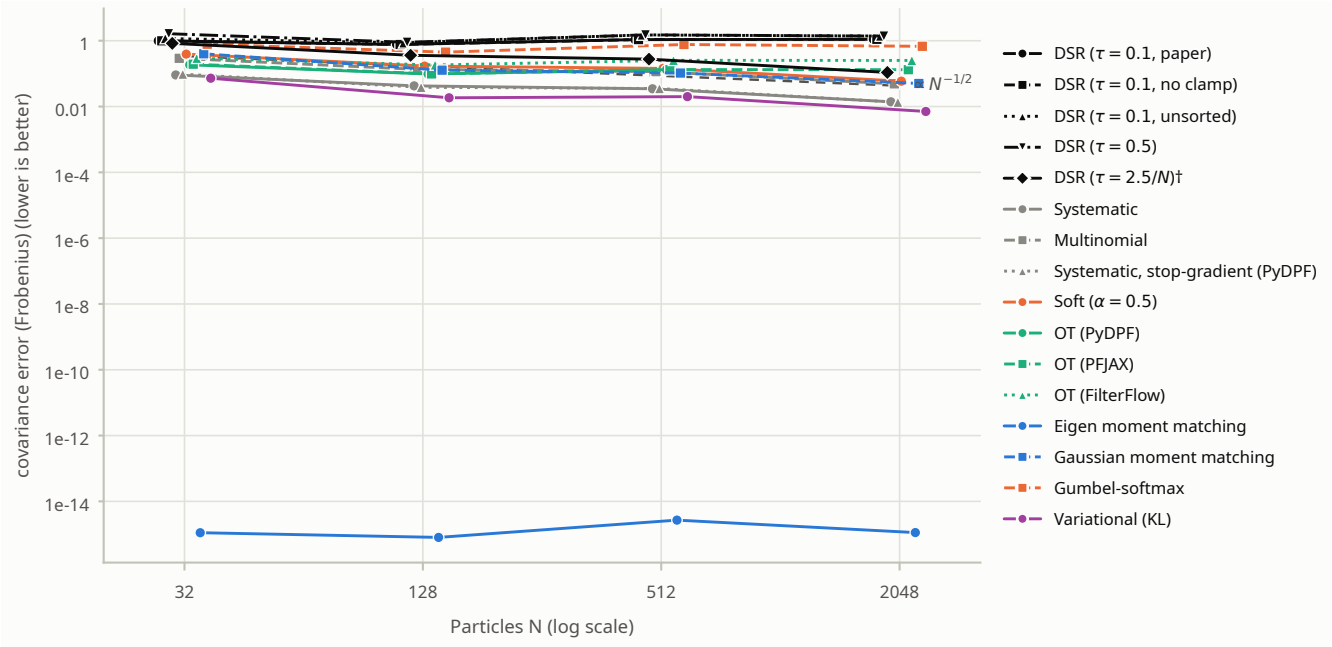


Figure 48: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 49: Skewed-weight cloud: MSE of $\mathbb{E}[x^2]$ (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.45 ± 0.00157	$0.277 \pm 1.25 \times 10^{-5}$	$0.575 \pm 3.78 \times 10^{-5}$	$0.564 \pm 1.22 \times 10^{-5}$
DSR (τ=0.1, no clamp)	0.45 ± 0.00157	$0.277 \pm 1.25 \times 10^{-5}$	$0.575 \pm 3.78 \times 10^{-5}$	$0.564 \pm 1.22 \times 10^{-5}$
DSR (τ=0.1, unsorted)	0.659 ± 0.00398	$0.357 \pm 2.45 \times 10^{-5}$	1.09	0.955
DSR (τ=0.5)	$1.29 \pm 5.55 \times 10^{-4}$	$0.384 \pm 6.25 \times 10^{-7}$	$1.11 \pm 3.27 \times 10^{-6}$	$0.947 \pm 1.98 \times 10^{-6}$
DSR (τ=2.5/N)†	0.32 ± 0.0019	$0.0603 \pm 3.38 \times 10^{-5}$	$0.0269 \pm 6.55 \times 10^{-5}$	$0.00483 \pm 5.03 \times 10^{-7}$
Systematic	$0.00164 \pm 2.04 \times 10^{-4}$	$4.25 \times 10^{-4} \pm 9.98 \times 10^{-5}$	$4.02 \times 10^{-4} \pm 7.96 \times 10^{-5}$	$7.2 \times 10^{-5} \pm 1.2 \times 10^{-5}$
Multinomial	0.0281 ± 0.00838	0.0105 ± 0.00217	0.0048 ± 0.00101	$9.07 \times 10^{-4} \pm 1.72 \times 10^{-4}$
Systematic, stop-gradient (PyDPF)	$0.00185 \pm 4.05 \times 10^{-4}$	$4.67 \times 10^{-4} \pm 1.27 \times 10^{-4}$	$4 \times 10^{-4} \pm 8.57 \times 10^{-5}$	$7.43 \times 10^{-5} \pm 1.49 \times 10^{-5}$
Soft (α=0.5)	0.0426 ± 0.00937	0.0194 ± 0.00388	0.00739 ± 0.00144	$0.00125 \pm 2.17 \times 10^{-4}$
OT (PyDPF)	0.0182	0.00472	0.00864	–
OT (PFJAX)	0.0187	0.00482	0.00869	0.00869
OT (FilterFlow)	0.043	0.0166	0.0319	0.0316
Eigen moment matching	$\mathbf{4.19 \times 10^{-31}}$	$\mathbf{1.11 \times 10^{-31}}$	$\mathbf{5.45 \times 10^{-30}}$	$\mathbf{1.26 \times 10^{-30}}$
Gaussian moment matching	0.0818 ± 0.0181	0.00762 ± 0.00217	$0.00438 \pm 8.69 \times 10^{-4}$	$0.00108 \pm 1.97 \times 10^{-4}$
Gumbel-softmax	0.289 ± 0.0819	0.0988 ± 0.0178	0.28 ± 0.0905	0.228 ± 0.0212
Variational (KL)	2.73×10^{-4}	6.02×10^{-5}	1.53×10^{-4}	2.86×10^{-6}

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

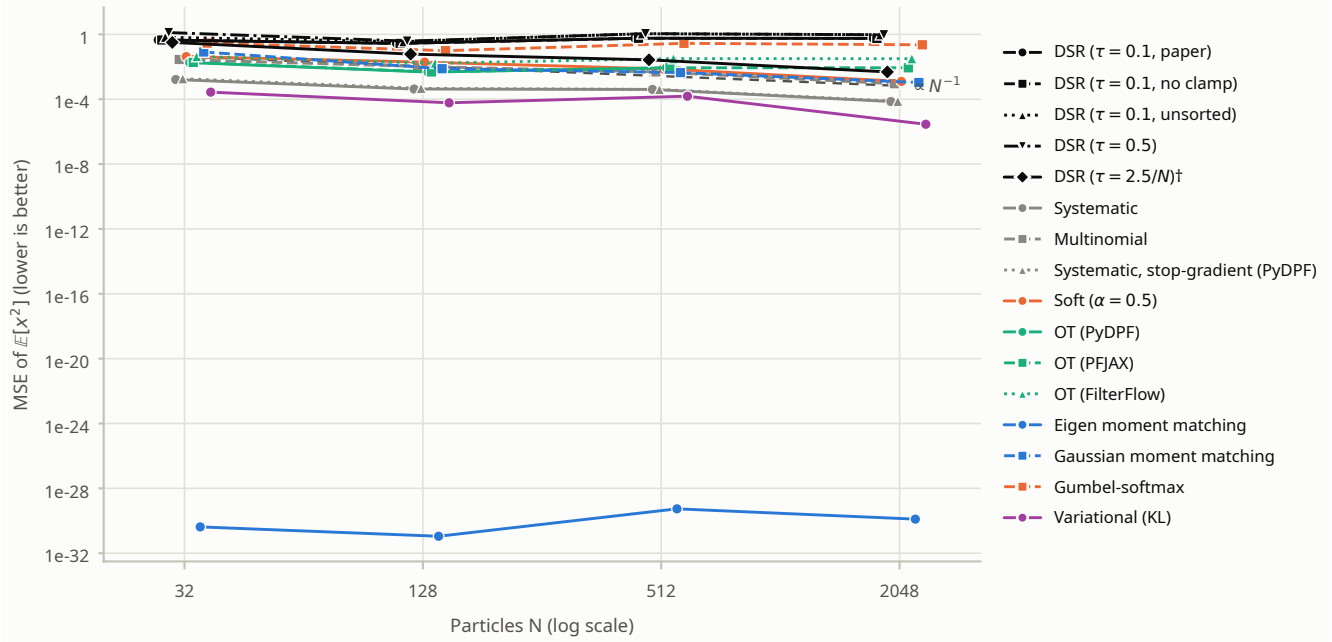


Figure 49: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 50: Fixed-grid cloud: sliced Wasserstein-1 (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.544 ± 0.00107	$0.453 \pm 1.36 \times 10^{-4}$	$0.479 \pm 1.54 \times 10^{-5}$	$0.424 \pm 1.91 \times 10^{-5}$
DSR (τ=0.1, no clamp)	0.544 ± 0.00107	$0.453 \pm 1.36 \times 10^{-4}$	$0.479 \pm 1.54 \times 10^{-5}$	$0.424 \pm 1.91 \times 10^{-5}$
DSR (τ=0.1, unsorted)	$0.318 \pm 8.51 \times 10^{-4}$	$0.349 \pm 1.9 \times 10^{-4}$	$0.39 \pm 4.21 \times 10^{-5}$	$0.382 \pm 1.25 \times 10^{-5}$
DSR (τ=0.5)	$0.883 \pm 8.9 \times 10^{-5}$	$0.887 \pm 3.55 \times 10^{-5}$	$0.92 \pm 6.68 \times 10^{-6}$	$0.909 \pm 1.69 \times 10^{-6}$
DSR (τ=2.5/N)†	0.467 ± 0.00156	$0.16 \pm 9.83 \times 10^{-5}$	$0.0811 \pm 2.42 \times 10^{-5}$	$0.0329 \pm 1.49 \times 10^{-5}$
Systematic	0.079 ± 0.00275	0.0358 ± 0.00137	$\mathbf{0.0189 \pm 5.66 \times 10^{-4}}$	$\mathbf{0.00798 \pm 2.35 \times 10^{-4}}$
Multinomial	0.249 ± 0.016	0.133 ± 0.00801	0.0679 ± 0.00371	0.0358 ± 0.00244
Systematic, stop-gradient (PyDPF)	$\mathbf{0.0769 \pm 0.00296}$	$\mathbf{0.0347 \pm 0.00136}$	$0.0191 \pm 5.84 \times 10^{-4}$	$0.00822 \pm 2.72 \times 10^{-4}$
Soft (α=0.5)	0.218 ± 0.013	0.128 ± 0.0083	0.067 ± 0.0032	0.0319 ± 0.00155
OT (PyDPF)	0.172	0.13	0.117	–
OT (PFJAX)	0.175	0.133	0.122	0.113
OT (FilterFlow)	0.233	0.195	0.197	0.189
Eigen moment matching	0.476	0.343	0.303	0.302
Gaussian moment matching	0.487 ± 0.0168	0.323 ± 0.00851	0.269 ± 0.00393	0.262 ± 0.00218
Gumbel-softmax	0.334 ± 0.0493	0.341 ± 0.0279	0.342 ± 0.0603	0.331 ± 0.0145
Variational (KL)	0.102	0.0957	0.0882	0.088

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

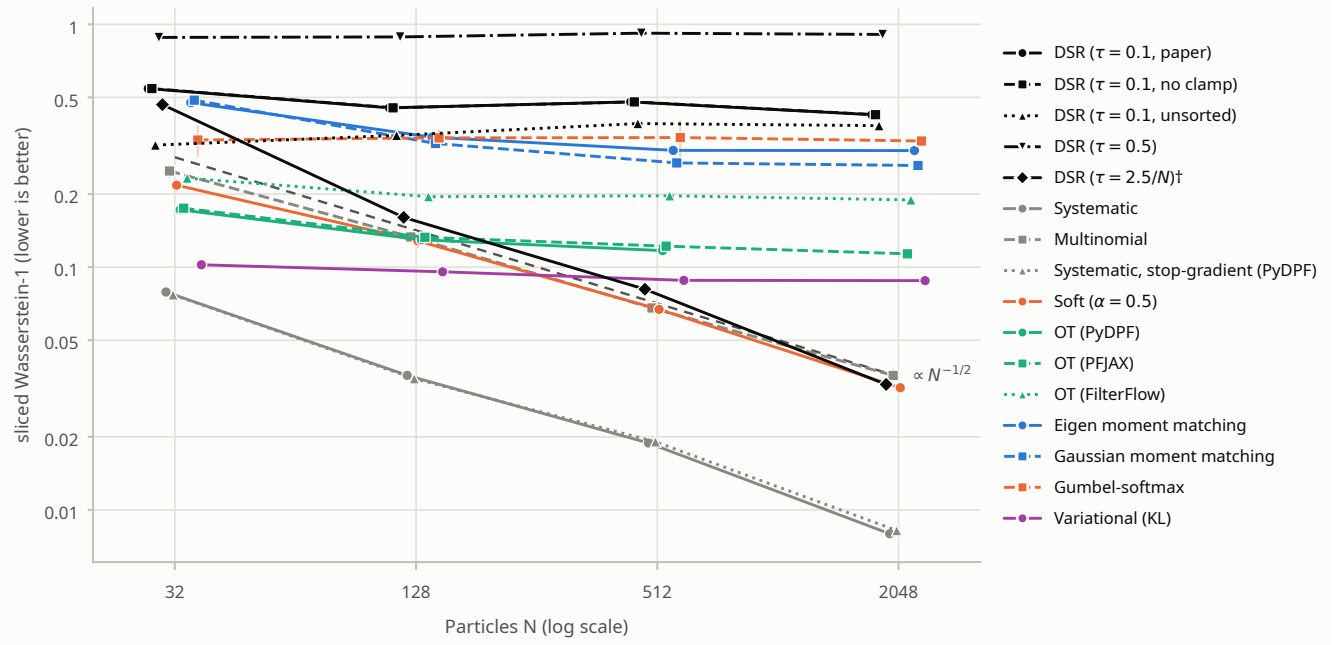


Figure 50: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 51: Fixed-grid cloud: Gaussian-kernel MMD^2 (random features) (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	$0.0993 \pm 7.53 \times 10^{-5}$	$0.06 \pm 1.74 \times 10^{-5}$	$0.0703 \pm 2.53 \times 10^{-6}$	$0.0574 \pm 9.08 \times 10^{-7}$
DSR (τ=0.1, no clamp)	$0.0993 \pm 7.53 \times 10^{-5}$	$0.06 \pm 1.74 \times 10^{-5}$	$0.0703 \pm 2.53 \times 10^{-6}$	$0.0574 \pm 9.08 \times 10^{-7}$
DSR (τ=0.1, unsorted)	$0.0231 \pm 2.27 \times 10^{-4}$	$0.0291 \pm 3.44 \times 10^{-5}$	$0.0376 \pm 3.76 \times 10^{-6}$	$0.0376 \pm 1.32 \times 10^{-6}$
DSR (τ=0.5)	$0.161 \pm 5.05 \times 10^{-5}$	$0.137 \pm 2.98 \times 10^{-5}$	$0.143 \pm 5.36 \times 10^{-6}$	$0.14 \pm 1.46 \times 10^{-6}$
DSR (τ=2.5/N)†	$0.0765 \pm 2.23 \times 10^{-4}$	$0.00558 \pm 2.35 \times 10^{-5}$	$8.6 \times 10^{-4} \pm 1.88 \times 10^{-6}$	$1.07 \times 10^{-4} \pm 7.47 \times 10^{-8}$
Systematic	$6.64 \times 10^{-4} \pm 6.21 \times 10^{-5}$	$1.63 \times 10^{-4} \pm 1.9 \times 10^{-5}$	$\mathbf{8.24 \times 10^{-5} \pm 1.04 \times 10^{-5}}$	$\mathbf{9.86 \times 10^{-6} \pm 1.51 \times 10^{-6}}$
Multinomial	0.0139 ± 0.00228	$0.00306 \pm 4.1 \times 10^{-4}$	$8.01 \times 10^{-4} \pm 1.07 \times 10^{-4}$	$2.17 \times 10^{-4} \pm 3.64 \times 10^{-5}$
Systematic, stop-gradient (PyDPF)	$\mathbf{6.35 \times 10^{-4} \pm 6.73 \times 10^{-5}}$	$\mathbf{1.59 \times 10^{-4} \pm 1.9 \times 10^{-5}}$	$8.27 \times 10^{-5} \pm 1.08 \times 10^{-5}$	$1.01 \times 10^{-5} \pm 1.34 \times 10^{-6}$
Soft (α=0.5)	0.0128 ± 0.00227	$0.00351 \pm 5.88 \times 10^{-4}$	$9.07 \times 10^{-4} \pm 1.36 \times 10^{-4}$	$1.83 \times 10^{-4} \pm 2.47 \times 10^{-5}$
OT (PyDPF)	0.00381	0.00314	0.00325	–
OT (PFJAX)	0.00413	0.00338	0.00353	0.00359
OT (FilterFlow)	0.0111	0.00911	0.0101	0.0104
Eigen moment matching	0.0221	0.0217	0.0174	0.0182
Gaussian moment matching	0.031 ± 0.00361	0.0191 ± 0.00147	$0.013 \pm 5.48 \times 10^{-4}$	$0.0127 \pm 2.95 \times 10^{-4}$
Gumbel-softmax	0.0357 ± 0.0155	0.0312 ± 0.00447	0.0267 ± 0.00938	$0.0267 \pm 7.66 \times 10^{-4}$
Variational (KL)	0.00196	0.00195	0.00211	0.00207

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

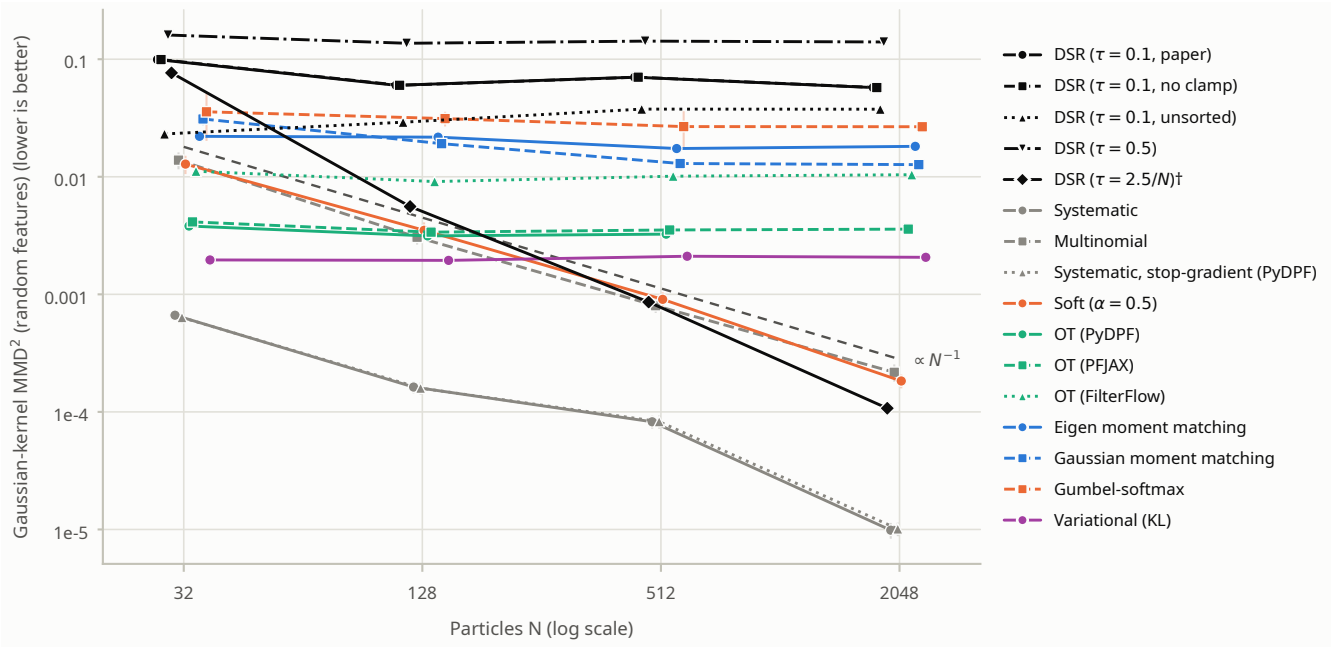


Figure 51: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 52: Fixed-grid cloud: covariance error (Frobenius) (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	2.19 ± 0.00304	$1.79 \pm 9.46 \times 10^{-4}$	$1.97 \pm 1.93 \times 10^{-4}$	$1.82 \pm 6.96 \times 10^{-5}$
DSR (τ=0.1, no clamp)	2.19 ± 0.00304	$1.79 \pm 9.46 \times 10^{-4}$	$1.97 \pm 1.93 \times 10^{-4}$	$1.82 \pm 6.96 \times 10^{-5}$
DSR (τ=0.1, unsorted)	1.24 ± 0.00495	1.32 ± 0.00119	$1.49 \pm 2.63 \times 10^{-4}$	$1.46 \pm 6.68 \times 10^{-5}$
DSR (τ=0.5)	$2.45 \pm 7.6 \times 10^{-6}$	$2.36 \pm 3.95 \times 10^{-6}$	2.45	2.43
DSR (τ=2.5/N)†	2 ± 0.00551	0.622 ± 0.00408	0.322 ± 0.00107	$0.125 \pm 3.1 \times 10^{-4}$
Systematic	0.348 ± 0.0321	0.145 ± 0.0147	0.0556 ± 0.00628	0.0255 ± 0.00282
Multinomial	0.752 ± 0.0811	0.39 ± 0.0417	0.192 ± 0.019	0.0961 ± 0.00969
Systematic, stop-gradient (PyDPF)	0.338 ± 0.0334	0.137 ± 0.0144	0.0587 ± 0.00659	0.0294 ± 0.00326
Soft (α=0.5)	0.485 ± 0.0598	0.28 ± 0.0278	0.139 ± 0.0111	0.0699 ± 0.00616
OT (PyDPF)	0.502	0.478	0.528	–
OT (PFJAX)	0.535	0.513	0.56	0.55
OT (FilterFlow)	0.812	0.797	0.884	0.871
Eigen moment matching	$\mathbf{1.08 \times 10^{-15}}$	$\mathbf{1.88 \times 10^{-15}}$	$\mathbf{9.16 \times 10^{-16}}$	$\mathbf{9.49 \times 10^{-16}}$
Gaussian moment matching	0.574 ± 0.0586	0.296 ± 0.0325	0.161 ± 0.0157	0.0841 ± 0.00765
Gumbel-softmax	1.05 ± 0.36	1.15 ± 0.138	1.25 ± 0.246	1.25 ± 0.0701
Variational (KL)	0.762	0.807	0.789	0.789

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

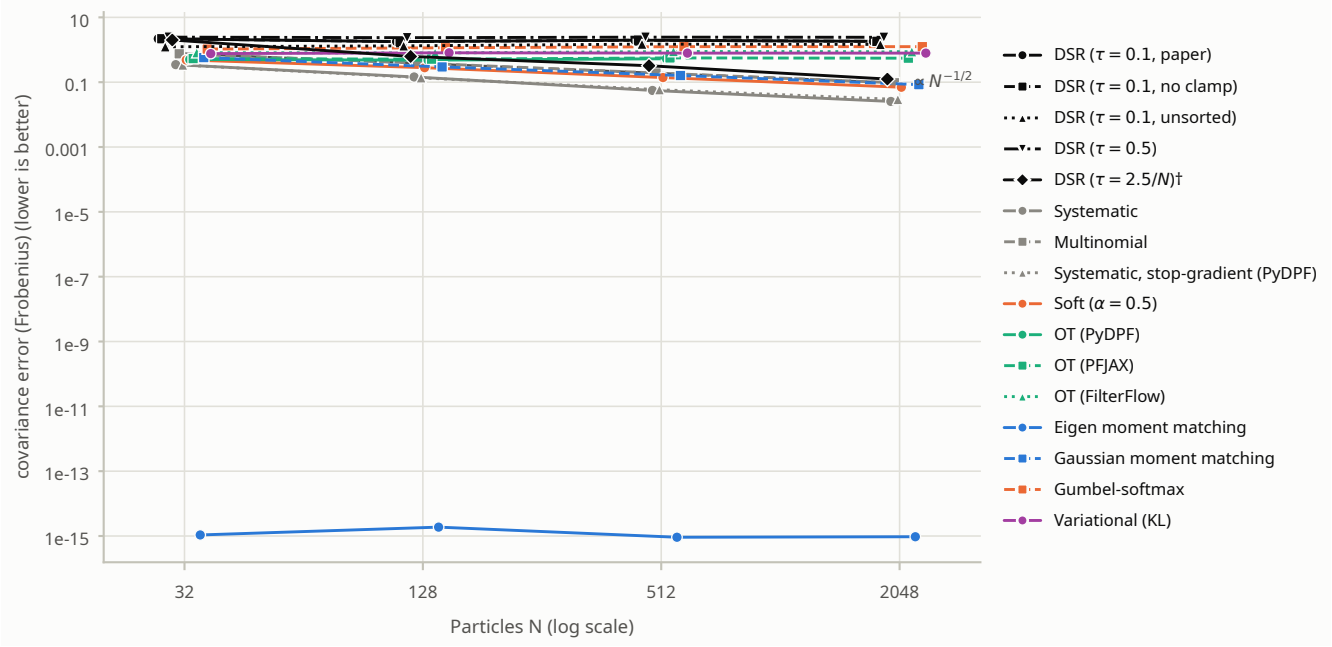


Figure 52: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 53: Fixed-grid cloud: MSE of $\mathbb{E}[x^2]$ (lower is better). D=2; mean ± 95% CI over 100 resampling draws (10 for OT and Gumbel at $N \leq 128$, 3 above).

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	1.44 ± 0.0184	1.15 ± 0.00725	1.36 ± 0.00177	$1.19 \pm 3.95 \times 10^{-4}$
DSR (τ=0.1, no clamp)	1.44 ± 0.0184	1.15 ± 0.00725	1.36 ± 0.00177	$1.19 \pm 3.95 \times 10^{-4}$
DSR (τ=0.1, unsorted)	0.489 ± 0.022	0.621 ± 0.00515	0.873 ± 0.00126	$0.833 \pm 3.14 \times 10^{-4}$
DSR (τ=0.5)	2.43 ± 0.00341	2.5 ± 0.00203	$2.74 \pm 4.43 \times 10^{-4}$	$2.55 \pm 1.15 \times 10^{-4}$
DSR (τ=2.5/N)†	1.23 ± 0.0195	0.166 ± 0.00292	$0.0394 \pm 2.96 \times 10^{-4}$	$0.00652 \pm 2.21 \times 10^{-5}$
Systematic	$0.00356 \pm 5.79 \times 10^{-4}$	$3.36 \times 10^{-4} \pm 7.87 \times 10^{-5}$	$2.85 \times 10^{-4} \pm 4.92 \times 10^{-5}$	$4.98 \times 10^{-5} \pm 1.26 \times 10^{-5}$
Multinomial	0.479 ± 0.133	0.102 ± 0.0286	0.0238 ± 0.00569	0.00727 ± 0.00216
Systematic, stop-gradient (PyDPF)	$0.00345 \pm 5.37 \times 10^{-4}$	$3.51 \times 10^{-4} \pm 7.61 \times 10^{-5}$	$3.38 \times 10^{-4} \pm 5.7 \times 10^{-5}$	$5.38 \times 10^{-5} \pm 1.25 \times 10^{-5}$
Soft (α=0.5)	0.522 ± 0.14	0.134 ± 0.0367	0.0347 ± 0.0078	0.00745 ± 0.00174
OT (PyDPF)	0.134	0.132	0.157	–
OT (PFJAX)	0.136	0.134	0.16	0.154
OT (FilterFlow)	0.311	0.317	0.391	0.38
Eigen moment matching	$\mathbf{1.6 \times 10^{-30}}$	$\mathbf{6.71 \times 10^{-30}}$	$\mathbf{6.41 \times 10^{-30}}$	$\mathbf{7.06 \times 10^{-30}}$
Gaussian moment matching	0.857 ± 0.243	0.286 ± 0.0949	0.0552 ± 0.0147	0.0153 ± 0.00475
Gumbel-softmax	1.09 ± 0.448	1.02 ± 0.345	0.738 ± 0.565	0.798 ± 0.34
Variational (KL)	0.0243	0.0144	0.0248	0.0229

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

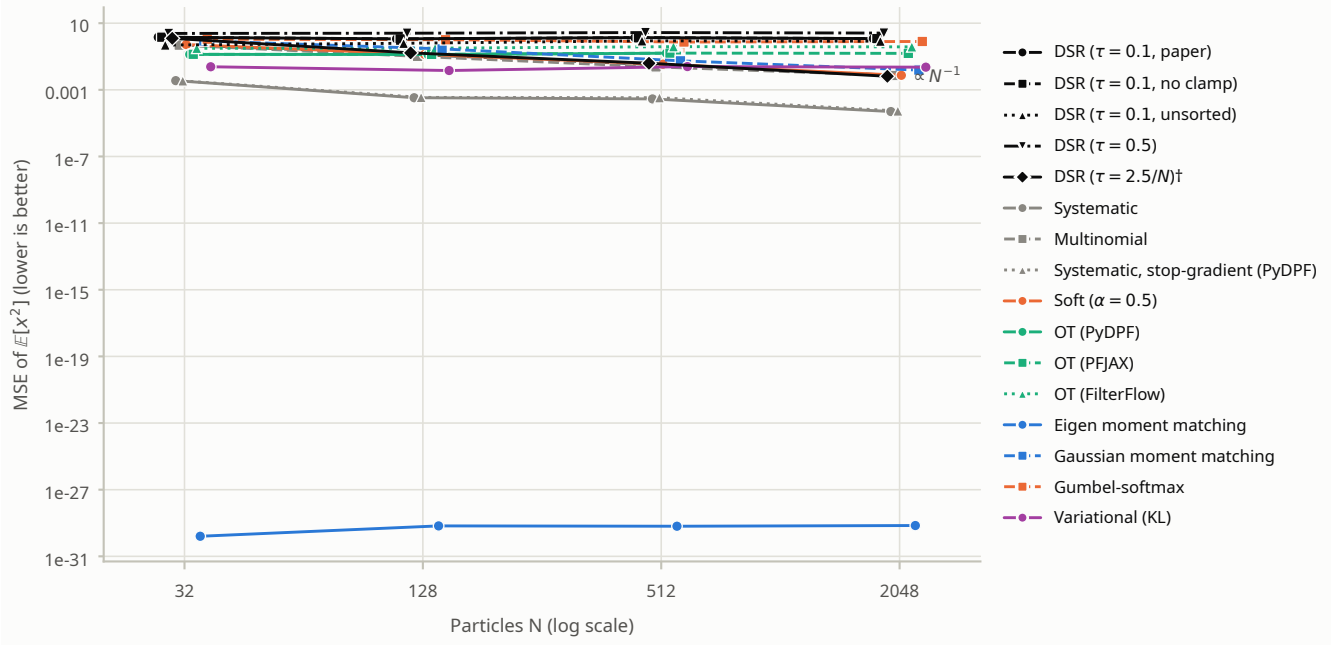


Figure 53: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Local gradients of a resampled expectation

Table 54: Local gradient, gaussian cloud, $f(x) = \|x\|^2$: gradient bias (lower is better). Gaussian cloud, D=2, $f(x) = \|x\|^2$; derivative of $\sum_j v_j f(y_j)$ w.r.t. a weight tilt, against the exact weighted-measure derivative; 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512
DSR (τ=0.1, paper)	0.891	0.226	0.0824
DSR (τ=0.1, no clamp)	0.891	0.226	0.0824
DSR (τ=0.1, unsorted)	1.11	0.374	0.225
DSR (τ=0.5)	1.17	0.372	0.198
DSR (τ=2.5/N)†	0.722	0.121	0.0131
Systematic (no gradient)	0.947	0.557	1.46
Systematic, stop-gradient (PyDPF)	0.00754	0.0187	6.07×10^{-4}
Systematic, stop-gradient (Cuthbert)	0.0296	0.0172	0.0038
Soft (α=0.5)	0.584	0.414	1.02
OT (PyDPF)	0.935	0.55	–
OT (PFJAX)	0.164	0.0284	–
OT (FilterFlow)	0.912	0.447	–
Eigen moment matching	$\mathbf{5.55 \times 10^{-16}}$	$\mathbf{2.22 \times 10^{-16}}$	$\mathbf{2.22 \times 10^{-16}}$
Gaussian moment matching	0.00132	0.0106	0.0015
Gumbel-softmax	0.513	0.174	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

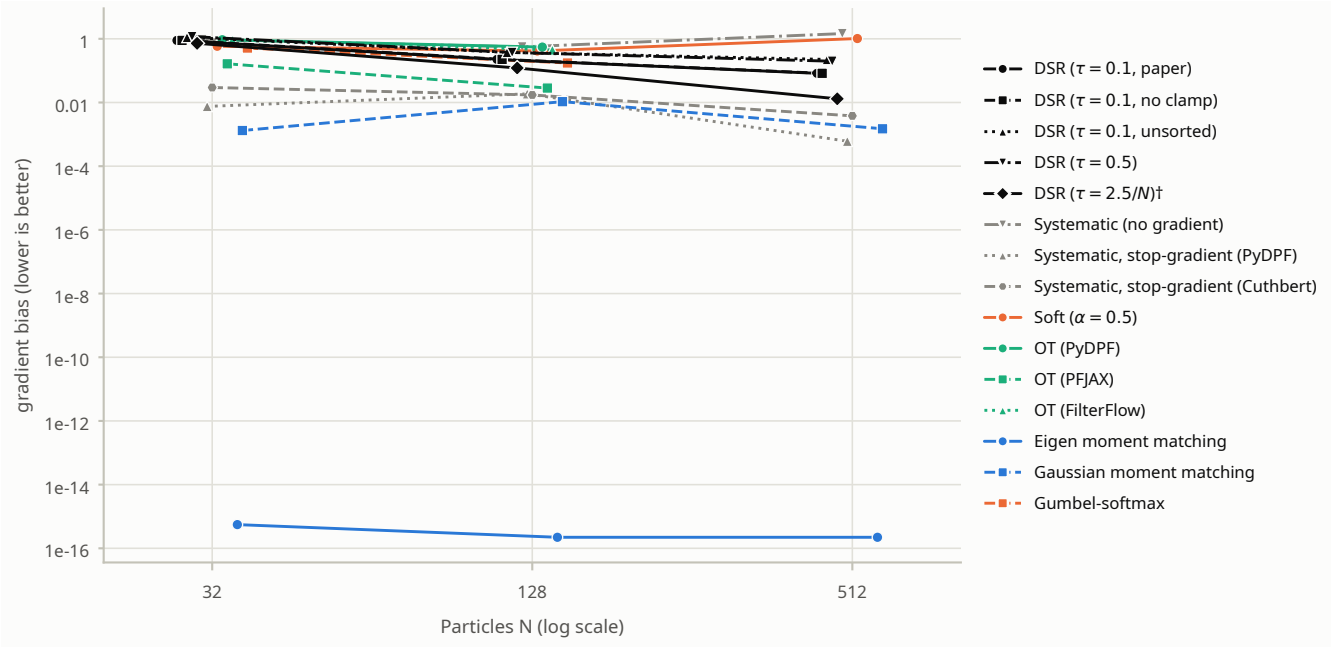


Figure 54: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis.

Table 55: Local gradient, gaussian cloud, $f(x) = \|x\|^2$: gradient variance (lower is better). Gaussian cloud, D=2, $f(x) = \|x\|^2$; derivative of $\sum_j v_j f(y_j)$ w.r.t. a weight tilt, against the exact weighted-measure derivative; 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512
DSR (τ=0.1, paper)	4.05×10^{-4}	2.67×10^{-5}	5.32×10^{-6}
DSR (τ=0.1, no clamp)	4.05×10^{-4}	2.67×10^{-5}	5.32×10^{-6}
DSR (τ=0.1, unsorted)	1.54×10^{-6}	1.73×10^{-7}	1.44×10^{-8}
DSR (τ=0.5)	2.51×10^{-6}	2.65×10^{-7}	1.03×10^{-7}
DSR (τ=2.5/N)†	8.93×10^{-4}	1.35×10^{-4}	2.34×10^{-6}
Systematic (no gradient)	**0**	**0**	**0**
Systematic, stop-gradient (PyDPF)	0.116	0.017	0.0028
Systematic, stop-gradient (Cuthbert)	0.117	0.0139	0.00262
Soft (α=0.5)	0.0329	0.00472	0.00347
OT (PyDPF)	**0**	**0**	–
OT (PFJAX)	**0**	**0**	–
OT (FilterFlow)	**0**	**0**	–
Eigen moment matching	**0**	**0**	**0**
Gaussian moment matching	0.126	0.0155	0.00779
Gumbel-softmax	0.133	0.00617	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

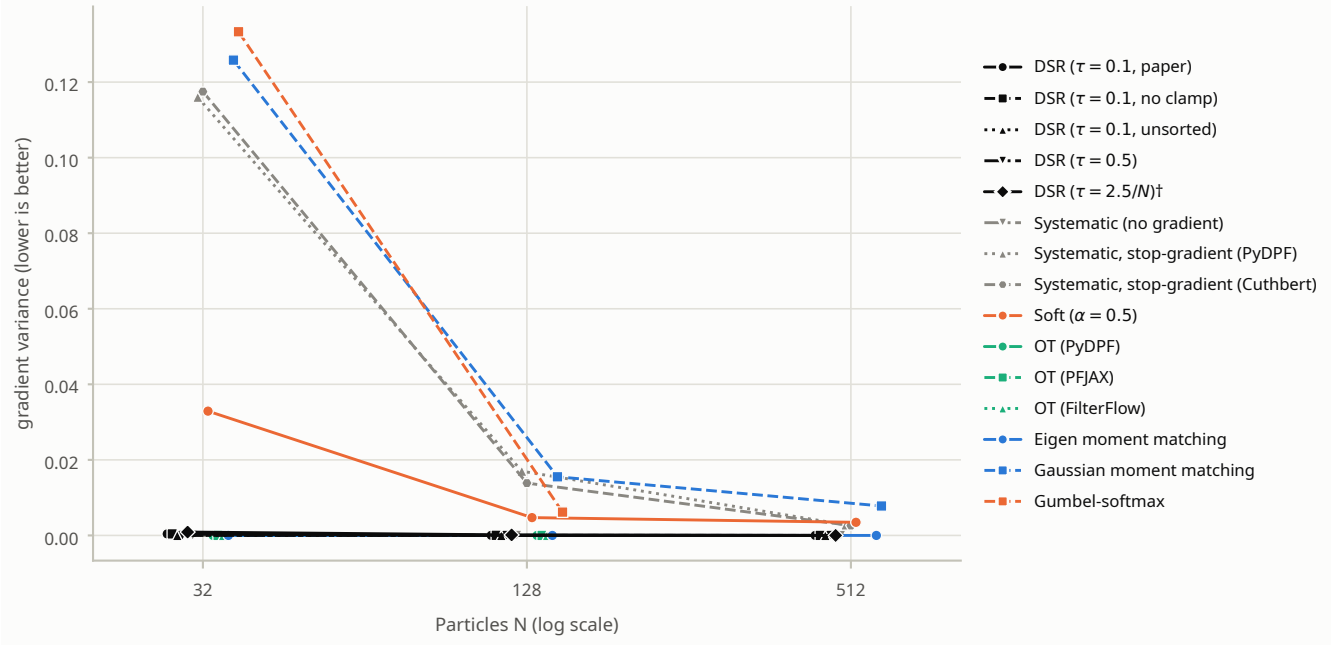


Figure 55: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family.

Table 56: Local gradient, gaussian cloud, $f(x) = \|x\|^2$: gradient MSE (lower is better). Gaussian cloud, D=2, $f(x) = \|x\|^2$; derivative of $\sum_j v_j f(y_j)$ w.r.t. a weight tilt, against the exact weighted-measure derivative; 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512
DSR (τ=0.1, paper)	0.795 ± 0.00709	$0.0511 \pm 4.62 \times 10^{-4}$	$0.0068 \pm 7.56 \times 10^{-5}$
DSR (τ=0.1, no clamp)	0.795 ± 0.00709	$0.0511 \pm 4.62 \times 10^{-4}$	$0.0068 \pm 7.56 \times 10^{-5}$
DSR (τ=0.1, unsorted)	$1.23 \pm 5.47 \times 10^{-4}$	$0.14 \pm 6.18 \times 10^{-5}$	$0.0506 \pm 1.07 \times 10^{-5}$
DSR (τ=0.5)	$1.37 \pm 7.37 \times 10^{-4}$	$0.139 \pm 7.6 \times 10^{-5}$	$0.0391 \pm 2.52 \times 10^{-5}$
DSR (τ=2.5/N)†	0.522 ± 0.00852	$0.0149 \pm 5.54 \times 10^{-4}$	$1.73 \times 10^{-4} \pm 7.85 \times 10^{-6}$
Systematic (no gradient)	0.896	0.31	2.13
Systematic, stop-gradient (PyDPF)	0.115 ± 0.0385	0.0172 ± 0.00502	$0.00277 \pm 4.99 \times 10^{-4}$
Systematic, stop-gradient (Cuthbert)	0.117 ± 0.0391	0.014 ± 0.00449	$0.00261 \pm 5.09 \times 10^{-4}$
Soft (α=0.5)	0.373 ± 0.0471	0.176 ± 0.0114	1.04 ± 0.0239
OT (PyDPF)	0.875	0.303	–
OT (PFJAX)	0.0269	8.06×10^{-4}	–
OT (FilterFlow)	0.832	0.2	–
Eigen moment matching	$\mathbf{3.08 \times 10^{-31}}$	$\mathbf{4.93 \times 10^{-32}}$	$\mathbf{4.93 \times 10^{-32}}$
Gaussian moment matching	0.125 ± 0.0361	0.0155 ± 0.00444	0.00772 ± 0.00204
Gumbel-softmax	0.383 ± 0.223	0.0358 ± 0.022	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

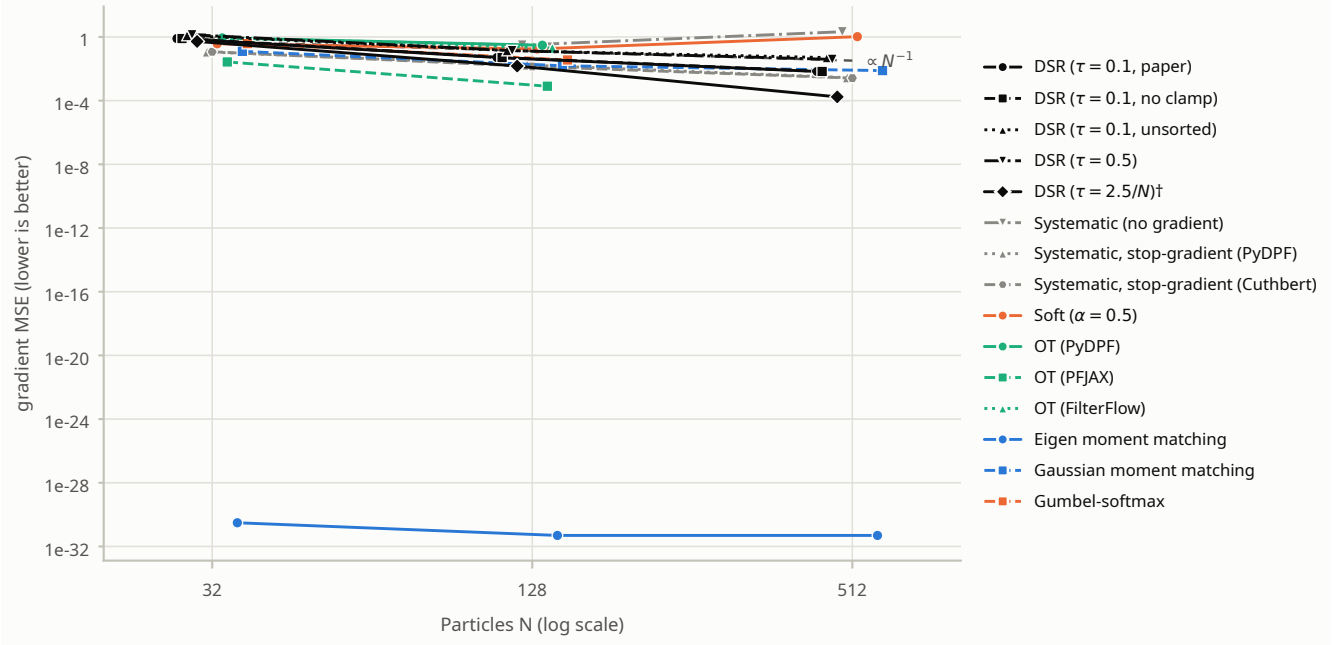


Figure 56: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 57: Local gradient, gaussian cloud, $f(x) = \sum_k \sin x_k$: gradient bias (lower is better). Gaussian cloud, D=2, $f(x) = \sum_k \sin x_k$; derivative of $\sum_j v_j f(y_j)$ w.r.t. a weight tilt, against the exact weighted-measure derivative; 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512
DSR (τ=0.1, paper)	0.0807	0.127	0.171
DSR (τ=0.1, no clamp)	0.0807	0.127	0.171
DSR (τ=0.1, unsorted)	0.0234	0.163	0.259
DSR (τ=0.5)	0.0121	0.153	0.265
DSR (τ=2.5/N)†	0.0711	0.0242	0.019
Systematic (no gradient)	0.497	0.412	0.553
Systematic, stop-gradient (PyDPF)	0.032	0.00462	0.00292
Systematic, stop-gradient (Cuthbert)	**0.0035**	0.00208	**0.00102**
Soft (α=0.5)	0.205	0.182	0.259
OT (PyDPF)	0.444	0.369	–
OT (PFJAX)	0.00945	0.0143	–
OT (FilterFlow)	0.386	0.256	–
Eigen moment matching	0.0141	0.00871	0.0273
Gaussian moment matching	0.0485	**0.00183**	0.0399
Gumbel-softmax	0.0097	0.0989	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

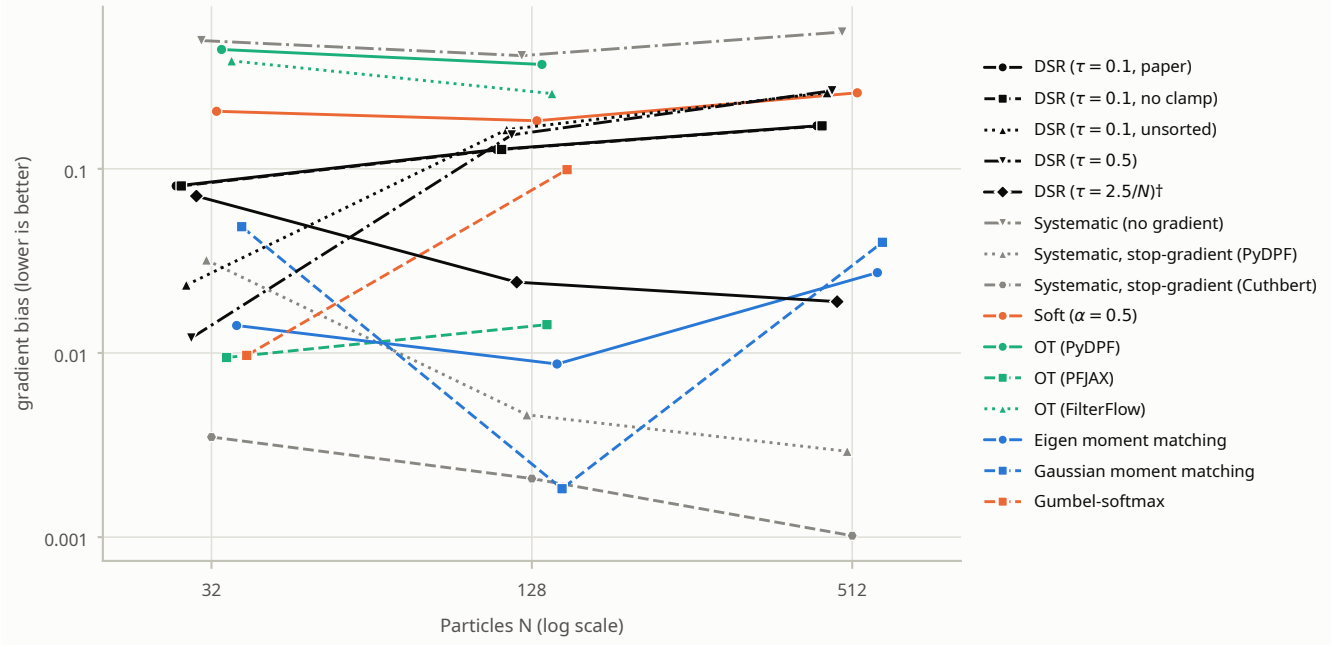


Figure 57: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis.

Table 58: Local gradient, gaussian cloud, $f(x) = \sum_k \sin x_k$: gradient variance (lower is better). Gaussian cloud, D=2, $f(x) = \sum_k \sin x_k$; derivative of $\sum_j v_j f(y_j)$ w.r.t. a weight tilt, against the exact weighted-measure derivative; 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512
DSR (τ=0.1, paper)	1.07×10^{-4}	4.37×10^{-6}	1.35×10^{-7}
DSR (τ=0.1, no clamp)	1.07×10^{-4}	4.37×10^{-6}	1.35×10^{-7}
DSR (τ=0.1, unsorted)	2.91×10^{-6}	3.4×10^{-8}	3.03×10^{-8}
DSR (τ=0.5)	5.09×10^{-6}	1.72×10^{-7}	3.56×10^{-9}
DSR (τ=2.5/N)†	1.07×10^{-4}	3.11×10^{-6}	1.68×10^{-7}
Systematic (no gradient)	**0**	**0**	**0**
Systematic, stop-gradient (PyDPF)	0.0143	0.00169	2.57×10^{-4}
Systematic, stop-gradient (Cuthbert)	0.0209	0.00149	2.36×10^{-4}
Soft (α=0.5)	0.00968	0.00133	3.54×10^{-4}
OT (PyDPF)	**0**	**0**	–
OT (PFJAX)	**0**	**0**	–
OT (FilterFlow)	**0**	**0**	–
Eigen moment matching	**0**	**0**	**0**
Gaussian moment matching	0.0125	0.00187	4.27×10^{-4}
Gumbel-softmax	0.0554	0.00884	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

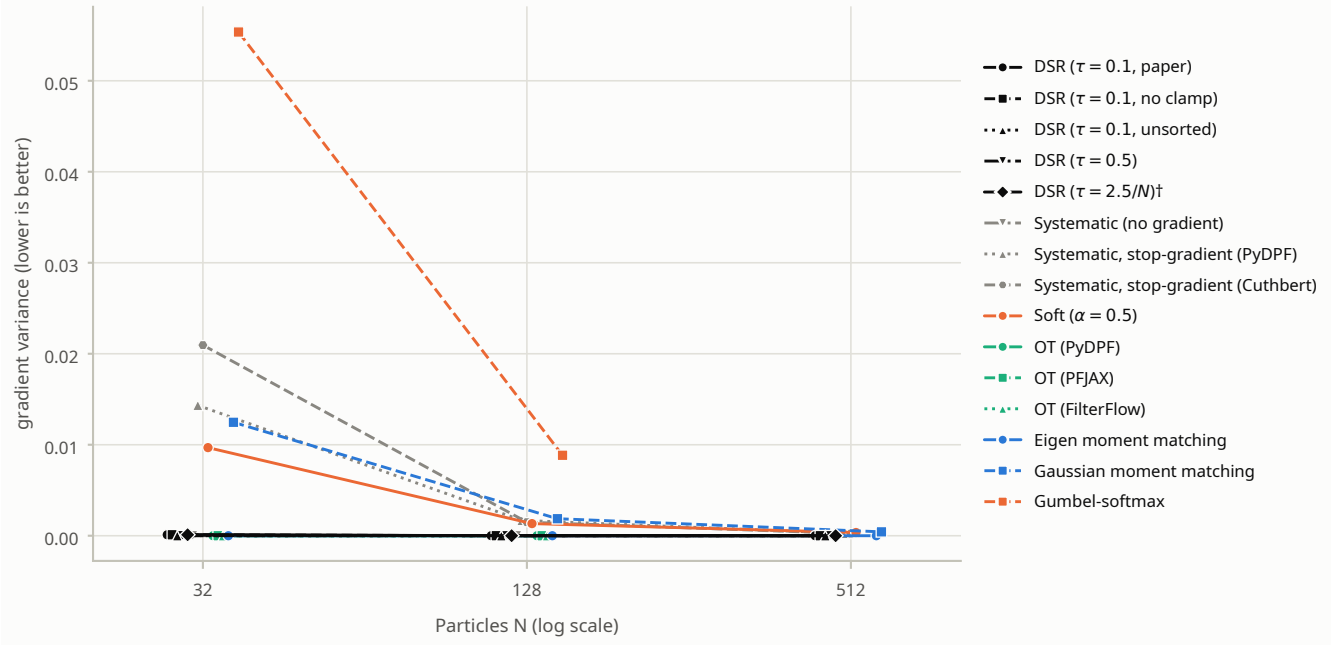


Figure 58: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family.

Table 59: Local gradient, gaussian cloud, $f(x) = \sum_k \sin x_k$: gradient MSE (lower is better). Gaussian cloud, D=2, $f(x) = \sum_k \sin x_k$; derivative of $\sum_j v_j f(y_j)$ w.r.t. a weight tilt, against the exact weighted-measure derivative; 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512
DSR (τ=0.1, paper)	$0.00662 \pm 3.36 \times 10^{-4}$	$0.0162 \pm 1.06 \times 10^{-4}$	$0.0293 \pm 2.5 \times 10^{-5}$
DSR (τ=0.1, no clamp)	$0.00662 \pm 3.36 \times 10^{-4}$	$0.0162 \pm 1.06 \times 10^{-4}$	$0.0293 \pm 2.5 \times 10^{-5}$
DSR (τ=0.1, unsorted)	$5.49 \times 10^{-4} \pm 1.57 \times 10^{-5}$	$0.0266 \pm 1.19 \times 10^{-5}$	$0.0671 \pm 1.79 \times 10^{-5}$
DSR (τ=0.5)	$1.51 \times 10^{-4} \pm 1.11 \times 10^{-5}$	$0.0233 \pm 2.51 \times 10^{-5}$	$0.0704 \pm 6.28 \times 10^{-6}$
DSR (τ=2.5/N)†	$0.00517 \pm 2.97 \times 10^{-4}$	$5.91 \times 10^{-4} \pm 1.71 \times 10^{-5}$	$3.61 \times 10^{-4} \pm 3.1 \times 10^{-6}$
Systematic (no gradient)	0.247	0.169	0.306
Systematic, stop-gradient (PyDPF)	0.0152 ± 0.00341	$0.00169 \pm 3.99 \times 10^{-4}$	$2.63 \times 10^{-4} \pm 4.92 \times 10^{-5}$
Systematic, stop-gradient (Cuthbert)	0.0208 ± 0.00437	$0.00148 \pm 3.5 \times 10^{-4}$	$\mathbf{2.35 \times 10^{-4} \pm 5.24 \times 10^{-5}}$
Soft (α=0.5)	0.0517 ± 0.00786	0.0346 ± 0.00263	0.0672 ± 0.00194
OT (PyDPF)	0.198	0.136	–
OT (PFJAX)	$\mathbf{8.93 \times 10^{-5}}$	2.03×10^{-4}	–
OT (FilterFlow)	0.149	0.0656	–
Eigen moment matching	1.99×10^{-4}	$\mathbf{7.59 \times 10^{-5}}$	7.45×10^{-4}
Gaussian moment matching	0.0147 ± 0.00468	$0.00186 \pm 5.64 \times 10^{-4}$	$0.00201 \pm 3.41 \times 10^{-4}$
Gumbel-softmax	0.0499 ± 0.0473	0.0177 ± 0.0152	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

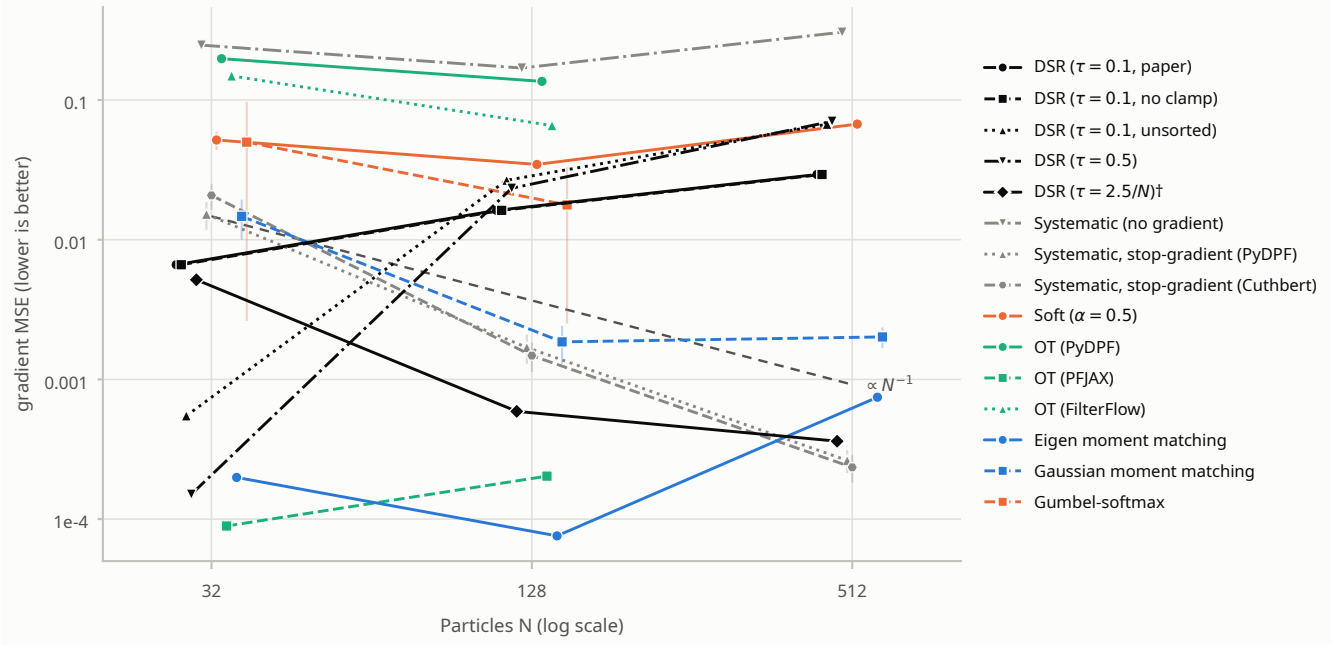


Figure 59: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 60: Local gradient, bimodal mixture cloud, $f(x) = \|x\|^2$: gradient bias (lower is better). Bimodal mixture cloud, D=2, $f(x) = \|x\|^2$; derivative of $\sum_j v_j f(y_j)$ w.r.t. a weight tilt, against the exact weighted-measure derivative; 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512
DSR (τ=0.1, paper)	1.73	1.6	0.934
DSR (τ=0.1, no clamp)	1.73	1.6	0.934
DSR (τ=0.1, unsorted)	12.4	15.6	14.4
DSR (τ=0.5)	13.9	16.8	15.4
DSR (τ=2.5/N)†	1.01	0.199	0.0626
Systematic (no gradient)	1.79	0.779	0.384
Systematic, stop-gradient (PyDPF)	0.157	0.0533	0.0447
Systematic, stop-gradient (Cuthbert)	0.157	0.0533	0.0447
Soft (α=0.5)	1.05	0.501	0.301
OT (PyDPF)	1.84	1.07	–
OT (PFJAX)	4.46	8.18	–
OT (FilterFlow)	4.54	2.4	–
Eigen moment matching	$\mathbf{1.13 \times 10^{-14}}$	$\mathbf{9.1 \times 10^{-15}}$	$\mathbf{3.16 \times 10^{-15}}$
Gaussian moment matching	0.109	0.379	0.0107
Gumbel-softmax	2.69	7.48	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

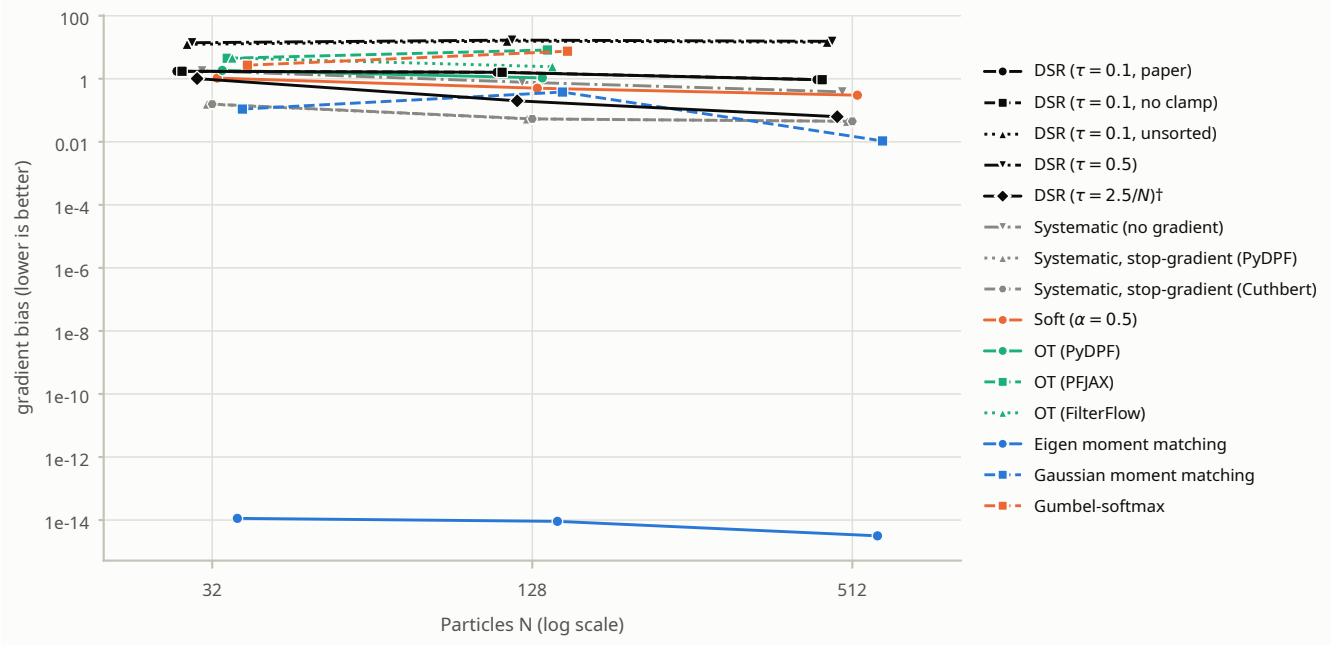


Figure 60: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis.

Table 61: Local gradient, bimodal mixture cloud, $f(x) = \|x\|^2$: gradient variance (lower is better). Bimodal mixture cloud, D=2, $f(x) = \|x\|^2$; derivative of $\sum_j v_j f(y_j)$ w.r.t. a weight tilt, against the exact weighted-measure derivative; 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512
DSR (τ=0.1, paper)	0.0498	0.00631	4.18×10^{-4}
DSR (τ=0.1, no clamp)	0.0498	0.00631	4.18×10^{-4}
DSR (τ=0.1, unsorted)	0.00123	2.39×10^{-6}	2.56×10^{-9}
DSR (τ=0.5)	0.0166	0.00135	7.33×10^{-5}
DSR (τ=2.5/N)†	0.0224	5.14×10^{-4}	4.18×10^{-5}
Systematic (no gradient)	**0**	**0**	**0**
Systematic, stop-gradient (PyDPF)	0.247	0.107	0.386
Systematic, stop-gradient (Cuthbert)	0.247	0.107	0.386
Soft (α=0.5)	0.725	0.11	0.0427
OT (PyDPF)	**0**	**0**	–
OT (PFJAX)	**0**	**0**	–
OT (FilterFlow)	**0**	**0**	–
Eigen moment matching	**0**	**0**	**0**
Gaussian moment matching	58.2	18.3	4.09
Gumbel-softmax	10.4	8.26	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

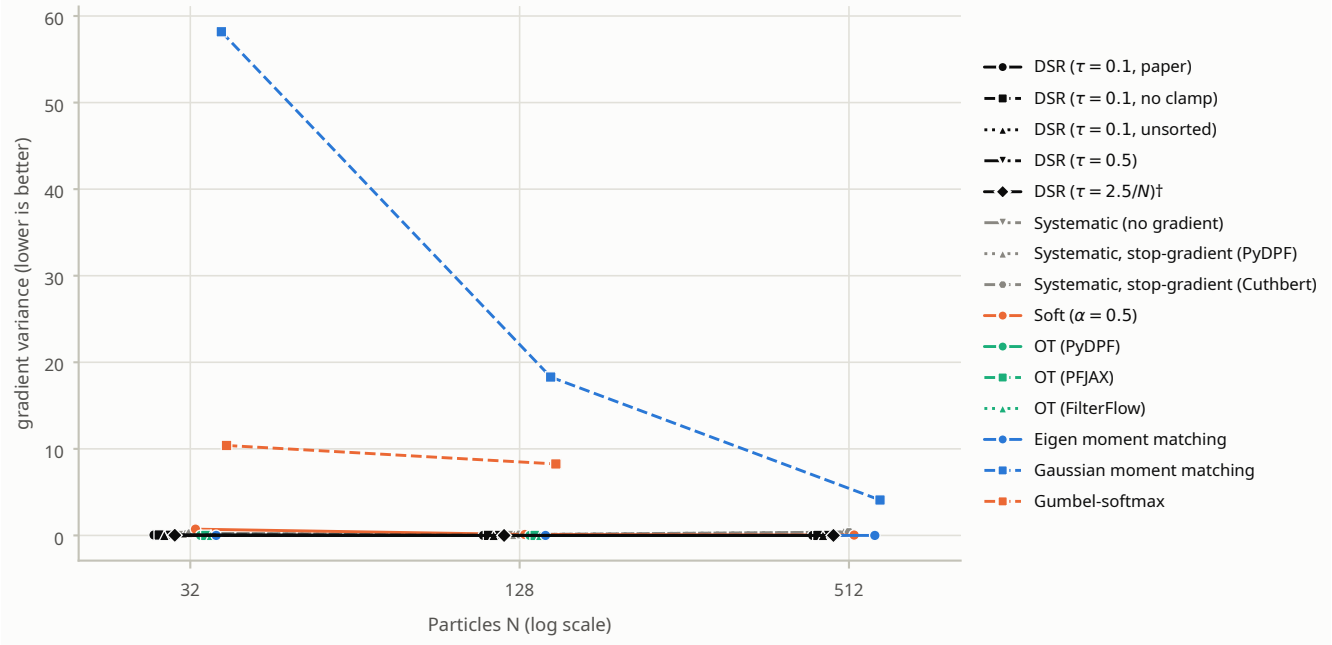


Figure 61: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family.

Table 62: Local gradient, bimodal mixture cloud, $f(x) = \|x\|^2$: gradient MSE (lower is better). Bimodal mixture cloud, D=2, $f(x) = \|x\|^2$; derivative of $\sum_j v_j f(y_j)$ w.r.t. a weight tilt, against the exact weighted-measure derivative; 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512
DSR (τ=0.1, paper)	3.04 ± 0.152	2.58 ± 0.0503	0.873 ± 0.00756
DSR (τ=0.1, no clamp)	3.04 ± 0.152	2.58 ± 0.0503	0.873 ± 0.00756
DSR (τ=0.1, unsorted)	153 ± 0.173	243 ± 0.00956	$208 \pm 2.89 \times 10^{-4}$
DSR (τ=0.5)	194 ± 0.71	284 ± 0.246	238 ± 0.0524
DSR (τ=2.5/N)†	1.05 ± 0.0594	0.0403 ± 0.00178	$0.00396 \pm 1.59 \times 10^{-4}$
Systematic (no gradient)	3.19	0.607	0.148
Systematic, stop-gradient (PyDPF)	0.27 ± 0.0067	0.109 ± 0.00939	0.385 ± 0.0699
Systematic, stop-gradient (Cuthbert)	0.27 ± 0.0067	0.109 ± 0.00939	0.385 ± 0.0699
Soft (α=0.5)	1.82 ± 0.423	0.36 ± 0.0725	0.133 ± 0.0276
OT (PyDPF)	3.39	1.14	–
OT (PFJAX)	19.9	66.9	–
OT (FilterFlow)	20.6	5.78	–
Eigen moment matching	$\mathbf{1.28 \times 10^{-28}}$	$\mathbf{8.29 \times 10^{-29}}$	$\mathbf{1 \times 10^{-29}}$
Gaussian moment matching	57.6 ± 14.9	18.3 ± 4.88	4.05 ± 1.13
Gumbel-softmax	16.6 ± 12.4	63.4 ± 36.7	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

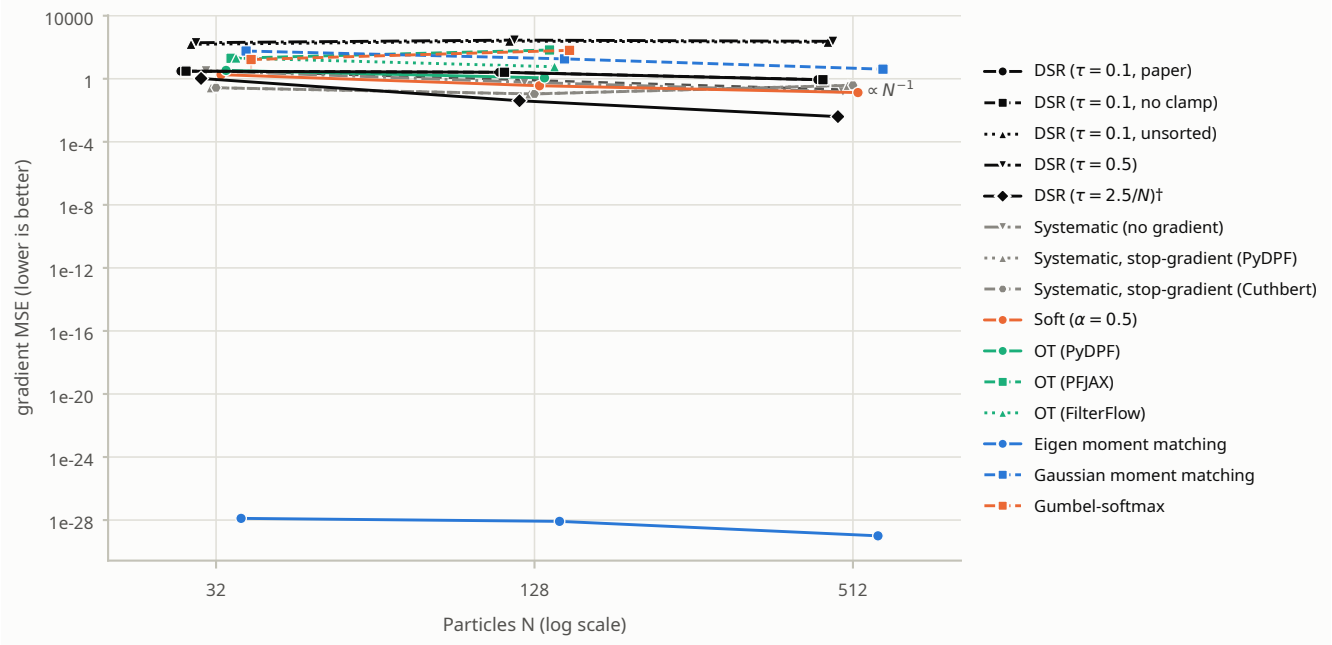


Figure 62: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 63: Local gradient, bimodal mixture cloud, $f(x) = \sum_k \sin x_k$: gradient bias (lower is better). Bimodal mixture cloud, D=2, $f(x) = \sum_k \sin x_k$; derivative of $\sum_j v_j f(y_j)$ w.r.t. a weight tilt, against the exact weighted-measure derivative; 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512
DSR (τ=0.1, paper)	0.0522	0.0801	0.146
DSR (τ=0.1, no clamp)	0.0522	0.0801	0.146
DSR (τ=0.1, unsorted)	4.45	5.14	5.02
DSR (τ=0.5)	4.04	4.76	4.68
DSR (τ=2.5/N)†	0.202	0.0296	0.0177
Systematic (no gradient)	0.232	0.223	0.179
Systematic, stop-gradient (PyDPF)	**0.0221**	0.0106	0.0945
Systematic, stop-gradient (Cuthbert)	**0.0221**	**0.0106**	0.0945
Soft (α=0.5)	0.166	0.115	0.0837
OT (PyDPF)	0.228	0.399	–
OT (PFJAX)	1.69	1.38	–
OT (FilterFlow)	3.86	2.44	–
Eigen moment matching	3.71	4.24	4.37
Gaussian moment matching	0.0694	0.248	**0.00858**
Gumbel-softmax	0.58	1.34	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

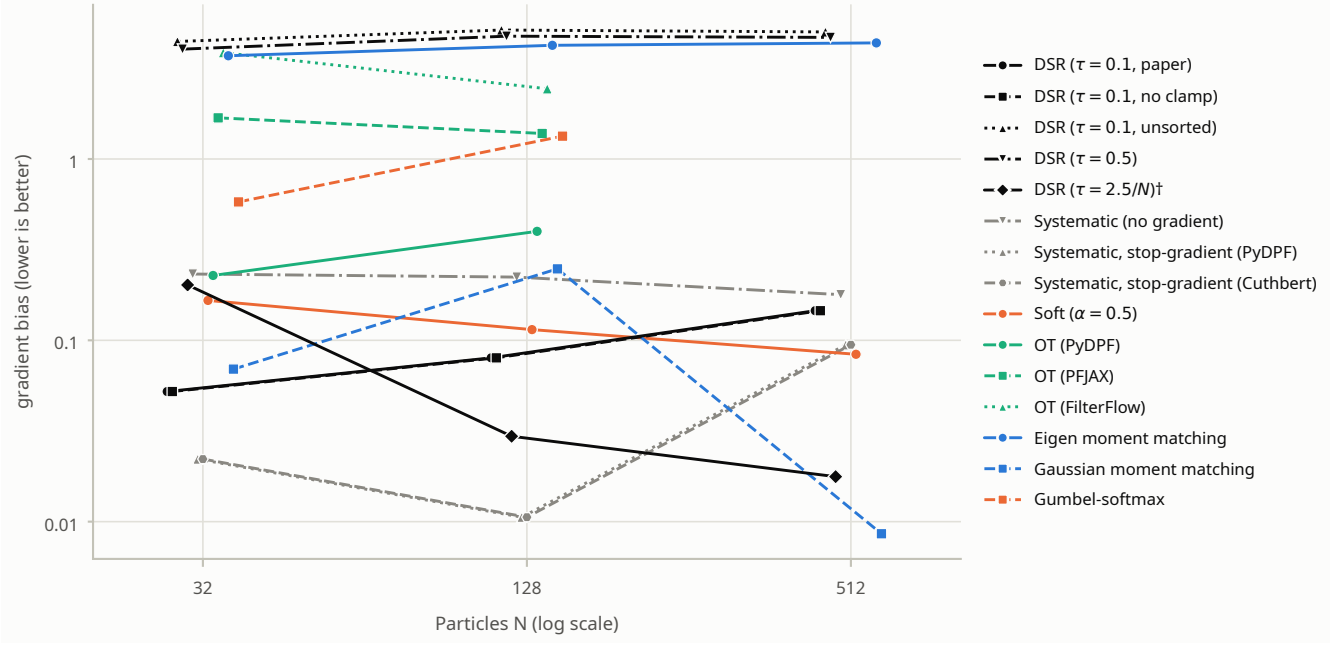


Figure 63: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis.

Table 64: Local gradient, bimodal mixture cloud, $f(x) = \sum_k \sin x_k$: gradient variance (lower is better). Bimodal mixture cloud, D=2, $f(x) = \sum_k \sin x_k$; derivative of $\sum_j v_j f(y_j)$ w.r.t. a weight tilt, against the exact weighted-measure derivative; 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512
DSR (τ=0.1, paper)	5.77×10^{-4}	5.84×10^{-5}	2.74×10^{-6}
DSR (τ=0.1, no clamp)	5.77×10^{-4}	5.84×10^{-5}	2.74×10^{-6}
DSR (τ=0.1, unsorted)	3.58×10^{-4}	5.12×10^{-8}	4.42×10^{-8}
DSR (τ=0.5)	0.00443	3.83×10^{-4}	1.96×10^{-5}
DSR (τ=2.5/N)†	4.18×10^{-5}	6.41×10^{-7}	1.58×10^{-7}
Systematic (no gradient)	**0**	**0**	**0**
Systematic, stop-gradient (PyDPF)	0.00848	2.23×10^{-7}	0.0392
Systematic, stop-gradient (Cuthbert)	0.00848	2.23×10^{-7}	0.0392
Soft (α=0.5)	0.0516	0.0102	0.00231
OT (PyDPF)	**0**	**0**	–
OT (PFJAX)	**0**	**0**	–
OT (FilterFlow)	**0**	**0**	–
Eigen moment matching	**0**	**0**	**0**
Gaussian moment matching	1.36	0.29	0.0612
Gumbel-softmax	1.17	0.204	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

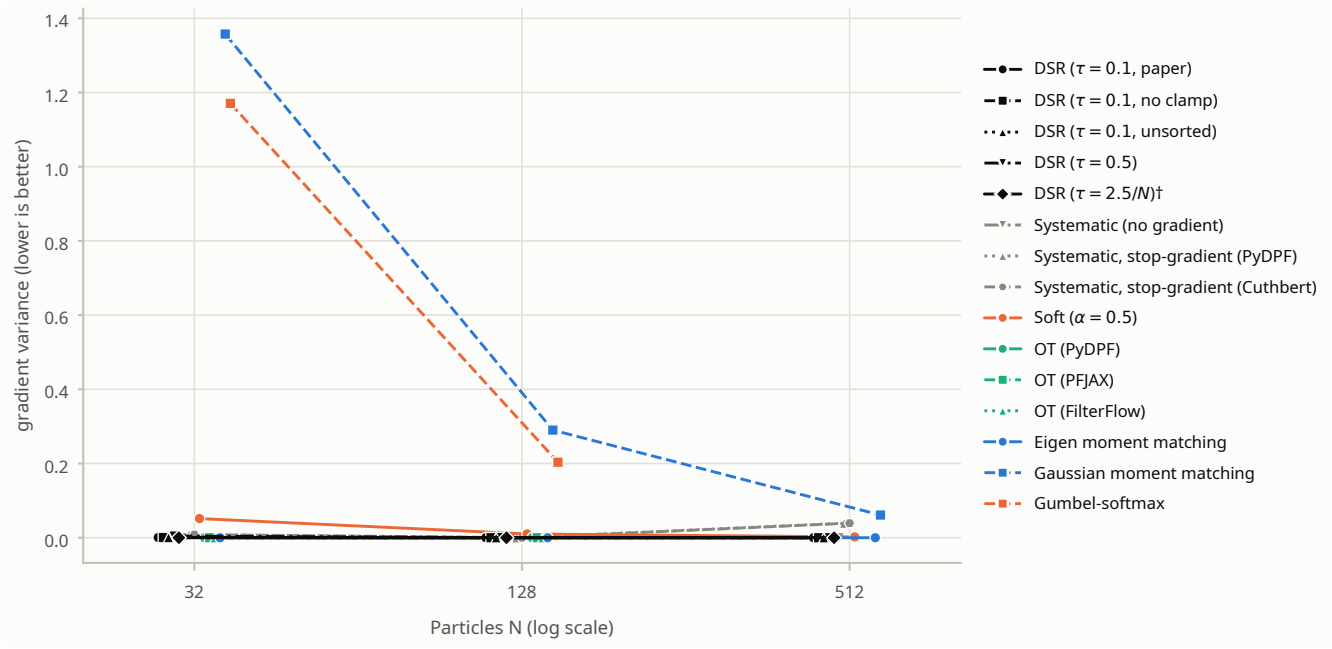


Figure 64: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family.

Table 65: Local gradient, bimodal mixture cloud, $f(x) = \sum_k \sin x_k$: gradient MSE (lower is better). Bimodal mixture cloud, D=2, $f(x) = \sum_k \sin x_k$; derivative of $\sum_j v_j f(y_j)$ w.r.t. a weight tilt, against the exact weighted-measure derivative; 100 seeds (10 for OT and Gumbel).

Method	N=32	N=128	N=512
DSR (τ=0.1, paper)	$\mathbf{0.0033 \pm 5.29 \times 10^{-4}}$	$0.00647 \pm 2.41 \times 10^{-4}$	$0.0212 \pm 9.56 \times 10^{-5}$
DSR (τ=0.1, no clamp)	$\mathbf{0.0033 \pm 5.29 \times 10^{-4}}$	$0.00647 \pm 2.41 \times 10^{-4}$	$0.0212 \pm 9.56 \times 10^{-5}$
DSR (τ=0.1, unsorted)	19.8 ± 0.0334	$26.5 \pm 4.62 \times 10^{-4}$	$25.2 \pm 4.19 \times 10^{-4}$
DSR (τ=0.5)	16.3 ± 0.107	22.6 ± 0.0369	21.9 ± 0.00822
DSR (τ=2.5/N)†	$0.0408 \pm 5.2 \times 10^{-4}$	$8.75 \times 10^{-4} \pm 9.46 \times 10^{-6}$	$\mathbf{3.14 \times 10^{-4} \pm 2.8 \times 10^{-6}}$
Systematic (no gradient)	0.0539	0.0498	0.0319
Systematic, stop-gradient (PyDPF)	0.00889 ± 0.0021	$\mathbf{1.12 \times 10^{-4} \pm 2.02 \times 10^{-6}}$	0.0478 ± 0.0134
Systematic, stop-gradient (Cuthbert)	0.00889 ± 0.0021	$\mathbf{1.12 \times 10^{-4} \pm 2.02 \times 10^{-6}}$	0.0478 ± 0.0134
Soft (α=0.5)	0.0787 ± 0.0204	0.0232 ± 0.0054	0.0093 ± 0.00162
OT (PyDPF)	0.0519	0.159	–
OT (PFJAX)	2.84	1.91	–
OT (FilterFlow)	14.9	5.95	–
Eigen moment matching	13.8	17.9	19.1
Gaussian moment matching	1.35 ± 0.354	0.349 ± 0.0909	0.0606 ± 0.0171
Gumbel-softmax	1.39 ± 0.862	1.97 ± 1.02	–

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

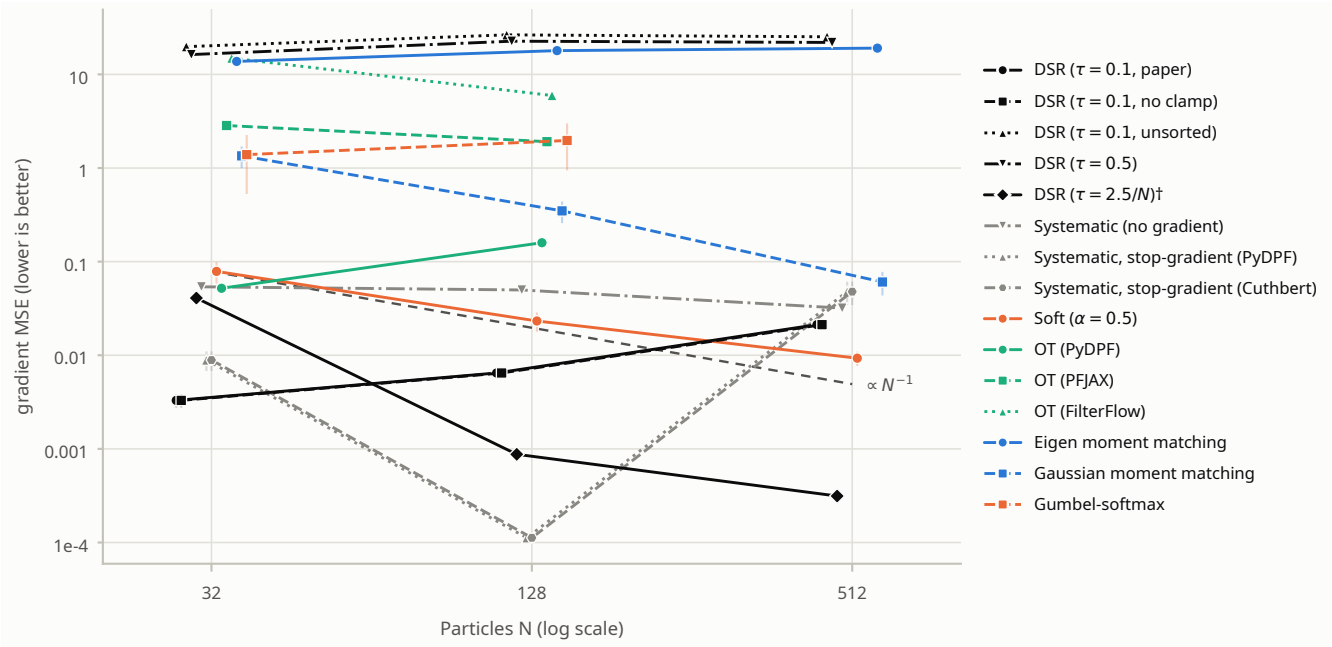


Figure 65: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Whiskers are the table's 95% CIs; points are shifted sideways so whiskers do not overlap. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Runtime

Table 66: Gaussian cloud, D=2: forward time per resampling call in ms (lower is better). Median over 100 calls (3–10 for OT and Gumbel), serial pass on one CPU thread (i7-6700HQ), fresh process per method; includes adapter conversions.

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.899	1.07	7.65	201
DSR (τ=2.5/N)†	0.908	1.08	7.76	201
Systematic	**0.174**	**0.182**	**0.212**	**0.318**
Multinomial	0.19	0.19	0.242	0.426
Systematic, stop-gradient (PyDPF)	0.778	0.785	0.839	1.01
Soft (α=0.5)	0.731	0.812	0.8	1.06
OT (PyDPF)	19.2	22.8	193	2.35×10^4
OT (PFJAX)	2.72	30.9	115	$2.3e+03$
OT (FilterFlow)	4.85	9.3	67	$1.25e+03$
Eigen moment matching	0.818	0.881	0.909	1.01
Gaussian moment matching	0.704	0.736	0.783	0.97
Gumbel-softmax	0.664	0.923	17.2	252
Variational (KL)	0.825	1.78	11.6	39.2

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

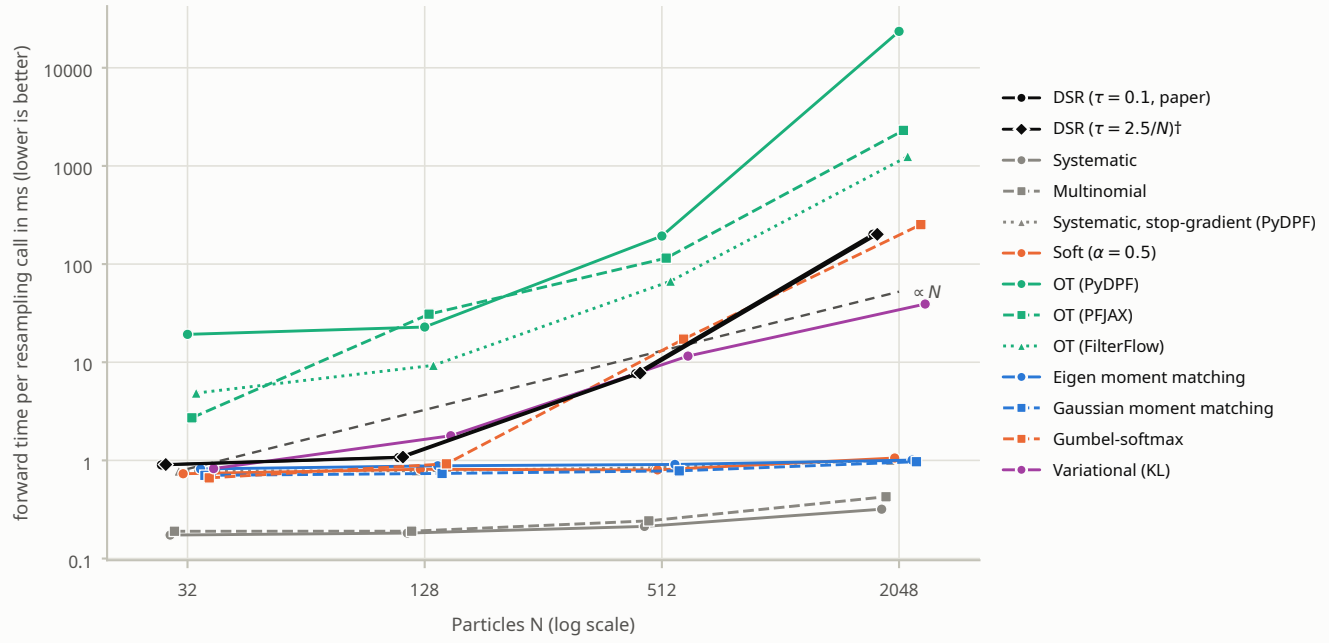


Figure 66: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Table 67: Bimodal mixture cloud, D=2: forward time per resampling call in ms (lower is better). Median over 100 calls (3–10 for OT and Gumbel), serial pass on one CPU thread (i7-6700HQ), fresh process per method; includes adapter conversions.

Method	N=32	N=128	N=512	N=2048
DSR (τ=0.1, paper)	0.929	1.07	7.7	201
DSR (τ=2.5/N)†	0.909	1.08	7.61	200
Systematic	**0.173**	**0.184**	**0.211**	**0.302**
Multinomial	0.181	0.276	0.258	0.435
Systematic, stop-gradient (PyDPF)	0.775	0.792	0.815	1.02
Soft (α=0.5)	0.721	0.754	0.818	1.06
OT (PyDPF)	20.8	32.9	321	–
OT (PFJAX)	3.29	26.7	99.3	$1.96e+03$
OT (FilterFlow)	5.7	11.5	101	–
Eigen moment matching	0.822	0.897	0.914	1.03
Gaussian moment matching	0.715	0.735	0.779	0.947
Gumbel-softmax	0.687	0.917	15.1	249
Variational (KL)	1.05	1.8	12.1	42.5

†Local diagnostic, not from the paper: τ rescaled so that τN equals the paper's value at N=25. DSR variants use Morton sorting unless marked unsorted. A dash means not run (dense methods stop at N=128). Bold marks the best mean per column, not a statistically significant difference.

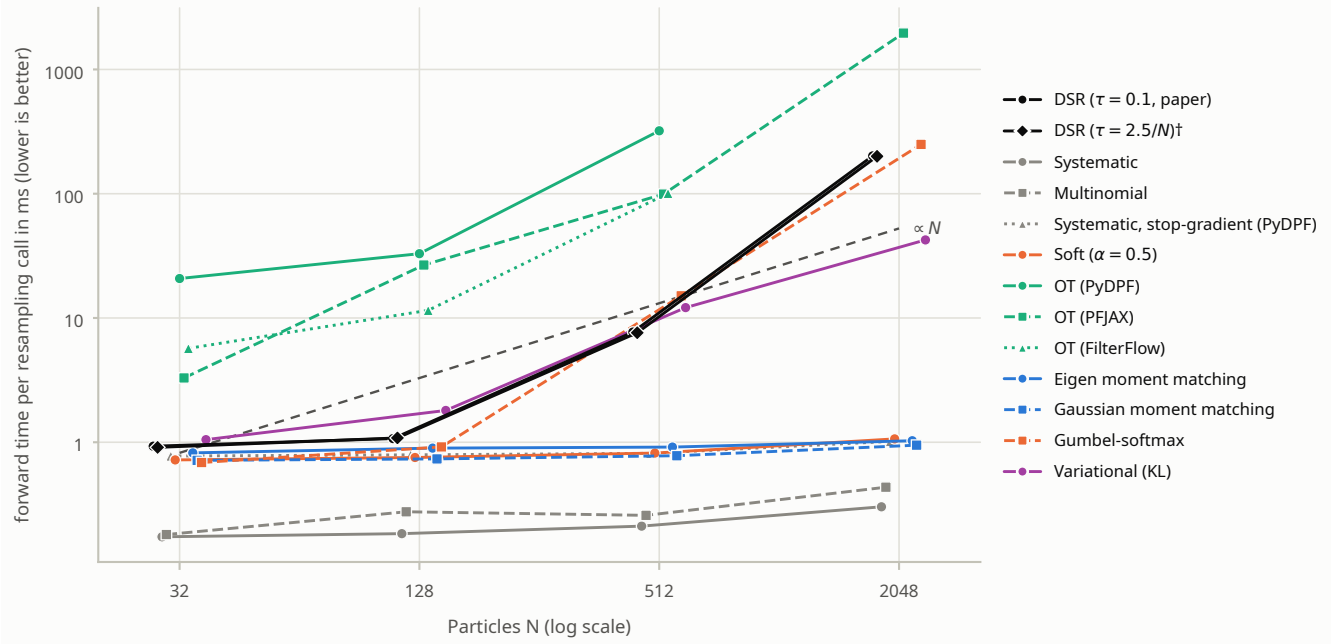


Figure 67: **(a)** The values of Table **??** against N (log scale), one line per row. Colour is the method family: black DSR, gray classical, blue moment matching, orange soft/Gumbel, green optimal transport, purple variational; line style and marker separate rows within a family. Log y axis. The dashed line is the Monte Carlo reference slope; a method whose line flattens has an error floor.

Discussion: trends in N and across tasks

Unless stated otherwise, a log–log slope is the least-squares slope of log(error) against $\log N$ over the available particle counts. A Monte Carlo resampler that copies ancestors should give a slope near -0.5 for distribution errors (rate $N^{-1/2}$). A slope near 0 means the error has a floor that more particles do not remove.

Trends in N, method by method

DSR, paper setting (τ=0.1, Morton; with or without gradient clamp). The clamp only acts in the backward pass, so the two variants are identical in every forward table. Their error does not decrease with N on any fixed cloud: the sliced-W1 slope is -0.02 (Gaussian), $+0.02$ (mixture), -0.03 (skewed) and -0.05 (fixed grid) (Tables **??**, **??**, **??**, **??**). The covariance error stays between 0.92 and 1.16 on the Gaussian cloud (Table **??**). The cause is that τ is measured in CDF units while the comb spacing is $1/N$. A fixed τ therefore averages over a number of ancestors that grows like τN, and the resampled cloud is contracted by a fixed amount whatever N is.

In the filters, the floor appears whenever the task is sensitive to the posterior spread:

- Linear Gaussian: the posterior-mean error against the Kalman filter goes 0.138 → 0.034, which is twice systematic's 0.017 at N=2048 (Table **??**).
- UNGM: the RMSE stops improving after N=512 (4.79 → 4.78, Table **??**).
- Bearing tracking: the RMSE gets *worse* from N=128 to N=2048 (4.83 → 5.19), and coverage stays at 0.06–0.09 (Tables **??**, **??**).

The likelihood-gradient bias in the differentiable PF falls from 19.3 to 7.1 and then plateaus, so its MSE settles at about 51 from N=512 on (Tables **??**, **??**).

The gradient clamp matters only at small N. Without it, the gradient MSE is 124 instead of 433 at N=32; from N=512 the two coincide. The log-likelihood variance falls at the normal $1/N$ rate (3.6 → 0.064, Table **??**). So the problem is bias, not noise.

DSR without Morton sorting, and DSR with τ=0.5. Both average over spatially unrelated ancestors and behave alike. They are the worst methods in almost every column and do not improve with N:

- On the mixture cloud every draw loses both modes (mode coverage 0, Table **??**), and the covariance error stays at about 8.4.
- The UNGM RMSE *increases* with N (7.5 → 8.7).
- The likelihood-gradient MSE plateaus near 400–440.

Morton sorting is therefore what keeps the paper-setting DSR usable.

DSR with τ=2.5/N (diagnostic, not from the paper). Holding τN constant restores consistency, and the rate is if anything slightly better than Monte Carlo:

- The sliced-W1 slopes are -0.56, -0.64, -0.59 and -0.62 on the four clouds.
- On the mixture cloud the covariance error falls with slope -0.79, and mode coverage stays at 1.
- The likelihood-gradient bias falls from 17.1 to 0.064 and the MSE from 351 to 0.31. From N=512 it beats stop-gradient systematic, whose MSE at N=2048 is 8.9.
- In the filters it tracks systematic resampling closely (UNGM RMSE 4.52 vs. 4.48 at N=2048).

The exception is bearing tracking, where coverage rises steadily (0.08 → 0.68) but stays below the copy-based methods ($\approx$ 0.90). Smoothing over about 2.5 CDF neighbours still under-disperses a 4-D cloud.

Systematic, multinomial and stop-gradient systematic. These are the textbook Monte Carlo rates. Sliced-W1 slopes are -0.47 to -0.54 (systematic) and -0.44 to -0.49 (multinomial). Systematic has a lower constant than multinomial: half or less of its W1 at equal N.

The single exception is the mixture cloud. There, systematic resampling (and every systematic implementation) has a flat W1 of about 0.89, and only 84–87% of draws keep both modes, at every N. The weights of that cloud alternate between two values in index order, and a single uniform comb aliases with that period. Multinomial and soft resampling, which draw independently, do not have this failure.

For gradients, stop-gradient systematic is close to unbiased in the local tests: its bias is at most 0.16 and is below 0.05 in most cells. On the Gaussian cloud its MSE falls steadily with N. In the PF-likelihood task it is unbiased but high-variance: the MSE falls from 561 to 8.9, driven by variance. Plain systematic resampling with no gradient path has a constant bias of about 12, because it misses the whole score term.

Soft resampling (α=0.5). Its distribution errors follow the Monte Carlo rate (slopes -0.44 to -0.48), with a constant similar to multinomial. In the filters it is close to systematic for $N \geq 512$. Its gradient is biased, and the bias does not vanish with N:

- The local gradient bias on the Gaussian cloud with $f = \|x\|^2$ is 0.58, 0.41 and 1.02 at $N = 32$, 128 and 512.
- The PF-likelihood gradient bias plateaus at about 5.5–6.0.

Entropic OT (PyDPF, PFJAX, FilterFlow). These run only to N=128, so the trends are short. Their distribution errors fall more slowly than Monte Carlo (sliced-W1 slopes -0.15 to -0.25 on the Gaussian cloud, about 0 on the mixture), the signature of entropic blur at fixed ε. As with fixed-τ DSR, the blur contracts the cloud: bearing coverage falls to 0.04–0.05 at N=128 and the bearing RMSE gets worse from N=32 to N=128. They have the largest log-likelihood error at small N on the linear model (-1.3 at N=32). The three implementations differ: PFJAX gives the most accurate local gradients on the Gaussian cloud (MSE 8×10^{-4}), while PyDPF and FilterFlow are biased by 0.4–0.55.

Eigen and Gaussian moment matching. These match the weighted mean and covariance by construction. Eigen matches them exactly (covariance error about 10^{-15}), and its local gradients for $f = \|x\|^2$ are exact. On Gaussian-shaped clouds their W1 falls at about the Monte Carlo rate (slopes -0.45 and -0.50). On the mixture and fixed-grid clouds W1 is nearly flat (slopes between -0.15 and $+0.02$), because a Gaussian map cannot reproduce non-Gaussian shape.

In the filters this trade-off pays off. They give the best-calibrated bearing filters at small N (coverage 0.51 and 0.36 at N=32, against 0.27 for systematic) and the most accurate PF-likelihood gradients at every N (eigen MSE $44 \to 0.27$, bias $3.7 \to 0.08$). On UNGM, whose posterior is bimodal, their RMSE improves more slowly than the copy-based methods (eigen 4.83 vs. systematic 4.48 at N=2048).

Gumbel-softmax. Like fixed-τ DSR this is a relaxation with a fixed temperature, and its static errors are flat in N (slopes about 0). It runs only to N=128, where its bearing coverage is 0.22 and its local and PF gradients are biased (PF gradient MSE 85).

Variational (KL) resampling. This is deterministic offspring allocation, so it has zero variance between draws but a rounding bias. On the skewed cloud it converges at the Monte Carlo rate (W1 slope -0.56) and is among the two most accurate methods at N=512 and 2048. On the Gaussian and fixed-grid clouds its W1 is almost flat (slopes -0.06 and -0.04): the deterministic rounding error does not average out over repeated draws. In the filters, the fresh propagation noise removes this effect, and it matches systematic resampling on every model.

Trends across tasks

Linear Gaussian SSM: least discriminative. The state RMSE is set by the observation noise. Every method lies between 0.519 and 0.530 at N=32 and between 0.503 and 0.507 at N=2048, inside each other's confidence intervals at every N (Table **??**). Differences appear only in the comparisons with the exact Kalman filter: the posterior-mean RMSE and the log-likelihood error. These separate consistent methods (floor $\to$ 0.017) from fixed-temperature relaxations (DSR 0.034, unsorted DSR 0.074).

Stochastic volatility: insensitive to resampling. All RMSEs lie within 0.41–0.43. Here the paper-setting DSR has the lowest RMSE for $N \geq 128$, and its coverage (0.95) is the one closest to nominal. The copy-based methods over-cover for $N \geq 512$ (0.97–0.98), so the slight contraction from DSR happens to help on this heavy-tailed, weakly observed model. Parameter learning shows the same flatness: estimated scale errors of 0.20–0.28 for every method, all within the confidence intervals (Table **??**).

UNGM: tests multimodality. The posterior is often bimodal, and methods that average particles pay for it. Fixed-τ DSR plateaus, unsorted DSR and τ=0.5 degrade with N, and eigen/Gaussian moment matching converge more slowly. Copy-based methods and scaled DSR keep improving down to RMSE ≈ 4.5.

Bearing-only tracking: the most discriminative task. The state is 4-D, the observations are sharp (0.05 rad), and there is a ten-step observation gap. Under-dispersion is punished hard. All methods that move particles towards local averages have very low coverage at every N where they ran: fixed-τ DSR (0.06–0.09), entropic OT (0.04–0.09) and Gumbel (0.20–0.22). The copy-based methods improve to ≈ 0.90, and moment matching is best calibrated at small N.

Fixed clouds: shape versus moments. The Gaussian, skewed and fixed-grid clouds rank methods mainly by Monte Carlo constant and rate. The mixture cloud separates methods that copy particles from methods that move them:

- Copy-based multinomial and soft resampling keep both modes.
- Systematic resampling suffers the periodic-weight aliasing described above.
- Moving methods either keep both modes (Morton-sorted DSR, OT, moment matching) or place all their mass between the modes (unsorted DSR, DSR with τ=0.5).

Gradients: bias is the axis that separates methods.

- Local expectation gradients reward exact moment maps: eigen is exact for $f = \|x\|^2$ but biased for $f = \sin$ on the mixture.
- Stop-gradient estimators are close to unbiased.
- In the PF-likelihood task, which accumulates resampling over 50 steps, only methods whose bias vanishes with N reach small MSE: eigen, Gaussian moment matching, scaled DSR and stop-gradient systematic.
- Methods whose bias does not vanish stay at large MSE: about 51 for paper-setting DSR and 38 for soft resampling at N=2048. Gumbel (85) and PyDPF OT (845) are also high at N=128, the largest N they ran.

Runtime. Classical copy-based resamplers, soft resampling and moment matching cost about 1 ms per call or less at every N (Tables **??**, **??**). Variational KL grows to about 42 ms at N=2048 (its allocation is $O(N^2)$). DSR and Gumbel build an $N \times N$ matrix. Their cost grows by about 7× from N=128 to 512 and 26× from 512 to 2048, reaching about 200 and 250 ms at N=2048. Entropic OT is a further 10–100× slower than DSR at large N (PFJAX about 2.3 s and PyDPF about 23 s at N=2048).

Stochastic Lorenz models (all methods)

These two tasks were run for every method of every repository (Tables **??**–**??**).

Lorenz-63 (camera observation, as in the DSR paper). The task is low-dimensional and well observed, so every copy-based method converges to the same answer. At N=2048, all classical resamplers, soft resampling, variational allocation and moment matching lie between 0.429 and 0.432 in RMSE, with coverage 0.945–0.953. The differences at N=32 (0.531–0.666) shrink steadily with N, and moment matching (PFJAX MVN 0.531, eigen 0.533) is best at small N.

The paper-setting DSR shows the same floor as on the earlier tasks:

- RMSE goes $0.708 \to 0.491 \to 0.486$ from N=32 to 512 to 2048.

- Coverage reaches only 0.68.
- Unsorted DSR and τ=0.5 are worse (0.514 and 0.520, coverage about 0.59).

With τ=2.5/N, DSR reaches 0.433 and coverage 0.93, close to the copy-based methods. At N=128, entropic OT (0.468–0.491) and Gumbel-softmax (0.484) trail the copy-based methods (0.456–0.469), and their coverage is lower (0.70–0.81 vs. 0.81–0.86).

Lorenz-96 (d=10, full observation). This is the hardest task in the suite: the mean ESS is only about 10% of N, and no method has converged by N=2048. The copy-based RMSE is still falling ($\approx 2.8 \to 1.06$–1.13) and their coverage is only 0.52–0.56.

Moment matching is clearly best at every N:

- PFJAX MVN reaches RMSE 0.735 (coverage 0.75), eigen 0.849 and Gaussian moment matching 0.865 (coverage about 0.71) at N=2048.
- In 10 dimensions, moving all particles to match the weighted mean and covariance fights weight degeneracy better than copying a few heavy particles.
- PFJAX MVN pays with numerical failures at small N: 9% of runs at N=32 and 3% at N=128, which are excluded from its means.

The paper-setting DSR fails here: RMSE 1.90 and coverage 0.01 at N=2048. It is indistinguishable from unsorted DSR (1.90), because Morton locality is diluted in 10 dimensions. Scaled DSR recovers the copy-based RMSE (1.08) but remains under-dispersed (coverage 0.27 vs. about 0.55). At N=128, entropic OT and Gumbel-softmax are again below the copy-based methods (RMSE 2.19–2.52 vs. 1.99–2.14, coverage at most 0.034).

What the Lorenz runs add. They confirm the earlier conclusions on two standard chaotic benchmarks:

- Fixed-τ DSR hits a floor, and $\tau \propto 1/N$ removes most of it.
- Methods that average particles under-cover.
- Copy-based resamplers are interchangeable once N is large.

They also show the one setting where moving particles helps: in higher-dimensional, degenerate filtering (Lorenz-96), *moment matching* beats every copy-based method. DSR does not share that benefit, because its CDF ordering loses locality as the dimension grows.

Conclusions

The main conclusion is that DSR as published fails at large N because its temperature τ is fixed, not because of the relaxation idea itself. The points below run roughly from strongest to most tentative.

About DSR.

1. *At the paper's τ=0.1, DSR is not a consistent resampler.* Its error has a floor that more particles do not remove. This shows up in the fixed clouds, in the filters' posterior error against the Kalman filter, in bearing calibration, and in a likelihood-gradient bias stuck at about 7. The cause is structural: a fixed τ averages over about τN ancestors, so the larger N is, the more particles are blended. The paper says DSR "scales gracefully" with τ fixed at 0.1. That claim concerns training NMSE up to N=200; these accuracy results do not support extending it.
2. *The relaxation itself works if τ shrinks with N.* With τ=2.5/N:
   - errors fall at or slightly faster than the Monte Carlo rate $N^{-1/2}$;
   - the likelihood-gradient bias goes to about 0 (MSE 0.31 at N=2048);
   - filtering matches systematic resampling.

   The published method therefore needs an N-dependent temperature, roughly $\tau \propto 1/N$. The constant 2.5 was set from the paper's N=25 operating point, not tuned, and bearing calibration still lags (0.68 vs. 0.90).

3. *Morton sorting is essential, and the gradient clamp is not useful.* Without sorting, DSR places all mass between the modes and gets worse with N. The clamp changes nothing at large N and raises the gradient MSE at small N (433 vs. 124 at N=32).

4. *The paper's bias-for-variance argument holds, but only at small N.* DSR's gradients have very low variance. That helps at $N \approx 25$–100, where unbiased estimators such as stop-gradient systematic are very noisy (MSE 561 at N=32). As N grows that variance disappears anyway and DSR's fixed bias dominates; stop-gradient systematic is already better from N=512. This fits the paper's own settings (N=25–200) and explains why its gains need not carry over to larger N.

Across methods.

5. *Moving particles systematically under-disperses them.* Fixed-τ DSR, entropic OT and Gumbel-softmax build each new particle from a weighted average of several ancestors. This shrinks the cloud, which is harmless on easy tasks but collapses calibration on sharp, higher-dimensional problems: in bearing tracking their coverage is at most 0.22, against about 0.90 for copy-based methods. Calibration, not RMSE, is the main risk of differentiable resamplers.

6. *Moment matching is the strongest differentiable baseline in this benchmark.* Eigen and Gaussian moment matching give the best likelihood gradients at every N, the best calibration at small N, and cost about 1 ms per call. Their weakness is shape: they cannot represent multimodal or non-Gaussian clouds, and they converge more slowly on UNGM. At large N, eigen matching beats paper-setting DSR on both cost and gradient accuracy.

7. *Bias, not variance, decides PF-likelihood gradients as N grows.* Only methods whose bias vanishes with N reach small MSE: moment matching, scaled DSR and stop-gradient systematic. Soft resampling, fixed-τ DSR, Gumbel-softmax and OT stay at MSE of about 40 or more.

8. *Classical resamplers remain the accuracy reference for forward filtering,* with one exception. Systematic resampling fails on the periodic-weight mixture cloud, where 13–16% of draws lose a mode; multinomial and soft resampling do not.

9. *Cost.* DSR is about 10–100× cheaper than entropic OT but is $O(N^2)$: about 1 ms at $N \leq 128$ and about 200 ms at N=2048, against about 1 ms for copy-based methods and moment matching.

About the benchmark.

10. *Linear-Gaussian RMSE and stochastic volatility barely separate resamplers.* Methods fall within each other's confidence intervals, and even SV parameter learning shows no significant differences with 10 datasets. The tasks that separate methods are bearing tracking, the bimodal cloud, UNGM, and the PF-gradient task against the exact Kalman gradient; future resampler evaluations should weight those.

Limits on these conclusions.

- Everything ran on CPU. OT and Gumbel-softmax only ran to N=128, so their large-N trends are extrapolated.
- The paper's training experiments were not reproduced, and that is where DSR's low-variance gradients are claimed to matter most. These results therefore do not refute its training results. They show that the method does not extend to accurate filtering or unbiased gradients at large N without rescaling τ.
- The scaled variant is a single, untuned diagnostic. A proper $\tau(N)$ study, for example τ=c/N over several values of c, is needed before recommending it.